



Birzeit University

Department of Computer Science

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BZU Assistant

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Abstract

Smartphones are now the leading computing devices for students and instructors, daily academic use of desktops and laptops being left far behind. At Birzeit University in particular this transition was a reason for the creation of BZU Assistant, an Android mobile app that erases paper ID cards and makes major campus services available in a single integrated system. The application has capabilities like taking attendance automatically, scheduling office hours, managing academic events, reserving classrooms and devices, viewing class schedules, and sharing individual timetables.

Attendance is recorded automatically using Bluetooth Low Energy (BLE) technology, where Raspberry Pi devices installed in classrooms detect nearby Android devices and register attendance without manual intervention. Office hours, devices, classrooms, and academic events are managed through QR code scanning or an in-app search interface. Instructors can generate events directly from the application, producing a unique QR code that students scan to instantly add the event to their mobile calendars.

The Android application provides real-time, personalized academic schedules directly to students and instructors thus removing paper-based methods and scattered communication channels. The project was focused on usability, reliability, and actual university environment for the implementation. The BZU Assistant illustrates that an Android first, mobile centric approach coupled with BLE, QR-based interactions, and calendar integration can facilitate academic workflows, enhance resource utilization, and support the creation of a more intelligent, efficient digital campus at Birzeit University.

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Chapter 1

Introduction

The BZU Assistant project was developed as a unified digital identification and service platform for Birzeit University, aiming to modernize and streamline essential academic and administrative processes across campus. The system is designed as an Android-based mobile application that consolidates multiple services previously handled through separate, manual, or paper-based systems into a single, coherent platform accessible by students, instructors, and secretaries.

The motivation for this project emerged from persistent inefficiencies experienced daily on campus. Traditional attendance recording methods rely heavily on manual verification, leading to time loss, inaccuracies, and administrative overhead. Similarly, device and classroom reservations are often handled through fragmented or informal mechanisms, while office hour coordination lacks a centralized scheduling solution. In addition, academic schedules and announcements are typically communicated through multiple disconnected channels, resulting in missed information and poor synchronization between instructors and students.

Recognizing these challenges, our team identified an opportunity to replace legacy workflows with a mobile first, automated solution aligned with Birzeit University's vision for a smart campus infrastructure. Smartphones already the primary computing devices for most users provide an ideal foundation for delivering real time, context aware academic services.

To realize this vision, the BZU Assistant was designed using a modular and domain-driven architecture, ensuring that each feature directly addresses a real campus need while remaining scalable and maintainable. The system integrates the following core functional domains:

1. Digital ID and Authentication

Secure user authentication and profile management for students, instructors, and secretaries, serving as the foundation for role-based access across the system.

2. Attendance Tracking

Automated student attendance recording using Bluetooth Low Energy (BLE), where Raspberry Pi devices installed in classrooms detect nearby student devices and register attendance without manual intervention.

3. Device and Classroom Reservation

QR code-based booking and check-in/check-out for laboratory equipment and classrooms, improving resource availability, traceability, and utilization.

4. Office Hours Management

Centralized appointment scheduling with instructors through QR code scanning or an in-app search interface, allowing students to book, or cancel meetings efficiently.

5. Academic Schedule Viewing and Sharing

Real-time access to personalized course schedules, with the ability to share timetable information using dynamically generated QR codes.

6. Academic Event Generation and Calendar Integration

Instructors can create academic events such as lectures, seminars, or announcements—directly within the application. Each event generates a unique QR code that students can scan to instantly add the event to their mobile calendars, ensuring accurate scheduling and reducing communication gaps.

1.1 Aim & Objectives

1.1.1 Aim

The aim of this project is to design and implement an integrated Android-based Digital ID mobile application for Birzeit University that digitizes and centralizes essential academic and administrative services. The system seeks to improve operational efficiency, reduce manual processes, and enhance the overall campus experience for students, instructors, and secretaries.

1.1.2 Objectives

The objectives of the BZU Assistant project are to:

1. Replace physical university ID cards with a secure, application-based digital identity system for all user roles.
2. Implement automated student attendance tracking using Bluetooth Low Energy (BLE) technology in conjunction with Raspberry Pi devices to enable accurate and real-time check-ins.

3. Develop device and classroom reservation mechanisms based on QR code technology, supporting efficient booking, check-in/check-out, and resource usage tracking.
4. Provide a centralized office hours management system that allows students to view instructor availability and schedule appointments through QR code scanning or an in-app search interface.
5. Enable real-time access to personalized course timetables, allowing users to view and manage daily and weekly academic schedules through a unified interface.
6. Support academic schedule and event sharing by generating dynamic QR codes, enabling coordination for lectures, meetings, and classes.
7. Allow instructors to create academic events within the application and generate QR codes that students can scan to instantly add events to their mobile calendars.
8. Reduce paperwork and manual administrative processes by digitizing routine academic interactions, thereby saving time and minimizing human errors.
9. Support Birzeit University's digital transformation initiatives by delivering a practical, scalable, and campus-ready technological solution aligned with real operational needs.

1.2 Overview of Technical Areas

The BZU Assistant system integrates a focused set of technical components to deliver a unified and efficient mobile-based campus management platform. The architecture is intentionally streamlined to support the project's core services attendance tracking, device and classroom reservations, office hour scheduling, academic event management, and schedule sharing—while remaining practical for real world deployment at Birzeit University. The system is built around three primary technical pillars: a Flutter-based mobile application, a Supabase backend platform, and Raspberry Pi devices for BLE-based attendance tracking.

The mobile application is developed using Flutter, which serves as the primary interface for all users, including students, instructors, and administrative staff. Flutter is responsible for rendering the user interface, managing user interactions, handling QR code scanning, communicating with backend services, and coordinating Bluetooth-based attendance detection. The application is designed specifically for Android devices to ensure stable BLE background execution and consistent hardware behavior.

Supabase is used as the backend platform for data storage, authentication, and real-time services. It provides a centralized database for managing user profiles, schedules, reservations, attendance records, and academic events. Supabase also supports real-time updates and push notifications, enabling instant synchronization of schedule changes, event announcements, and reservation confirmations across the system.

Attendance tracking is implemented using Bluetooth Low Energy (BLE) technology in combination with Raspberry Pi devices installed in classrooms and laboratories. Each Raspberry Pi functions as a BLE scanning node, continuously detecting nearby student smartphones during lecture sessions. When a student's device is detected within range, attendance data is securely transmitted to the backend for processing and storage, eliminating the need for manual check-ins.

The Raspberry Pi devices run lightweight Python-based BLE scanning scripts, enabling efficient and continuous device detection. This approach ensures reliable attendance recording while maintaining low power consumption and minimal hardware overhead.

By integrating Flutter for the mobile application layer, Supabase for backend services and notifications, and Raspberry Pi devices for BLE-based automation, the BZU Assistant system delivers a cohesive, scalable, and campus-ready technical solution. This architecture supports real-time interaction, minimizes manual intervention, and provides a solid foundation for future system expansion.

1.3 Overview of the Report

This report is organized into a series of chapters that present the BZU Assistant project in a logical and progressive manner, starting from background concepts and moving through system design, implementation, evaluation, and final conclusions.

Chapter 2 presents the background and related work relevant to the project. It reviews existing digital ID systems, mobile-based campus solutions, and Bluetooth Low Energy (BLE) techniques for automated attendance tracking, providing the theoretical foundation for the proposed system.

Chapter 3 focuses on the system design of the BZU Assistant. This chapter describes the overall system architecture, major system components, data flow, and user interactions, highlighting how the mobile application, backend services, and hardware elements integrate to support the system's functionality.

Chapter 4 details the implementation of the system. It explains how the proposed design was realized using Flutter for the mobile application, Supabase for backend services and notifications, and Raspberry Pi devices for BLE-based attendance tracking. Key implementation challenges and design decisions are also discussed.

Chapter 5 presents the testing and evaluation of the system. This chapter outlines the testing strategy, test cases, and validation methods used to verify system functionality, reliability, and performance under real-world conditions.

Chapter 6 concludes the report by summarizing the achieved results, discussing system limitations, and identifying potential directions for future improvements and extensions.

Chapter 2

Background

The chapter presents the theoretical foundations and prior research that inform the design decisions of the BZU Assistant system. It provides the necessary background on digital identity systems, Bluetooth Low Energy (BLE) communication, QR code technology, and mobile-based campus automation, which together form the conceptual basis of the proposed solution.

The chapter begins by introducing the core theoretical concepts relevant to the project, focusing on mobile digital identification, Bluetooth Low Energy (BLE) based proximity detection, and QR code driven interactions. These technologies are discussed from a conceptual perspective to explain their suitability for academic environments and large scale campus deployment, without delving into implementation-specific details.

In addition, this chapter reviews related academic and commercial work in the areas of BLE-based attendance systems, indoor positioning approaches, and QR code-based academic services. The discussion highlights existing solutions, their advantages, and their limitations, providing context for the design choices made in the BZU Assistant system. Advanced indoor positioning techniques, such as fingerprinting and machine-learning-based localization, are presented as related work to contrast them with the proximity-based approach adopted in this project.

Finally, the chapter outlines the main tools and technologies used in the system at a high level, including the mobile application framework, backend services, and hardware components. Together, this background establishes the technical feasibility of the system and provides a clear rationale for the selected technologies and architectural approach, paving the way for the system design and implementation chapters that follow.

2.1 Details & Relevant Theory

2.1.1 Digital Identity System

Identification is a fundamental requirement in university environments, where students, instructors, and staff must be verified to access academic and administrative services such as course registration, attendance tracking, and resource reservations. Traditionally, these functions rely on physical identification cards. While physical authentication is generally reliable, it is prone to loss, impersonation, and misuse, which can compromise both security and operational efficiency.

With the rapid advancement of information technology, research efforts have increasingly focused on digitizing identity management systems. In digital environments, identity management systems are typically structured around three core functions: identification, authentication, and authorization, which together ensure that users are correctly recognized and granted appropriate access [5]. These systems rely on digital credentials such as usernames, passwords, cryptographic tokens, or biometric data to verify identity.

Several architectural models for digital identity have been proposed. Centralized identity systems store and manage identity data under the control of a trusted authority, such as a university. Federated identity systems enable users to access multiple services using a single identity provider. More recently, Self-Sovereign Identity (SSI) has emerged as a user-centric model in which individuals retain control over their identity credentials and selectively disclose attributes as needed [1]. While SSI offers strong privacy and autonomy, it often introduces additional technical and organizational complexity.

Privacy-preserving identity solutions have been further explored through attribute-based credential systems, such as those proposed by the ABC4Trust project [2]. These systems emphasize data minimization by allowing users to reveal only the required identity attributes during authentication, reducing unnecessary exposure of personal information.

The importance of digital identity systems in education became especially evident during the COVID-19 pandemic, which accelerated the transition to remote and hybrid learning environments. Digital student ID systems enabled universities to authenticate users, monitor academic participation, and provide access to services without physical presence [6]. Such systems, often deployed as mobile applications, reduce dependency on physical infrastructure and increase flexibility.

Modern identity solutions are increasingly mobile-based, leveraging smartphones as secure carriers of identity credentials. Mobile identity platforms integrate both physical and digital identity verification and support real-time authentication [3]. Strong authentication mechanisms, including multi-factor and SIM-based authentication, have also been explored to enhance security in digital identity systems [5]

Despite their advantages, the adoption of digital identity systems presents challenges related to security, privacy, user acceptance, and institutional integration. These challenges

have been analyzed in recent studies examining large-scale identity deployment across sectors such as education, banking, and public services [4]

Based on these considerations, a centralized digital identity model is well suited for institutional environments such as universities, where trusted infrastructure and existing authentication systems already exist. This approach balances security, usability, and scalability while enabling seamless integration with academic services.

2.1.2 Bluetooth Low Energy (BLE) for Classroom Attendance

Attendance tracking in university classrooms has been implemented using multiple technologies, each offering different trade-offs between accuracy, cost, scalability, and usability. Biometric systems based on fingerprint or facial recognition provide high precision and effectively prevent proxy attendance due to the uniqueness of physiological features [7]. However, their deployment in large classes is often limited by high implementation cost, time consumption, hygiene concerns, and privacy implications.

RFID-based attendance systems improve speed and ease of use by allowing students to tap identification cards, yet they remain vulnerable to card sharing and impersonation unless additional authentication layers are applied [8]. QR code-based systems represent a low-cost and easily deployable solution, leveraging students' smartphones; however, they rely on manual interaction and are susceptible to misuse when QR codes are shared remotely [9]. These limitations motivate the exploration of more automated and contactless attendance solutions.

Bluetooth Low Energy (BLE) has emerged as a promising alternative for smart campus attendance systems. BLE-based attendance enables automatic detection of student devices within a classroom without physical interaction, reducing disruption during lectures. When combined with device authentication mechanisms, BLE systems exhibit increased resistance to manipulation and scale effectively to large classroom environments [10]. Despite challenges related to signal variability and the requirement for enabled Bluetooth, BLE remains a strong candidate for real-time, contactless attendance tracking.

BLE and RSSI Fundamentals

Bluetooth Low Energy (BLE), introduced as part of the Bluetooth 4.0 specification, is designed for low-power, short-range communication, making it suitable for indoor environments with dense device populations. Modern smartphones natively support BLE, allowing participation without additional hardware requirements.

BLE-based positioning relies on the Received Signal Strength Indicator (RSSI), which represents the power level of a received wireless signal. In theory, RSSI decreases as the distance between transmitter and receiver increases. However, in indoor environments such as classrooms, RSSI values are highly unstable due to multipath propagation, signal attenuation

caused by walls and furniture, and interference from human movement. These factors significantly degrade the reliability of direct distance estimation based on RSSI alone [15, 16].

Indoor Positioning Techniques Using BLE

Two primary techniques are commonly used for BLE-based indoor localization: trilateration and fingerprinting. Trilateration estimates a device's position by converting RSSI values into distance measurements from multiple reference points. While conceptually straightforward, this approach is highly sensitive to environmental noise and path loss variations, leading to reduced accuracy in complex indoor settings such as classrooms.

Fingerprinting offers a more robust alternative by capturing the unique radio frequency characteristics of indoor environments. During an offline phase, RSSI measurements are collected at predefined reference locations to construct a fingerprint database, also known as a radio map. In the online phase, real time RSSI measurements are compared against this database to estimate the device's location. Fingerprinting has been shown to outperform trilateration in indoor environments due to its ability to implicitly model environmental effects [19, 17].

Enhancements to Fingerprinting Accuracy

To further improve the accuracy and robustness of fingerprint-based indoor positioning systems, several enhancement techniques have been proposed.

Artificial Neural Networks (ANNs) Artificial Neural Networks (ANNs) are capable of modeling nonlinear relationships within RSSI data. By training ANNs on large fingerprint datasets collected from indoor environments, localization accuracy can be significantly improved. Hybrid ANN models, combining Feedforward Multilayer Perceptrons (MLP) and Radial Basis Function (RBF) networks, have been successfully applied to refine position estimation in BLE-based systems [11].

Self-Adaptive K-Nearest Neighbors (SAKNN) The K-Nearest Neighbors (KNN) algorithm is widely used in fingerprint-based localization by identifying the K reference points whose RSSI patterns most closely match the current measurement. The Self-Adaptive KNN (SAKNN) approach enhances traditional KNN by dynamically adjusting the value of K and the weighting of neighbors based on signal conditions and reference point density. This adaptability improves localization stability across heterogeneous indoor environments [14, 12, 18].

Kalman Filtering RSSI measurements are inherently noisy in indoor environments. The Kalman Filter is a recursive estimation algorithm that smooths RSSI sequences by combining

prior predictions with current observations. Applying Kalman filtering to RSSI data reduces short-term fluctuations and improves the reliability of subsequent fingerprint matching and distance estimation processes [15].

Proximity Estimation Proximity estimation provides coarse-grained localization by determining whether a device is near a particular reference point or region based on relative RSSI strength. This technique often uses predefined RSSI thresholds or selects the strongest received signal to infer proximity. Proximity estimation can function as a standalone localization method or as a preprocessing step to reduce the fingerprint search space, thereby improving computational efficiency [13].

Related Work

A representative BLE-based classroom attendance system is presented by Puckdeevongs *et al.* [11], where BLE sensors are deployed within classrooms to collect RSSI data from students' smartphones. The system employs fingerprinting enhanced with artificial neural networks to estimate student positions and automatically record attendance. The study demonstrates the effectiveness of combining BLE indoor positioning with machine learning techniques in smart campus environments.

2.1.3 Quick Response (QR) Codes

Quick Response (QR) codes are two-dimensional barcodes capable of encoding a wide range of data, including URLs, identifiers, and structured information. Due to their simplicity, low deployment cost, and universal compatibility with modern smartphones, QR codes have become widely adopted in educational and administrative environments. Unlike hardware-dependent identification technologies, QR codes do not require specialized readers and can be displayed digitally or printed, making them a flexible and scalable solution for university campuses.

Secure Data Sharing and Authentication

QR codes have been increasingly explored as lightweight and secure mechanisms for authentication and short-term data sharing. A randomized QR code attendance system proposed by Imanullah and Reswan [22] introduces dynamically generated QR codes that are refreshed periodically to prevent reuse and unauthorized sharing. This approach significantly reduces the risk associated with static QR codes, such as code capture or remote distribution.

The proposed system relies on QR randomization rather than persistent identifiers, enabling secure attendance verification with minimal user interaction. Experimental evaluation reported in [22] demonstrates that randomized QR scanning provides a low cost and efficient

alternative to traditional attendance systems, achieving improved security without requiring biometric sensors or additional hardware.

QR Codes in Educational and Administrative Systems

Beyond authentication and attendance, QR codes have proven effective in enhancing accessibility and user engagement in educational contexts. Ibrahim [23] highlights the use of QR codes in academic libraries to bridge physical and digital spaces by enabling instant access to electronic resources, multilingual materials, audio content, and interactive services. These applications improve inclusivity and usability for diverse user groups, including individuals with visual or physical impairments.

QR codes have also been successfully applied in administrative systems for inventory management and asset tracking. Rizqi *et al.* [21] present an Android-based inventory management system that uses QR codes to monitor equipment location, condition, and status within an academic department. The system supports both online and offline QR scanning, enabling efficient updates and reducing administrative overhead. Such applications demonstrate the practicality of QR codes for managing shared university resources.

Comparison with RFID and NFC Technologies

Compared to RFID and Near Field Communication (NFC) systems, QR codes offer several advantages in terms of deployment flexibility and cost efficiency. RFID-based systems depend on physical cards or tags that can be shared among users, increasing the risk of misuse unless additional security mechanisms are employed [22]. Furthermore, RFID and NFC solutions require dedicated readers and physical tokens, which increase infrastructure and maintenance costs.

In contrast, QR codes do not require physical cards or specialized hardware, as they can be displayed on screens or printed when needed. They can be easily regenerated or randomized, enhancing security while maintaining simplicity. These characteristics make QR codes a practical and scalable solution for large academic environments where ease of deployment and adaptability are critical.

2.2 Related Past / Reported Work

2.2.1 Transact Mobile Credential

Transact Mobile Credential is a commercial digital identification solution widely adopted by universities to provide students with mobile-based campus credentials. The system enables students to use their smartphones as official university IDs through integration with Apple

Wallet and Google Wallet. This approach replaces traditional physical ID cards with a secure, digital alternative that is convenient and easy to deploy at scale.

The platform primarily focuses on access control and financial services. It supports secure entry to campus facilities, cashless payments, and service usage across a range of university amenities, including dining services, vending machines, and laundry facilities. By leveraging mobile wallet technologies, Transact Mobile Credential offers a robust and secure solution for identity verification and transactional services within campus environments.

Despite its strengths in access management and payment processing, Transact Mobile Credential is not designed to support specialized academic functionalities. Features such as automated attendance tracking, course scheduling, or academic workflow integration are outside the scope of the system. As a result, while it serves effectively as a digital campus ID and access solution, it does not address academic process automation requirements.

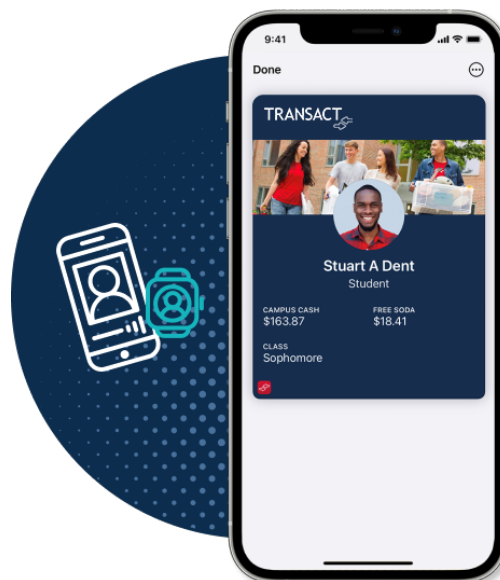


Figure 2.1: Transact Mobile Credential application home screen

2.2.2 CBORD Mobile ID

CBORD Mobile ID is a mobile-based identity and access management solution designed for universities that utilize CBORD campus systems. The application enables students and staff to use their smartphones as digital credentials for accessing campus facilities and participating in dining and event services.

The system primarily focuses on physical access control and student life services, offering real time interactions and limited reservation-related functionalities. While CBORD Mobile ID effectively supports secure identity verification and campus access, it is not designed to provide academic oriented features such as automated attendance tracking, course

scheduling, or office hour management. Consequently, its functionality is largely confined to access management and non-academic campus services.



Figure 2.2: CBORD Mobile ID application home screen

2.2.3 TouchNet OneCard Mobile

TouchNet OneCard Mobile is a component of TouchNet’s campus commerce ecosystem, providing mobile identification, account management, and payment services for university environments. The application enables students and staff to access digital campus credentials, view account balances, review transaction histories, and interact with access-controlled facilities and services.

The system is primarily oriented toward financial operations and identity-based access management. While it is integrated with various campus services, TouchNet OneCard Mobile does not offer academic oriented functionalities such as automated attendance tracking, course schedule visualization, or instructional workflow support. As a result, its scope is largely limited to campus commerce and access-related use cases rather than academic process automation.



Figure 2.3: TouchNet OneCard Mobile application home screen

2.2.4 Ellucian Mobile

Ellucian Mobile is a customizable mobile platform designed for higher education institutions that utilize Ellucian's Banner or Colleague enterprise systems. The application provides students and staff with access to academic information, including class schedules, grades, notifications, and university announcements, through a centralized mobile interface. While some institutions integrate digital identification features within Ellucian Mobile, the primary focus of the platform is the delivery of academic data and institutional communication. The system does not natively support real-time attendance tracking, physical access control, or automated academic workflow management. Consequently, Ellucian Mobile functions primarily as an information and communication platform rather than a comprehensive academic service or identity management solution.

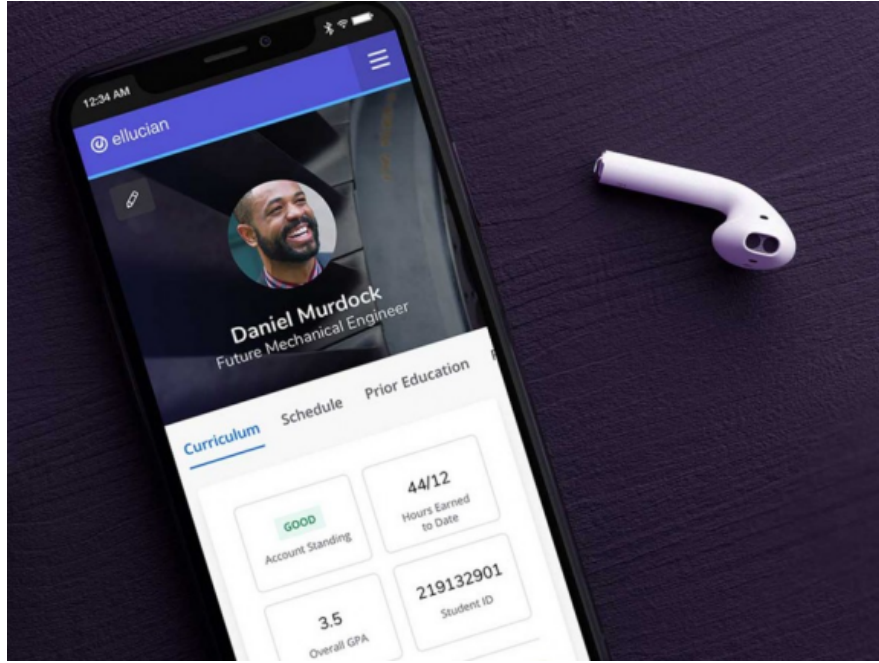


Figure 2.4: Ellucian Mobile application home screen

Table 2.1: Comparison of Existing Campus Mobile Applications

Feature	Transact	CBORD	TouchNet	Ellucian	BZU Assistant
Digital ID	✓	✓	✓	Partial	✓
Physical Access Control	✓	✓	✓	×	✓
Cashless Payments	✓	✓	✓	×	×
Attendance Tracking	×	×	×	×	✓
Office Hour Management	×	×	×	×	✓
Schedule Visualization	×	×	×	✓	✓
Device Reservation	×	Limited	Limited	×	✓
Academic Workflow Integration	×	×	×	Partial	✓

Table 2.1 highlights a clear functional gap between existing commercial campus applications and the requirements of academic process automation. Commercial solutions such as Transact Mobile Credential, CBORD Mobile ID, and TouchNet OneCard Mobile primarily focus on identity management, physical access control, and campus commerce. While effective in these domains, they lack support for core academic functionalities such as attendance tracking, office hour management, and integrated academic workflows.

Ellucian Mobile provides access to academic information such as schedules and grades; however, it functions mainly as an information delivery platform and does not support real time academic interactions or automation. In contrast, the BZU Assistant application is designed as an academic centric platform that integrates digital identification with attendance tracking, schedule visualization, office hour coordination, and resource reservation within a

unified system.

This comprehensive integration addresses the limitations observed in existing solutions by aligning identity management with instructional and administrative workflows. As a result, BZU Assistant fills a critical gap by supporting both operational efficiency and academic engagement, making it more suitable for smart campus environments focused on educational process optimization.

2.3 Tools and Technologies

The section provides a high-level overview of the main tools and technologies used in the development of the BZU Assistant system. The selected technologies were chosen to support a mobile first architecture, enable automated attendance tracking, and ensure scalability and reliability within a university campus environment.

2.3.1 Flutter Mobile Framework

The BZU Assistant mobile application is developed using *Flutter*, a modern UI framework for building natively compiled applications from a single codebase. Flutter is used to design and render the user interface, manage user interactions, and coordinate communication with backend services. Its reactive architecture enables responsive user experiences and consistent behavior across supported devices. In this project, Flutter serves as the primary interaction layer through which students, instructors, and administrative staff access digital identity services, schedules, reservations, and notifications.

2.3.2 Supabase Backend Platform

Supabase is used as the backend platform for data management, authentication, and real time services. It provides a centralized database for storing user profiles, academic schedules, reservations, attendance records, and event information. Supabase also supports secure authentication mechanisms and real time data synchronization, allowing updates to be reflected instantly across the system. Additionally, Supabase is utilized for delivering notifications related to schedule changes, reservations, and academic events, supporting timely communication between system users.

2.3.3 Raspberry Pi and BLE Infrastructure

Raspberry Pi devices are deployed within classrooms and laboratories to support Bluetooth Low Energy (BLE) based attendance tracking. Each Raspberry Pi functions as a BLE scanning node, detecting nearby student smartphones during lecture sessions. The collected

proximity data is transmitted to the backend system for processing and storage. This infrastructure enables automated and contactless attendance registration without requiring manual interaction from students or instructors.

2.3.4 QR Code Technology

Quick Response (QR) codes are used as a lightweight interaction mechanism across multiple system features. QR codes enable efficient access to services such as office hour booking, device and classroom reservations, schedule sharing, and academic event registration. By scanning a QR code using the mobile application, users can quickly initiate actions without navigating complex interfaces, improving usability and reducing interaction time.

2.3.5 Integrated System Overview

Together, Flutter, Supabase, Raspberry Pi devices, BLE communication, and QR code technology form a cohesive technical ecosystem. This combination supports secure digital identity management, real-time data exchange, automated attendance tracking, and efficient academic service delivery. The selected tools provide a practical balance between functionality, scalability, and ease of deployment, making them suitable for a real world university environment.

Chapter 3

System Analysis and Design

The chapter presents the analysis and design of the BZU Assistant system. It translates the project requirements and background concepts into a structured system design, defining how the system is organized, how its components interact, and how functional and non-functional requirements are addressed.

The chapter begins with a product description that outlines the system objectives, main features, operating environment, and constraints. It then presents the functional and non-functional requirements that guide the system behavior. Following this, functional decomposition and system models are introduced using standard UML diagrams. Finally, the chapter describes the overall system architecture and data management design, providing a comprehensive blueprint for system implementation.

3.1 Product Description

The section provides an overview of the BZU Assistant system from a functional and operational perspective. It defines the system objectives, highlights the main features offered to users, describes the operating environment and constraints, and specifies the functional and non-functional requirements that shape the system design.

3.1.1 System Objectives

The primary objective of the BZU Assistant system is to provide a unified, mobile-based digital platform that enhances academic and administrative operations at Birzeit University. The system aims to replace fragmented and manual processes with automated, secure, and efficient digital services accessible through a single Android application.

The specific objectives of the system are as follows:

- To provide a secure digital identity mechanism that replaces physical ID cards and enables authenticated access to university services.

- To automate student attendance tracking using Bluetooth Low Energy (BLE) technology and Raspberry Pi devices, reducing manual intervention and improving accuracy.
- To support efficient reservation and management of devices and classrooms through QR code-based interactions.
- To facilitate centralized office hours management, allowing students to view instructor availability and schedule appointments easily.
- To enable real-time access to academic schedules, including daily and weekly timetables, through a unified interface.
- To allow instructors to create academic events and share them with students using QR codes that integrate directly with mobile calendars.
- To improve communication and coordination between students, instructors, and administrative staff by providing timely notifications and real-time updates.

These objectives guide the system design decisions and ensure that the BZU Assistant platform aligns with the operational needs and digital transformation goals of Birzeit University.

3.1.2 System Main Features

The BZU Assistant system provides a set of integrated features designed to support core academic and administrative activities within Birzeit University. These features are implemented through a unified Android mobile application and are accessible to students, instructors, and administrative staff based on their roles. The main system features are summarized as follows:

- **Digital Identity and Authentication:** The system provides a secure digital identity for each user, enabling authenticated access to university services through the mobile application. This feature replaces physical ID cards and serves as the foundation for all system interactions.
- **Automated Attendance Tracking:** Student attendance is recorded automatically using Bluetooth Low Energy (BLE) technology in combination with Raspberry Pi devices installed in classrooms. Attendance is registered when a student's smartphone is detected within the classroom range, eliminating the need for manual check-ins.
- **Device and Classroom Reservation:** The system enables users to reserve devices and classrooms through QR code-based interactions. Reservations support check-in and check-out functionality, allowing efficient tracking and utilization of university resources.

- **Office Hours Management:** Students can view instructor office hour availability and schedule appointments using either QR code scanning or an in-app search interface. Instructors can manage their availability and appointment schedules through the system.
- **Academic Schedule Viewing and Sharing:** Users can access personalized academic schedules, including daily and weekly timetables, directly from the application. Schedules can also be shared with other users using dynamically generated QR codes.
- **Academic Event Creation and Calendar Integration:** Instructors can create academic events such as lectures, seminars, or announcements within the system. Each event generates a QR code that allows students to add the event directly to their mobile calendars.
- **Notifications and Real-Time Updates:** The system provides real-time notifications to inform users of reservation confirmations, office hour updates, and academic event announcements, improving communication and coordination across the campus.

These features collectively aim to streamline university operations, reduce administrative overhead, and enhance the overall campus experience by providing a centralized, mobile-first solution.

3.1.3 Operating Environment

The BZU Assistant system is designed to operate within the technical and physical environment of Birzeit University. The operating environment defines the hardware, software, and network conditions under which the system is expected to function reliably.

The mobile application operates on **Android based smartphones** used by students, instructors, and secretaries. These devices are required to support Bluetooth Low Energy (BLE), camera access for QR code scanning, and internet connectivity to communicate with backend services.

The backend services are hosted on a cloud based platform and are accessible through standard internet connections. The system relies on continuous network availability to support user authentication, data synchronization, reservation management, and real-time notifications. Temporary network interruptions are handled by synchronizing data once connectivity is restored.

For attendance tracking, **Raspberry Pi devices** equipped with BLE capabilities are installed in classrooms and laboratories across the campus. These devices operate continuously during lecture sessions and communicate with the backend system over the university network to report proximity data.

QR code interactions require camera functionality on the mobile device and are designed to function in indoor campus environments under typical lighting conditions. The system assumes the presence of standard university infrastructure, including Wi-Fi coverage in classrooms, laboratories, and public campus areas.

Overall, the operating environment supports a mobile first architecture that integrates Android devices, cloud services, and campus-deployed hardware to enable automated and real-time academic services.

3.1.4 Constraints

The design and development of the BZU Assistant system are subject to several technical, operational, and institutional constraints. These constraints influence system architecture, technology selection, and feature implementation.

- **Platform Constraint:** The system is designed exclusively for Android devices. This decision is driven by the need for reliable Bluetooth Low Energy (BLE) background operation and consistent hardware support. As a result, users of other mobile platforms are not supported.
- **Bluetooth Dependency:** Automated attendance tracking depends on students enabling Bluetooth on their smartphones. If Bluetooth is disabled or restricted by the operating system, attendance detection may be affected.
- **Signal Interference and Environmental Factors:** BLE signal strength can be influenced by environmental conditions such as classroom layout, walls, furniture, and human movement. These factors may impact proximity detection accuracy and require careful calibration of detection thresholds.
- **Network Availability:** The system relies on internet connectivity for authentication, data synchronization, reservations, and notifications. Limited or unstable network access may temporarily affect system responsiveness, although data synchronization can occur once connectivity is restored.
- **Hardware Deployment Constraints:** Attendance tracking requires the installation and maintenance of Raspberry Pi devices in classrooms and laboratories. Hardware availability, power supply, and physical placement may limit deployment scalability.
- **Privacy and Data Protection:** The system must comply with institutional policies and data protection regulations related to user privacy. This constraint restricts the collection and storage of sensitive personal information and influences system design choices.

- **Institutional Integration:** The system must integrate with existing university authentication and administrative workflows. Limitations imposed by legacy systems and institutional policies may affect the extent of automation and data access.

3.1.5 Functional Requirements

The section defines the functional requirements of the BZU Assistant system. Each system feature is presented using a structured format that includes a brief description, priority level, stimulus–response interactions, and a list of functional requirements. This approach ensures clarity, traceability, and consistency across all system functionalities.

Digital Identity and Authentication

Description: The Digital Identity and Authentication feature provides secure user authentication and digital identity management for students, instructors, and secretaries. Users are authenticated using university-issued credentials and granted role-based access to system features. The system maintains secure sessions, supports biometric auto-login, and provides a digital ID card with a verifiable QR code for identity confirmation across university services. **Priority:** High

Table 3.1: Stimulus–Response for Digital Identity and Authentication

Stimulus	System Response
User launches the application	The system displays the authentication interface.
User submits valid credentials	The system authenticates the user and navigates to the suitable role-based dashboard.
User submits invalid credentials	The system denies access and displays an error message.
User access the digital ID card	The system displays the user’s digital ID card with personal information and a QR code generation option.
Authority scans the user’s QR code	The system validates the QR code, extracts identity data, and displays a verification dialog with full user details.
User logs out	The system terminates the session, clears stored credentials, and returns to the login screen.

Table 3.2: Functional Requirements for Digital Identity and Authentication

FR-01	The system shall authenticate users using university-issued credentials consisting of a personal ID and password.
FR-02	The system shall provide role-based access for students, instructors, and secretaries, each with a dedicated dashboard and permitted feature set..
FR-03	The system shall validate authentication status for every protected feature and enforce row-level security policies at the database level.
FR-04	After successful login, the system shall display a digital ID card containing the user's photo, English and Arabic names, personal ID, issue year, study level, and a QR code generation button.
FR-05	The system shall generate a scannable QR code encoding the complete digital identity in JSON format.
FR-06	The QR code shall include a type identifier, version number, user ID, full name, Arabic name, Personal ID, role, issue year, study level (English and Arabic), profile picture URL, and generation timestamp.
FR-07	The system shall validate all authentication attempts and display specific error messages for invalid credentials, missing data, or unverified accounts.
FR-08	The system shall provide a logout function that terminates the authentication session, clears all stored credentials, resets in-memory data, and navigates to the login screen.

Automated Attendance Tracking

Description: This feature enables automatic student attendance recording using BLE and Raspberry Pi devices. Instructor enables Bluetooth Low Energy (BLE) attendance tracking at the start of a class. Students with the mobile application and BLE enabled on their devices are automatically detected and marked present through Raspberry Pi devices installed in classrooms. **Priority:** High

Table 3.3: Stimulus–Response for Automated Attendance Tracking

Stimulus	System Response
Lecture session starts	The system activates attendance scanning.
Student enters class-room	The system detects the device via BLE.
Device detected within range	The system records attendance.

Table 3.4: Functional Requirements for Automated Attendance Tracking

FR-09	The system shall enable the instructor to start BLE attendance tracking for a session using the mobile app.
FR-10	The system shall initiate BLE scanning on the classroom's Raspberry Pi device upon instructor request.
FR-11	The system shall detect student devices using BLE during lectures.
FR-12	The system shall register attendance automatically without manual input.
FR-13	The system shall associate attendance with the correct course and session.
FR-14	The system shall prevent duplicate attendance records per session.

Device Reservation

Description: The Device Reservation System enables students and instructors to reserve university-owned equipment such as laptops, projectors, cameras, and tablets through a mobile application. The system supports device discovery, real-time availability validation, conflict detection, and a mandatory secretary approval workflow. All reservations are protected against race conditions at the database level to prevent double booking and ensure data consistency. The reservation process follows three stages: user request submission, system availability validation, and secretary approval or denial. **Priority:** Medium

Table 3.5: Stimulus–Response for Device Reservation

Stimulus	System Response
User searches for devices by name, type, or scans a QR code	The system displays matching devices with device details.
User selects a device and specifies a reservation date and time range	The system validates the input and checks real-time availability for the selected time slot.
The selected device is unavailable for the requested time	The system rejects the request and displays alternative available time slots and similar devices.
User confirms a valid reservation request	The system stores the reservation with status "Pending", notifies the secretary, and confirms submission to the user.
Two users attempt to reserve the same device simultaneously	The system accepts the first request and rejects the second with a conflict notification and alternatives.
Secretary opens the reservation management interface	The system displays all pending reservations sorted by submission date.
Secretary approves or denies a reservation	The system updates the reservation status, updates the device availability if approved, and notifies the requesting user.
The reservation time period ends	The system automatically completes the reservation and returns the device to an available state.

Table 3.6: Functional Requirements for Device Reservation

FR-15	The system shall enable authenticated users to search for devices by name, or type using case-insensitive matching.
FR-16	The system shall provide filtering of devices by category, including Laptop, Projector, Camera, and Tablet.
FR-17	The system shall support QR code scanning to instantly identify a device.
FR-18	The system shall display device details including name, and type.
FR-19	The system shall verify device availability for the requested date and time range before accepting any reservation request.
FR-20	The system shall detect reservation conflicts using overlapping time range validation and prevent overlapping reservations.
FR-21	When a device is unavailable, the system shall suggest available alternative time slots on the same date.
FR-22	When a device is unavailable, the system shall suggest alternative available devices of the same type for the requested time.
FR-23	The system shall allow students and instructors to submit reservation requests specifying device, date, start time, end time, and optional purpose.
FR-24	The system shall validate reservation inputs, including future dates only, start time before end time, and maximum reservation duration.
FR-25	All submitted reservations shall be created with an initial status of "Pending" and shall not block device availability until approved.
FR-26	Before final submission, the system shall display a confirmation summary of the reservation details.
FR-27	Upon submission, the system shall notify secretaries of new pending reservation requests.
FR-28	The system shall allow secretaries to view all pending reservations with full device and requester details.
FR-29	The system shall allow secretaries to approve reservations, changing the reservation status to "Approved" and the device status to "Reserved".
FR-30	The system shall allow secretaries to deny reservations with an optional reason, maintaining device availability.
FR-31	The system shall notify users when their reservation is approved or denied.
FR-32	The system shall automatically complete reservations when the reserved time period ends and update the device status to "Available".

Classroom Reservation

Description: The Classroom Reservation System enables students and instructors to reserve a classrooms and labs for exams, makeup lectures, and special events. The system supports classroom discovery, real-time availability validation, conflict detect such as classrooms is already used for teaching courses, or booked by other user, and mandatory secretary approval/denial workflow. The reservation process follows three stages: user request submission, system validate availability, and secretary approval or denial. **Priority:** Medium

Table 3.7: Stimulus–Response for Classroom Reservation

Stimulus	System Response
User searches for a classroom by selecting the building then the classroom, or scans QR code	The system displays classroom details such as type, capacity, floor, and available equipment.
User selects a classroom and specifies a date and time range	The system validates the input and checks real-time availability for the selected classroom.
Selected classroom is unavailable for the requested time	The system rejects the request and shows message that the classroom is already reserved.
User confirms a valid classroom reservation request	The system stores the request with status "Pending", notifies the secretary, and confirms submission.
Multiple users attempt to reserve the same classroom simultaneously	The system accepts the first valid request and rejects subsequent requests with conflict notifications.
Secretary accesses classroom reservation management	The system displays all pending classroom reservations sorted by submission date.
Secretary approves or denies a classroom reservation	The system updates reservation status, updates classroom availability, and notifies the requester.
Reserved time slot ends	The system automatically completes the reservation and releases the classroom.

Table 3.8: Functional Requirements for Classroom Reservation

FR-33	The system shall allow authorized users to search classrooms by building.
FR-34	The system shall assign a unique QR code to each classroom for quick identification and reservation initiation.
FR-35	After selecting classroom, the system shall display detailed classroom information including capacity, floor, and availability equipment.
FR-36	The system shall verify classroom availability for the requested date and time range before accepting any reservation request.
FR-37	The system shall detect overlapping classroom reservations using time-range conflict validation and prevent double booking.
FR-38	The system shall allow the students and instructors to submit classroom reservation requests specifying classroom, date, time range, and purpose.
FR-39	The system shall validate reservation inputs, including future dates only, valid time ranges, and maximum reservation duration limits.
FR-40	All classroom reservation requests shall be created with an initial status of "Pending" awaiting secretary approval.
FR-41	The system shall display a confirmation summary of classroom reservation details prior to final submission.
FR-42	Upon submission, the system shall notify the secretary of new pending classroom reservation requests.
FR-43	The system shall allow secretaries to view all pending classroom reservations with requester and classroom details.
FR-44	The system shall allow secretaries to approve classroom reservations, changing the status to "Approved" and marking the classroom as reserved.
FR-45	The system shall allow secretaries to deny classroom reservations with an optional reason while keeping the classroom available.
FR-46	The system shall notify users when their classroom reservation is approved or denied.
FR-47	The system shall automatically release classrooms when the reservation time period ends.

Office Hours Management

Description: The Office Hours Reservation System enables students to book one-on-one meetings with instructors during predefined office hours. The system provides a structured workflow for instructor discovery, time slot selection, request submission, approval or denial by instructors, and meeting completion. It enforces strict conflict prevention rules to avoid double bookings and ensures real-time notifications are delivered to both students and instructors throughout the reservation lifecycle. The reservation process consists of student request submission, system-level availability validation, instructor approval or denial, and

meeting execution. **Priority:** Medium

Table 3.9: Stimulus–Response for Office Hours Management

Stimulus	System Response
Student searches for instructor or scan instructor's QR code	The system displays the selected instructor's time
Student selects a time slot	The system validates slot availability and enables meeting request submission.
Student submits a meeting request	The system validates inputs, creates a meeting request with status "Pending", and notifies the instructor.
Instructor reviews a pending meeting request	The system allows the instructor to approve or decline the request.
Instructor approves or declines a request	The system updates the request status and notifies the student.
The scheduled meeting time ends	The system marks the meeting as completed and releases the time slot

Table 3.10: Functional Requirements for Office Hours Management

FR-48	The system shall allow students to browse and search instructors who offer office hours.
FR-49	The system shall support scanning instructor QR codes to initiate meeting reservations with the instructor pre-selected.
FR-50	The system shall display office-hour time slots for each instructor based on predefined schedules.
FR-51	The system shall validate selected time slots against existing pending and approved meetings before accepting requests.
FR-52	The system shall allow students to submit meeting requests specifying instructor, date, time slot, and optional purpose.
FR-53	All new meeting requests shall be created with an initial status of "Pending" awaiting instructor approval.
FR-54	The system shall display a confirmation summary before final submission of a meeting request.
FR-55	Upon submission of a meeting request, the system shall notify the instructor in real time.
FR-56	The system shall allow students to view all their meeting requests with current status and details.
FR-57	The system shall allow instructors to view all pending meeting requests submitted by students.
FR-58	The system shall allow instructors to approve meeting requests, changing the status to "Approved".
FR-59	The system shall allow instructors to decline meeting requests with an optional reason.
FR-60	Before final approval, the system shall revalidate time slot availability to prevent race conditions.
FR-61	The system shall notify students when their meeting requests are approved, declined, or completed.
FR-62	The system shall automatically mark meetings as completed when the scheduled time period ends.
FR-63	The system shall enforce transactional constraints to prevent double booking of office-hour time slots.

Academic Schedule Viewing and Sharing

Description: The Academic Schedule Viewing and Sharing feature allows students and instructors to view their personalized academic timetables within the application. The feature supports daily and weekly schedule views and enables users to share their current course schedules with others through dynamically generated QR codes. When a QR code is scanned, the recipient can view the most up-to-date version of the sender's schedule. All schedule sharing actions are logged for audit and traceability purposes. **Priority:** Low

Table 3.11: Stimulus–Response for Academic Schedule Viewing and Sharing

Stimulus	System Response
User opens the schedule view.	User opens the schedule view.
User switches between daily and weekly views.	The system updates the timetable display according to the selected view.
User requests to share other user's schedule	The system generates a QR code encoding the user's current academic schedule that located in ID card.
Another user scans the shared schedule QR code.	The system decodes the QR code and displays the sender's latest academic schedule.

Table 3.12: Functional Requirements for Academic Schedule Viewing and Sharing

FR-64	The system shall display a personalized academic schedule for each authenticated student and instructor.
FR-65	The system shall support a daily schedule view showing all scheduled courses and activities for a selected day.
FR-66	The system shall support a weekly schedule view showing scheduled courses and activities across the academic week.
FR-67	The system shall generate a QR code representing the user's current academic schedule in the ID card.
FR-68	The system shall allow users to scan a shared schedule QR code and view the sender's latest academic schedule.

Academic Event Creation and Calendar Integration

Description: The Academic Event Creation and Calendar Integration feature allows instructors to create academic events such as assignments, quizzes, exams, or announcements and share them with students using dynamically generated QR codes. Instructors define event details including title, description, and start and end times. Students scan the event QR code using the application to automatically add the event to their personal device calendars. This feature improves event dissemination and ensures students receive accurate scheduling information. **Priority:** Low

Table 3.13: Stimulus–Response for Academic Event Creation

Stimulus	System Response
Instructor creates a new academic event	The system validates the input and saves the event with its associated details.
Student scans an academic event QR code	The system decodes the event information and adds the event to the student’s device calendar.

Table 3.14: Functional Requirements for Academic Event Creation

FR-69	The system shall allow instructors to create academic events by providing a title, description, start time, and end time.
FR-70	The system shall generate a dynamic QR code for each academic event to enable sharing with students.
FR-71	The system shall support integration with student device calendars to add scanned academic events automatically.

Notifications and Real-Time Updates

Description: The Notifications and Real-Time Updates feature provides timely system-generated notifications to keep users informed about important activities and changes. Notifications are delivered in real time for schedule updates, reservation status changes, and newly created academic events. This feature ensures users remain aware of relevant system actions without the need to manually check each module. **Priority:** Low

Table 3.15: Stimulus–Response for Notifications

Stimulus	System Response
A user's academic schedule is updated.	The system sends a notification informing the user of the schedule change.
A reservation request is approved, denied, or completed.	The system sends a notification to the affected user with the updated reservation status.
A new academic event is created.	The system broadcasts a notification to relevant users about the new event.

Table 3.16: Functional Requirements for Notifications

FR-72	The system shall notify users when changes occur in their academic schedules.
FR-73	The system shall notify users when the status of their reservations changes, including approval, denial, or completion.
FR-74	The system shall notify relevant users when new academic events are created.

3.1.6 Non-Functional Requirements

The section defines the non-functional requirements of the BZU Assistant system. These requirements describe the quality attributes, constraints, and performance characteristics that the system must satisfy to ensure reliability, usability, security, and scalability within a real university environment.

Performance Requirements

- The system shall authenticate users and display the main interface within an acceptable response time under normal network conditions.
- The system shall record attendance data without noticeable delay during lecture sessions.
- The system shall handle concurrent access from multiple users without performance degradation.

Reliability and Availability

- The system shall be available during standard university operating hours.
- The system shall maintain data consistency in the event of temporary network interruptions.
- The system shall prevent data loss during normal operation and unexpected failures.

Security Requirements

- The system shall restrict access to services based on authenticated user roles.
- The system shall protect user data against unauthorized access.
- The system shall ensure secure communication between the mobile application, back-end services, and Raspberry Pi devices.

Usability Requirements

- The system shall provide an intuitive and user-friendly interface suitable for students, instructors, and administrative staff.
- The system shall minimize the number of steps required to perform common tasks such as viewing schedules or booking appointments.
- The system shall provide clear feedback to users for system actions and errors.

Scalability

- The system shall support an increasing number of users without requiring major architectural changes.
- The system shall allow additional classrooms, devices, and users to be added with minimal configuration.

Maintainability

- The system shall be designed using a modular architecture to simplify maintenance and future enhancements.
- The system shall allow updates and improvements without disrupting existing services.

Compatibility and Portability

- The system shall operate on Android devices that support Bluetooth Low Energy (BLE).
- The system shall function across different Android versions commonly used by university members.

Privacy and Data Protection

- The system shall collect only the minimum data required to perform its functions.
- The system shall store and process user data in accordance with institutional policies and applicable data protection regulations.
- The system shall ensure transparency in how user data is used within the platform.

3.2 Functional Decomposition

The section decomposes the BZU Assistant system into its primary functional elements by identifying the system actors and their interactions with the system. Functional decomposition provides a structured view of system behavior and serves as a foundation for defining use cases and system models.

3.2.1 Actors

The BZU Assistant system involves multiple actors that interact with the system based on their roles and responsibilities within the university environment. Table ?? summarizes the identified actors and their descriptions.

Table 3.17: Primary and Secondary Actors of the BZU Assistant System

Actor	Type	Description
Student	Primary	Initiates system interactions such as authentication, attendance participation, reservations, schedule viewing, and calendar integration.
Instructor	Primary	Initiates teaching-related operations including attendance activation, office hour management, event creation, and schedule sharing.
Secretary	Secondary	Initiates administrative operations related to monitoring and managing device and classroom reservations.
Authentication Service	Secondary	Validates user credentials during the login process and supports secure authentication and authorization.
Android Calendar	Secondary	Receives academic event data from the system and stores it in the user's device calendar.
Raspberry Pi (BLE Node)	Secondary	Detects student devices via Bluetooth Low Energy during lecture sessions and transmits detection data to the BZU Assistant system for attendance recording.

3.2.2 Use Cases

The section describes the system use cases that define the interactions between actors and the BZU Assistant system. Each use case is documented using a structured template to clearly specify actors, triggers, flows, and system behavior.

Table 3.18: UC-01 Authenticate and Manage Digital ID

Use Case ID	UC-01
Use Case Name	Authenticate and Manage Digital ID
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Students, Instructors, Secretaries
Secondary Actors	Authentication Service
Trigger	User opens the mobile application and attempts to log in.
Description	BZU Assistant enables users to securely authenticate into the BZU Assistant system using valid credentials. Upon successful authentication, the system establishes a secure session, determines the user role, and displays the user's digital ID within the application.
Preconditions	1. The user is registered in the system. 2. The mobile application is installed on a compatible Android device.
Postconditions	1. The user is authenticated and redirected to the main interface. 2. The digital ID is available for use in attendance tracking, QR code scanning, and reservation services.
Normal Flow	1. The user launches the BZU Assistant mobile application. 2. The user enters login credentials (personal ID and password). 3. The system validates the credentials. 4. The system generates a secure session token. 5. The system determines the user role. 6. The system loads the appropriate dashboard and displays the digital ID for the student and instructor.
Alternative Flows	A1. Unregistered User 1. The user attempts to log in without an existing account. 2. The system displays an error message and redirects the user to a help or registration page.
Exceptions	E1. System Unavailable The system displays a connectivity error message. E2. Session Token Failure The system prompts the user to retry or notifies the administrator.
Priority	High
Other Information	1. The system uses token-based authentication. 2. The digital ID may be represented as a QR code for identity-related services.
Assumptions	1. The user device supports the required Android OS version and has internet access. 2. Backend authentication services are operational.

Table 3.19: UC-02 Register Attendance Automatically

Use Case ID	UC-02
Use Case Name	Register Attendance Automatically
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Instructor, Student
Secondary Actors	Raspberry Pi (BLE Node)
Trigger	The instructor activates attendance for a lecture session, and students enter the classroom with Bluetooth enabled.
Description	BZU Assistant enables automatic attendance registration for students using Bluetooth Low Energy (BLE). Before attendance tracking begins, the instructor activates the Raspberry Pi device assigned to the classroom. Once activated, the Raspberry Pi scans for nearby student smartphones and registers attendance automatically without requiring manual interaction from students. The system also allows the instructor to register attendance manually by selecting the lecture session and marking students as present or absent when automatic attendance is unavailable.
Preconditions	1. The instructor is authenticated in the BZU Assistant application. 2. The instructor has activated attendance for the lecture session. 3. The student is authenticated in the BZU Assistant application. 4. The lecture session is active. 5. Bluetooth is enabled on the student's Android device. 6. The classroom Raspberry Pi device is operational. 7. The instructor has permission to perform manual attendance.
Postconditions	1. The student is marked as present for the lecture session. 2. The attendance record is stored in the backend database.
Normal Flow	1. The instructor opens the BZU Assistant application. 2. The instructor selects the lecture session and activates attendance tracking. 3. The system enables the corresponding Raspberry Pi device for scanning. 4. Students enter the classroom with Bluetooth enabled. 5. The Raspberry Pi device scans for nearby BLE-enabled student devices. 6. The system verifies the student identity and lecture context. 7. The system records attendance for detected students.

Alternative Flows	A1. Manual Attendance 1. The instructor selects the lecture session. 2. The instructor chooses the manual attendance option. 3. The system displays the list of students enrolled in the course. 4. The instructor marks students as present or absent. 5. The system records the attendance and stores it in the database.
Exceptions	E1. Bluetooth Disabled Attendance is not recorded and the student is prompted to enable Bluetooth. E2. Raspberry Pi Failure Attendance scanning is unavailable and the system logs the error.
Priority	High
Other Information	1. Attendance registration is instructor-controlled and automated. 2. Duplicate attendance records are prevented for the same lecture session. 3. Manual attendance overrides or complements automatic attendance when needed.
Assumptions	1. Instructors activate attendance at the start of each lecture. 2. Student devices support Bluetooth Low Energy (BLE). 3. Classroom Raspberry Pi devices are correctly configured and connected.

Table 3.20: UC-03 Reserve Device use case description

Use Case ID	UC-03
Use Case Name	Reserve Device
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Student, Instructor
Secondary Actors	Secretary
Trigger	The user scans a QR code associated with a device, or searches for an available resource in the application.
Description	BZU Assitant enables the students and instrutors to reserve university devices such as laptops, projectors, and cameras through the BZU Assistant system. Reservations can be initiated by scanning a QR code attached to the device or by selecting a device through the in-app search interface. The system validates availability, records the reservation, and tracks usage through check-in and check-out actions.
Preconditions	1. The user is authenticated in the BZU Assistant application. 2. The device is registered in the system. 3. The device is available for reservation during the selected time period.
Postconditions	1. The device reservation is successfully recorded in the system. 2. The device status is updated to reflect the reservation.
Normal Flow	1. The user opens the device reservation interface. 2. The user scans the QR code of a device, or searches for a resource. 3. The system displays device details such as the name and device's type. 4. The user selects a reservation date and time. 5. The system validates the reservation request and checks for time conflicts. 6. The use confirms the reservation request. 7. The system stores the reservation with status Pending and notifies the secretary. 8. The secretary reviews and approves the reservation request. 9. The system updates the reservation status to Approved and reserves the device.

Alternative Flows	A1. Device Unavailable 1. The user selects a device that is already reserved. 2. The system displays an unavailability message and suggests alternative times or devices with the same category. A2. Reservation Cancellation 1. The user selects an existing reservation. 2. The user cancels the reservation. 3. The system updates the device status to available. A3. Reservation Denied 1. The secretary denies the reservation request. 2. The system updates the reservation status to Denied. 3. The system notifies the user of the decision.
Exceptions	E1. QR Code Invalid or Unreadable The system displays an error message and prompts the user to retry or search manually. E2. Backend Service Unavailable The reservation request fails and the system notifies the user to try again later.
Priority	Medium
Other Information	1. Reservations support devices using a unified workflow. 2. Conflicting reservations are prevented by the system.
Assumptions	1. QR codes are correctly assigned to devices.

3.2.3 Use Case Diagram

Figure 3.1 illustrates the interaction between the actors and a system. The figure below shows a use case diagram of the BZU Assistant system depicts how primary and secondary actors interact with the system's core services. Students and instructors are modeled as primary actors who initiate academic functions such as authentication, attendance tracking, scheduling, and event management, while the secretary is modeled as a secondary actor responsible for reviewing and managing device and classroom reservations. Automated attendance is initiated by instructors and supported by Raspberry Pi BLE nodes that detect student devices without direct interaction. External systems, including the authentication service and Android Calendar, are represented as secondary actors to illustrate integration boundaries and supporting functionality.

3.3 System Models

3.3.1 Class Diagram

Figure 3.2 presents the overall structure of the system by modeling the core entities, their attributes, and the relationships that coordinate academic, administrative, and resource-management workflows. At the center is the User entity, which represents all system actors and connects to role-specific and functional components such as enrollments, meetings, reservations, schedules, and attendance records. Academic organization is handled through Courses, Course Schedules, and Enrollments, linking users to instructional activities over time. Physical resources are modeled using Buildings, Classrooms, and their related reservation and event entities, enabling structured classroom management, availability tracking, and approval workflows. Operational features such as Meetings, Meeting Reservations, Events, Devices, and Notifications are supported through dedicated entities that capture status, timestamps, approvals, and metadata to ensure traceability and lifecycle control. Supporting tables like saved schedules, subscriptions, logs, and debug records enhance system reliability, auditing, and user experience, resulting in a cohesive and extensible system design. The attendance feature is part of this model and will be discussed in detail in a later section.

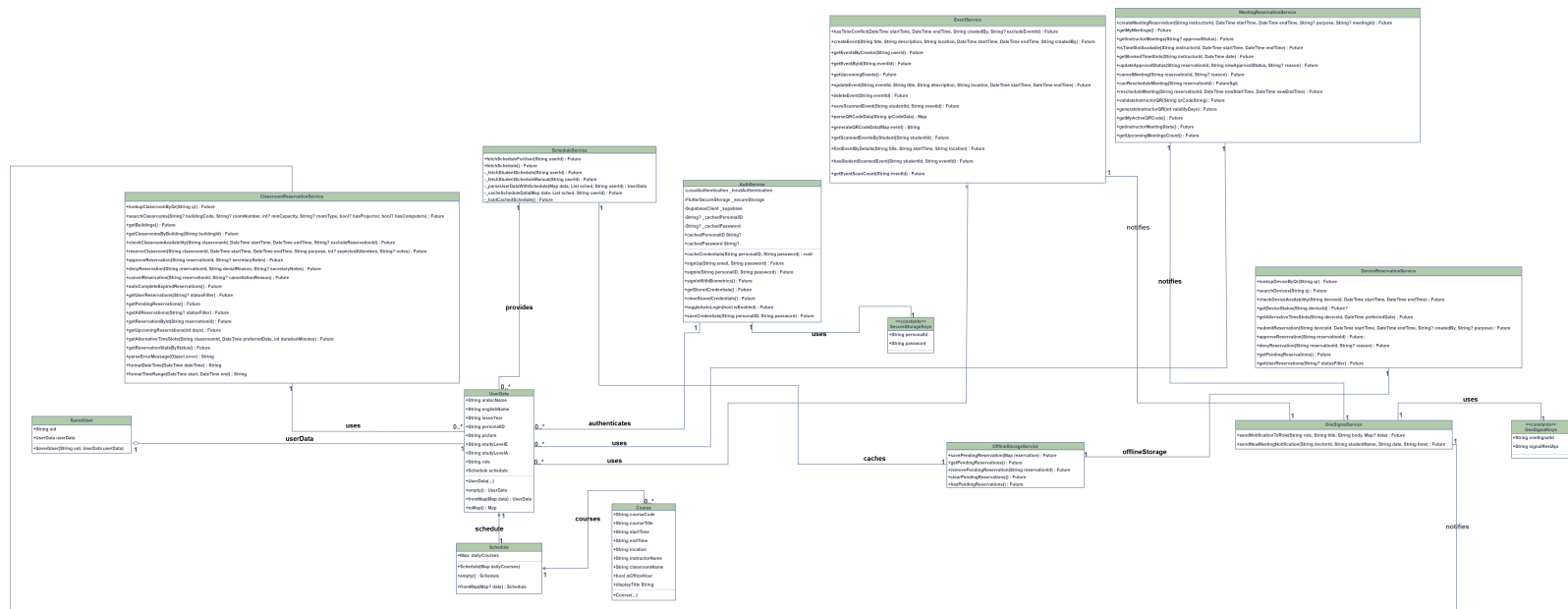


Figure 3.2: Class Diagram for BZU Assistant Application

3.3.2 Sequence Diagrams

A sequence diagram is used to model the dynamic behavior of the system by illustrating the interactions between actors and system components over time for a specific use case. It focuses on the order of messages exchanged rather than the system structure. To create a sequence diagram, a single use case is selected, the involved participants are identified, and the main interaction flow is represented as a sequence of method calls and responses. Sequence diagrams are particularly useful for clarifying system logic, validating requirements, and demonstrating how different components collaborate to accomplish a task.

Authentication and Digital ID Sequence Diagram

Figure 3.3 illustrates the authentication and digital ID initialization workflow in the BZU Assistant system. The process begins when a student or instructor opens the mobile application and submits their personal ID and password through the login interface. The Mobile App forwards the credentials to the `AuthenticationService`, which validates them and, upon success, issues a secure session token.

After successful authentication, the application requests the user's schedule from the `ScheduleService`, which loads cached schedule data and constructs the required domain objects by mapping raw data into `UserData` and `Schedule` instances. Once the user profile and schedule are prepared, the system displays the main dashboard containing the user's digital ID. If the submitted credentials are invalid, the `AuthenticationService` returns a failure response and the application displays an error message. In the case of backend or connectivity issues, a service error is returned and an appropriate connectivity error message is presented to the user.

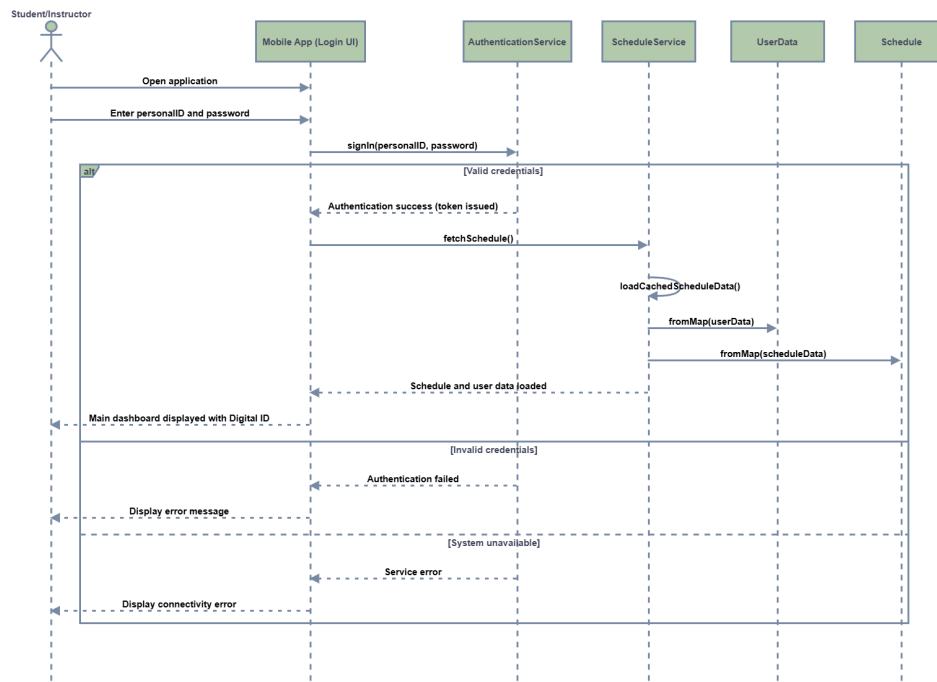


Figure 3.3: Sequence diagram for UC-01 – Authenticate and Manage Digital ID

Reserved Device Sequence Diagram

Figure 3.4 illustrates the workflow of the Reserve Device use case in the BZU Assistant system. The process begins when a student or instructor opens the device reservation interface and either scans a device QR code or searches for an available device. Once the device is identified, the system retrieves and displays its details to the user.

The user then selects the desired reservation date and time, after which the `DeviceReservationService` validates device availability and checks for scheduling conflicts. If the device is available, the user confirms the reservation request and the system records it with a `Pending` status while notifying the secretary for review. The secretary evaluates the request and either approves or denies it, leading the system to update the reservation status accordingly and notify the user of the decision.

Alternative execution paths handle scenarios such as device unavailability, invalid QR codes, and reservation cancellation, ensuring that conflicts and exceptional conditions are properly managed.

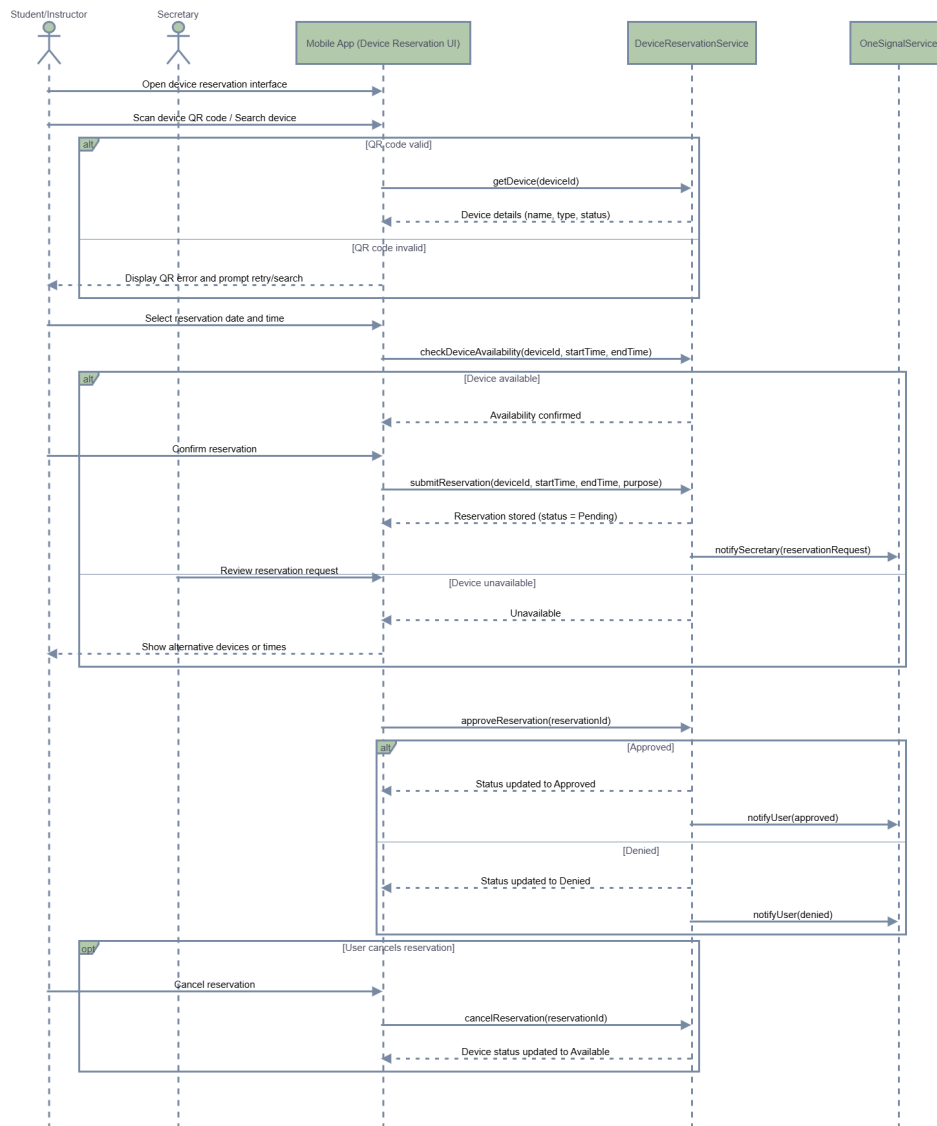


Figure 3.4: Sequence diagram for UC-03 – Reserve Device

Office Hours Management Sequence Diagram

Figure 3.5 presents the sequence diagram for the Office Hours Management use case, illustrating the interaction flow between instructors, students, and backend services during office hour scheduling and management.

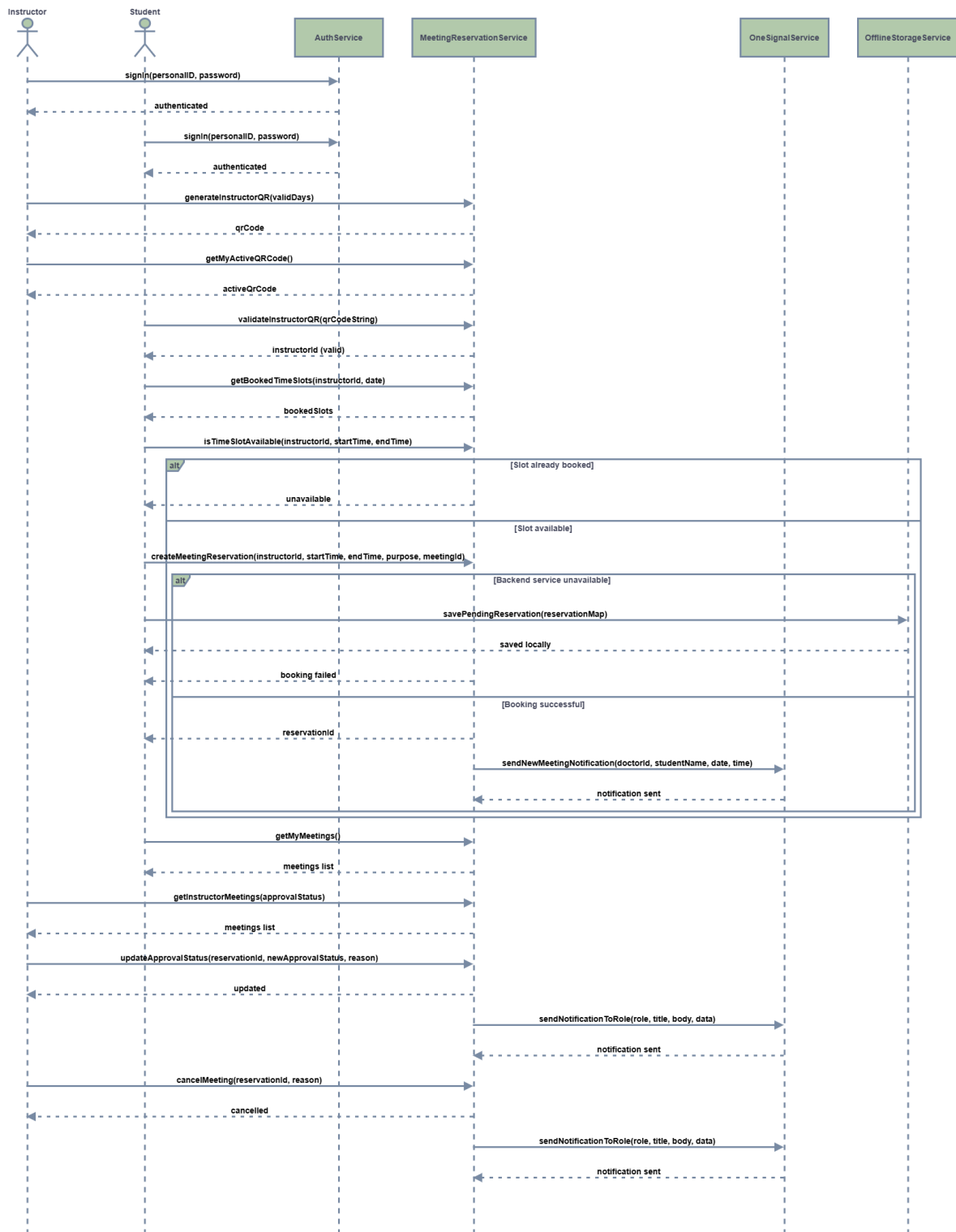


Figure 3.5: Sequence diagram for the Office Hours Management use case

The sequence starts with both the instructor and the student authenticating through the `AuthService`. The instructor then publishes office hour availability by generating an instructor-specific QR code using the `MeetingReservationService`. This QR code enables students to access the instructor's available office hours either by scanning it or by selecting the instructor within the application.

Once the instructor is identified, the system retrieves already booked time slots and validates the selected time interval to prevent scheduling conflicts. If the selected slot is unavailable, the system rejects the request and prompts the student to choose another slot. If the slot is available, the student submits a meeting reservation request, which is processed and stored by the system.

In cases where backend services are temporarily unavailable, the reservation request is cached locally using the `OfflineStorageService` to ensure reliability. After a successful booking, notification messages are dispatched through the `OneSignalService`. Both instructors and students can later view their scheduled appointments, while instructors retain the ability to approve or cancel meetings, with all updates propagated through system notifications.

3.3.3 Activity Diagrams

Activity diagrams are used to describe the dynamic behavior of a system by illustrating the flow of activities from start to finish. They focus on how a process is carried out, showing the sequence of actions, decision points, and possible paths that may occur during execution. By representing both normal and alternative flows, activity diagrams help clarify complex workflows and make it easier to understand how different roles and system components interact to achieve a specific goal. In this project, activity diagrams are used to model key system processes in a clear and intuitive way, providing a high-level view of system behavior that complements the class and sequence diagrams.

Device Reservation Activity Diagram

Figure 3.6 illustrates the complete workflow of the Device Reservation process in the BZU Assistant system, highlighting the interactions between the Student/Instructor, the BZU Assistant System, and the Secretary.

The process starts when the student or instructor opens the device reservation interface. The user can initiate the reservation either by scanning the QR code attached to a device or by searching for a device through the application. If a QR code is scanned, the system validates its correctness; in the case of an invalid or unreadable QR code, an error message is displayed and the user is prompted to retry the scanning process. Alternatively, the user may search for a device and select it from the search results. In both paths, the system displays the selected device's details.

After selecting the device, the user specifies the desired reservation date and time. The reservation request is then processed by the BZU Assistant system, which validates the request and checks device availability and potential time conflicts. If the device is unavailable, the system suggests alternative available time slots or other devices within the same category. If the device is available, the user confirms the reservation request.

Once confirmed, the system stores the reservation with a Pending status and notifies the secretary. The secretary reviews the request and either approves or denies it. If approved, the system updates the reservation status to Approved, updates the device status to Reserved, and notifies the user. If denied, the system updates the reservation status to Denied and informs the user. The activity concludes after the user receives the final notification, ensuring that device availability and reservation integrity are consistently maintained.

The activity concludes after the user receives the final notification, ensuring that device availability and reservation integrity are consistently maintained.

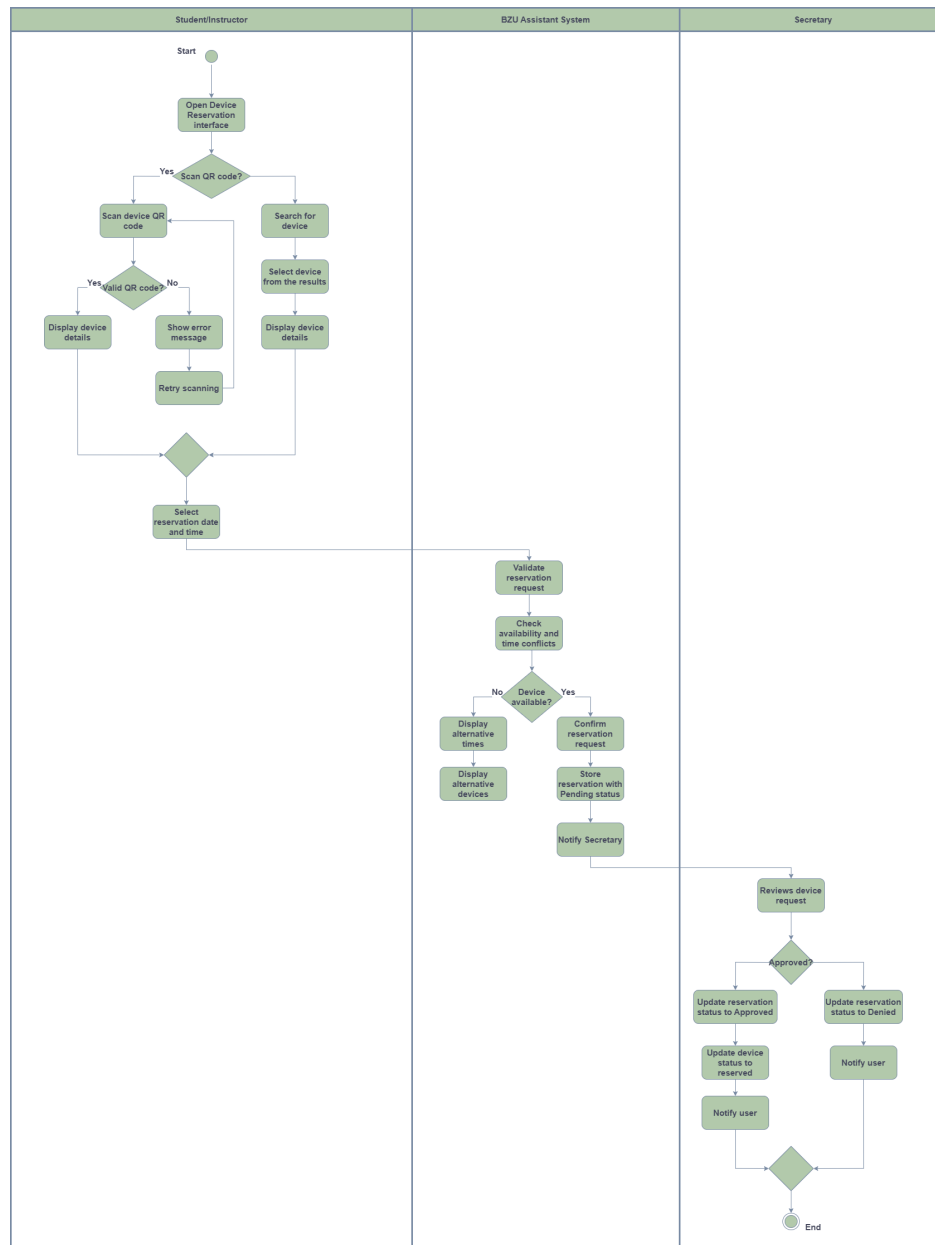


Figure 3.6: Device Reservation Activity Diagram illustrating the reservation workflow, validation process, and secretary approval in the BZU Assistant system

Register Attendance Activity Diagram

Figure 3.7 illustrates the workflow of the attendance registration process in the BZU Assistant system, covering both automatic and manual attendance modes.

The process begins when the instructor opens the BZU Assistant application and selects an active lecture session. The instructor then activates attendance, prompting the system to enable the classroom Raspberry Pi device to scan for nearby student devices using Bluetooth Low Energy (BLE). As students enter the classroom, the system detects their devices, verifies their identity and lecture context, and records attendance while preventing duplicate entries and handling late arrivals when applicable.

When automatic attendance is unavailable, the activity diagram shows an alternative execution path in which the instructor switches to manual attendance. In this mode, the instructor views the list of enrolled students and records attendance manually. This fallback mechanism ensures that attendance can still be registered accurately in the event of technical issues or connectivity limitations.

The activity concludes after attendance records are successfully stored, demonstrating clear instructor control, automated system behavior, and robust fallback support to maintain reliable attendance tracking.

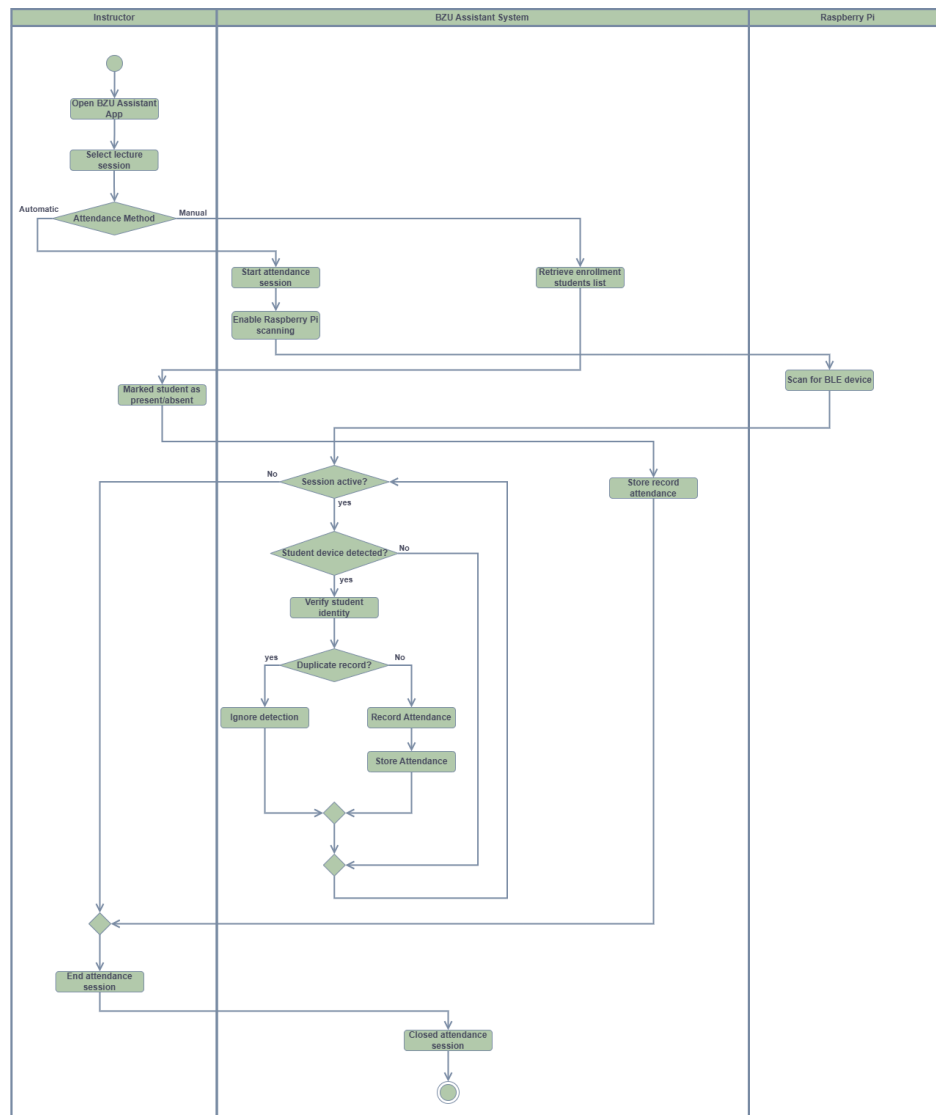


Figure 3.7: Activity Diagram for Register Attendance

3.3.4 State Chart Diagram

State chart diagrams are used to model the dynamic behavior of system entities whose functionality depends on their current state and the events that occur over time. Unlike sequence or activity diagrams, state diagrams focus on the lifecycle of a single object and clearly define how it transitions between states in response to internal or external triggers. In the BZU Assistant system, state diagrams are particularly useful for representing components that enforce strict operational rules, such as attendance sessions and resource reservations, where certain actions are only valid in specific states. These diagrams help clarify system constraints, prevent invalid operations, and ensure consistent behavior throughout the system.

Device Reservation State Diagram

Figure 3.8 illustrates the lifecycle of a reservation request in the resource reservation feature, focusing on how availability checks and approval decisions control state transitions. The process begins when the system is in the Available state, indicating that the resource is free for the requested time period. Upon submitting a reservation request, the system evaluates the resource availability using the `isAvailable(start, end)` operation. If the resource is available, the request transitions to the Pending state, where it awaits review by the secretary. From this state, the request may either be approved or denied. An approved request moves to the Approved state, granting temporary access to the resource, and is automatically transitioned to the Completed state when the reservation time ends through the `autoReleased()` operation. If the request is denied, it transitions to the Denied state and terminates without activating the reservation. The diagram clearly separates decision points, approval actions, and time-based transitions, providing a precise view of how reservation requests are processed and finalized.

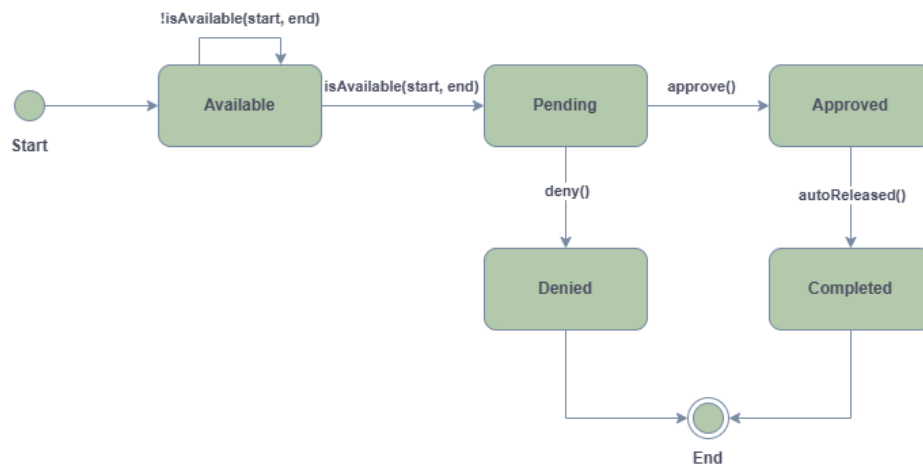


Figure 3.8: State chart diagram illustrating the lifecycle of a device reservation request, including availability verification, approval or denial decisions, and automatic completion after the reservation period ends.

3.4 System Architecture

This section describes the overall architecture of the BZU Assistant system and explains how its components are structured and interact with one another. The system architecture provides a high-level view of the system, focusing on the organization of functionality, separation of responsibilities, and interaction between software components and external devices. Defining the system architecture helps ensure scalability, maintainability, and a clear mapping between system requirements and implementation.

3.4.1 Sub-Systems

The BZU Assistant system is decomposed into several sub-systems, each responsible for a distinct functional area. This modular structure supports separation of concerns, simplifies system maintenance, and allows individual sub-systems to evolve independently. The sub-systems are identified based on the system requirements, use cases, and UML models, ensuring full coverage of the system functionality.

User Management Sub-System

This sub-system manages user authentication, authorization, and role-based access control for all system users, including students, instructors, and secretaries. It ensures that users can only access functionalities permitted by their roles and provides secure access to the system.

Course and Academic Structure Sub-System

The Course and Academic Structure sub-system manages academic entities such as courses, lecture sessions, enrollments, and academic events. It provides the academic context required by attendance tracking and resource reservation processes and ensures consistency across academic data.

Attendance Management Sub-System

This sub-system is responsible for registering student attendance during lecture sessions. It supports both automatic attendance using Bluetooth Low Energy (BLE) through Raspberry Pi devices and manual attendance performed by instructors. It handles attendance sessions, identity verification, duplicate prevention, and attendance record storage.

Hardware Integration Sub-System

The Hardware Integration sub-system manages interaction between the BZU Assistant system and physical classroom devices, particularly Raspberry Pi BLE nodes. It enables device scanning, detects student smartphones, and handles hardware-related failures while isolating device-specific logic from core system functionality.

Reservation and Resource Management Sub-System

This sub-system handles the reservation of university resources, including devices and classrooms. It manages resource availability checks, reservation creation, conflict prevention, and status updates using a unified reservation workflow.

Approval and Administration Sub-System

The Approval and Administration sub-system allows secretaries to review, approve, or reject reservation requests submitted by students or instructors. It enforces administrative control over resource allocation and ensures compliance with university policies.

QR Code and Identification Sub-System

This sub-system manages the generation and scanning of QR codes used to identify academic events and physical resources. QR codes provide a fast and reliable mechanism for initiating system actions such as reservations.

Notification Sub-System

The Notification sub-system is responsible for delivering system-generated notifications to users. These notifications inform users about attendance registration, reservation status changes, and administrative decisions, ensuring consistent and timely communication.

Data Persistence Sub-System

The Data Persistence sub-system handles the storage and retrieval of system data from the backend database. It supports create, read, update, and delete operations for all core entities and separates data access logic from business logic to improve system maintainability and scalability.

Table 3.21: Sub-System Summary of the BZU Assistant System

Sub-System	Description
User Management	Handles user authentication, authorization, and role-based access control for students, instructors, and secretaries.
Course and Academic Structure	Manages courses, lecture sessions, enrollments, and academic events, providing the academic context required by other sub-systems.
Attendance Management	Controls automatic and manual attendance registration, including BLE-based detection, duplicate prevention, and attendance record storage.
Hardware Integration	Manages communication with physical devices such as Raspberry Pi BLE nodes, including scanning operations and hardware error handling.
Reservation and Resource Management	Handles device and classroom reservations, including availability checks, reservation creation, conflict prevention, and status updates.
Approval and Administration	Allows secretaries to review, approve, or reject reservation requests and ensures administrative control over resource allocation.
QR Code and Identification	Manages QR code generation and scanning for identifying resources and academic events and initiating related system actions.
Notification	Delivers system notifications related to attendance, reservations, and administrative decisions to users.
Data Persistence	Handles database access and storage for all system entities, supporting create, read, update, and delete operations.

3.4.2 Software Architecture

The BZU Assistant system follows a layered and modular software architecture that separates concerns between presentation, business logic, and data persistence. This architectural approach improves maintainability, scalability, and clarity by ensuring that each layer has a well-defined responsibility and interacts with other layers through clear interfaces.

At the presentation layer, the mobile application provides user interfaces for students, instructors, and secretaries. This layer handles user interactions such as activating attendance, reserving resources, and reviewing requests, while delegating all processing logic to the underlying services.

The application and business logic layer contains the core services responsible for im-

plementing system functionality. This layer includes attendance management, reservation handling, user management, notification services, QR code processing, secretary approval, and course and academic structure. It coordinates system workflows, enforces business rules, and integrates with external components such as Raspberry Pi BLE devices.

The data persistence layer manages access to the backend database and is responsible for storing and retrieving system data. It encapsulates all database operations and ensures that business logic remains independent of data storage details.

In addition, the architecture integrates a hardware interaction component that enables communication with Raspberry Pi BLE nodes deployed in classrooms. This component operates as part of the business logic layer but remains logically isolated to handle device scanning, detection, and error handling.

Overall, Figure 3.9 below ensures a clear separation of responsibilities, supports system extensibility, and provides a solid foundation for implementing the functional requirements of the BZU Assistant system.

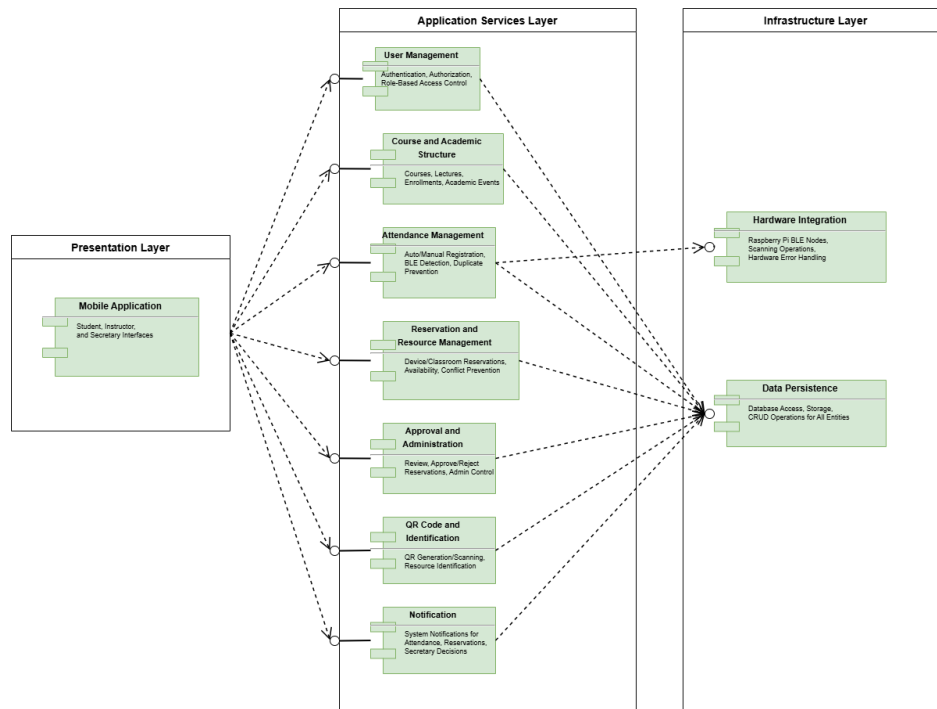


Figure 3.9: Component diagram showing the three-layer architecture of the BZU Assistant system with sub-system decomposition and interface dependencies.

3.4.3 Deployment Diagram

Figure 3.10 illustrates the physical deployment of the BZU Assistant system, showing how its software components are distributed across different hardware nodes and how they communicate at runtime. It provides a clear view of the system's execution environment, highlighting the interaction between client devices, backend services, database infrastructure, and external hardware components involved in attendance tracking and resource management. In

this deployment, the BZU Assistant Mobile App runs on users' mobile devices and communicates with the Application Server over secure HTTPS connections. The Application Server hosts all core backend services, including user management, academic structure handling, attendance management, reservation and resource management, approval workflows, QR code processing, and notification delivery. Persistent data is stored in the BZU Assistant Database Server, which is accessed through SQL using Supabase. For automated attendance tracking, the Application Server also communicates over IP/TCP with a Raspberry Pi BLE Node that runs a BLE scanning service to detect nearby devices. This configuration ensures secure data exchange, clear separation of responsibilities, and reliable integration between mobile clients, backend logic, database storage, and IoT hardware.

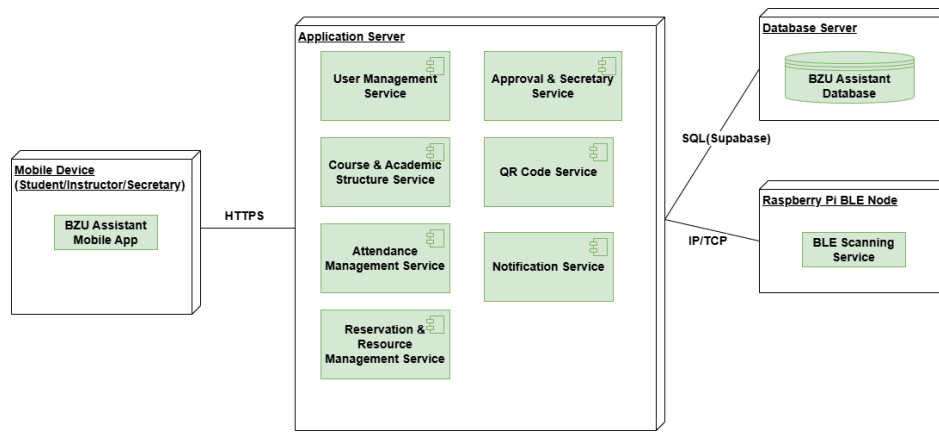


Figure 3.10: Deployment diagram of the BZU Assistant system showing mobile clients, application server services, database server, and Raspberry Pi BLE integration.

3.5 Data Management and ERD

Figure 3.11 illustrates the overall data architecture of the BZU Assistant system and defines how its core subsystems are structured and interconnected. At the center of the model is the Users entity, which represents all system actors, including students, instructors, and secretaries, and serves as the primary reference for authentication, authorization, and activity tracking across the system.

The academic domain is modeled through entities such as Courses, Enrollments, Course Schedule, and Course Attendance, which together manage course offerings, user participation, lecture scheduling, and attendance recording. Attendance data is captured through both manual and automated mechanisms, supported by Attendance Logs for BLE-based detection and Course Attendance for finalized attendance records linked to specific courses and students.

The physical infrastructure of the university is represented by Buildings and Classrooms, where classrooms are associated with buildings and enhanced with metadata such as capacity, equipment availability, QR codes, and beacon configuration. Classroom usage is man-

aged through Classroom Reservations and Classroom Overrides, allowing scheduled bookings, administrative approval, and temporary room changes when required.

Resource management is handled through the Devices and Reservations entities, enabling users to reserve university equipment with full lifecycle tracking, including pending approval, approval, denial, and cancellation states. Similar approval-based workflows are applied in the Meeting Reservations and Meetings entities, which support office hour scheduling and instructor–student meetings.

Event-related functionality is modeled using the Events and Student Scanned Events entities, supporting QR-based event dissemination and student participation tracking. User personalization and system services are supported through entities such as Saved Schedules, User Push Subscriptions, and Trigger Debug, which handle schedule bookmarking, notification delivery, and system diagnostics, respectively.

Overall, the ERD reflects a modular yet integrated design, where academic processes, resource reservations, attendance tracking, and administrative workflows are consistently linked through well-defined relationships. This structure ensures data integrity, prevents conflicts, and supports the scalable operation of the BZU Assistant system.

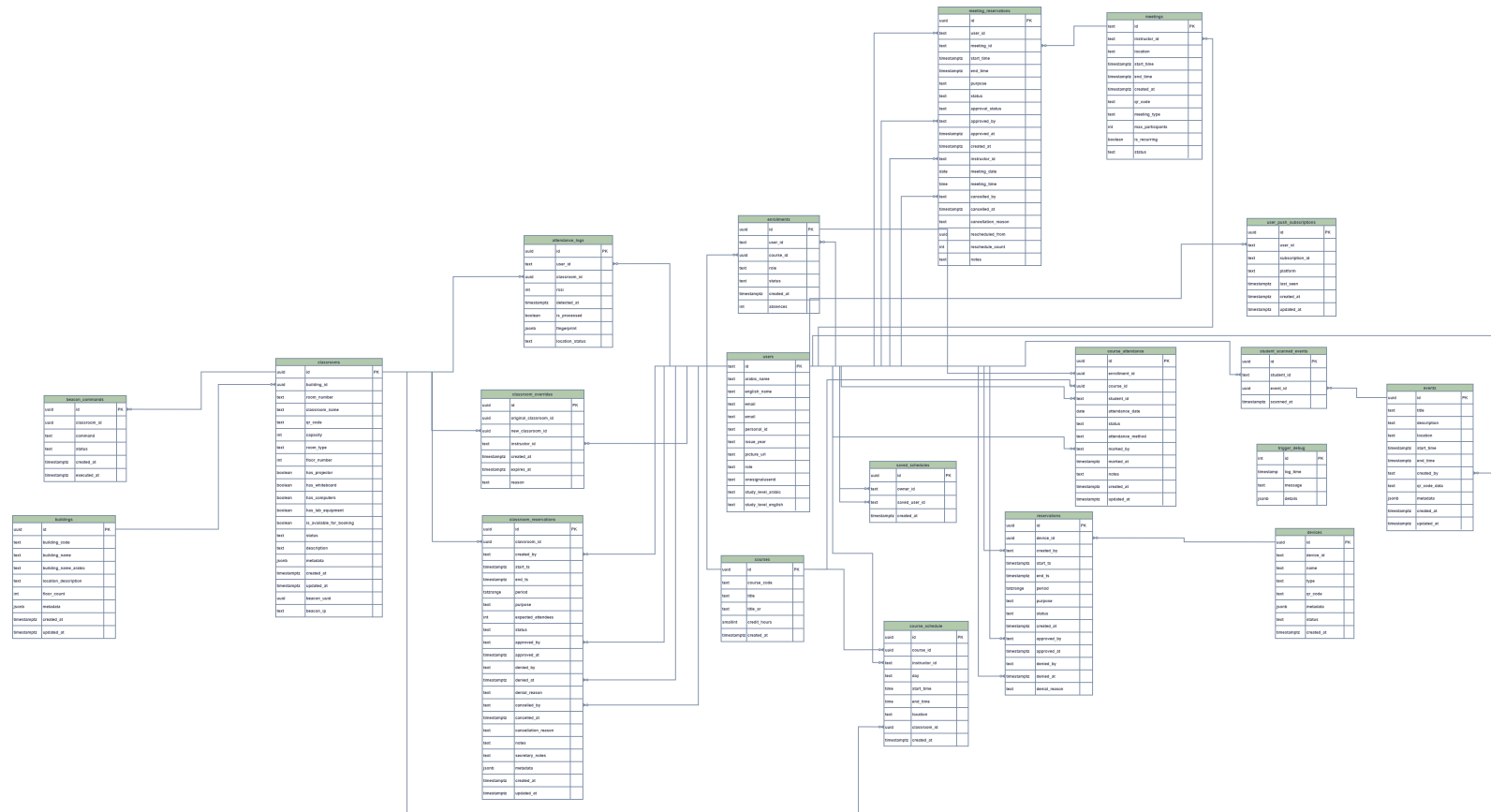


Figure 3.11: Entity–Relationship Diagram (ERD) illustrating the complete database schema and core data relationships of the BZU Assistant system.

Chapter 4

Implementation

Chapter 4 presents the implementation of the BZU Assistant system, translating the proposed design and architectural decisions into a working software solution. The chapter explains how the system components were developed and integrated using Flutter for the Android mobile application, Supabase for backend services and real-time data management, and Raspberry Pi devices for BLE-based attendance tracking. It details the implementation of the core functional modules—including digital identity authentication, automated attendance, device and classroom reservation, office hours management, schedule viewing, and academic event creation—while highlighting key technical decisions and challenges encountered during development. The goal of this chapter is to demonstrate how the system design was realized in practice within a real university environment.

4.1 Design Philosophy

The BZU Assistant application adopts a simple and clean user interface to ensure clarity, ease of use, and broad accessibility. This design choice follows common UI standards widely used in Palestinian applications, where users value straightforward layouts over complex visuals. The interface relies on clear navigation, readable typography, and minimal visual elements to reduce cognitive load and allow users to complete tasks quickly without confusion. Core features are presented directly, unnecessary animations are avoided, and screens follow a consistent structure, making the application intuitive for users with different technical backgrounds. This approach ensures the system remains practical, familiar, and efficient in everyday academic use.

4.2 Implementation of Backend Services

The section explains how the BZU Assistant Flutter application is organized using the Model–View–Controller (MVC) pattern. MVC separates data representation (Models), user inter-

face (Views), and business logic (Controllers) to keep the codebase modular, maintainable, and easy to extend.

4.2.1 Models

The `models` layer defines the core data structures used across the application. It contains three main entities: `UserData` (user profile details such as names, personal ID, picture, study level, and role), `UserSchedules` (daily and weekly schedule structures), and `UserCourses` (individual course information). These model classes handle serialization/deserialization and provide factory constructors to build objects directly from Supabase responses, ensuring consistent and validated data mapping throughout the system.

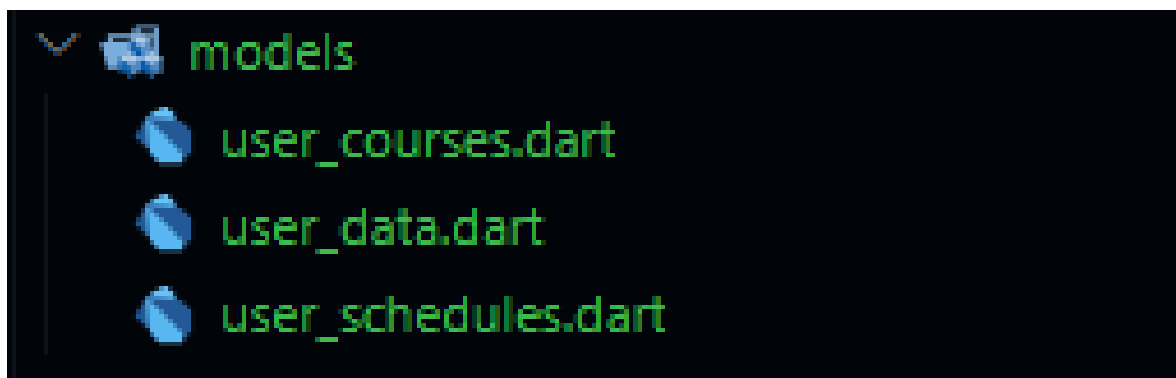


Figure 4.1: Models layer structure (`UserData`, `UserSchedules`, and `UserCourses`) with serialization and Supabase mapping.

4.2.2 Views

The `views` layer focuses only on presentation and user interaction. It is split into `pages` (27 full-screen screens such as login, dashboards, reservations, QR scanning, attendance, and event screens) and `reusable widgets` (9 UI components including buttons, text components, navigation drawers, daily/weekly schedule widgets, and responsive layouts). Views capture user inputs and forward actions to controllers without embedding business logic, keeping the UI clean and consistent.

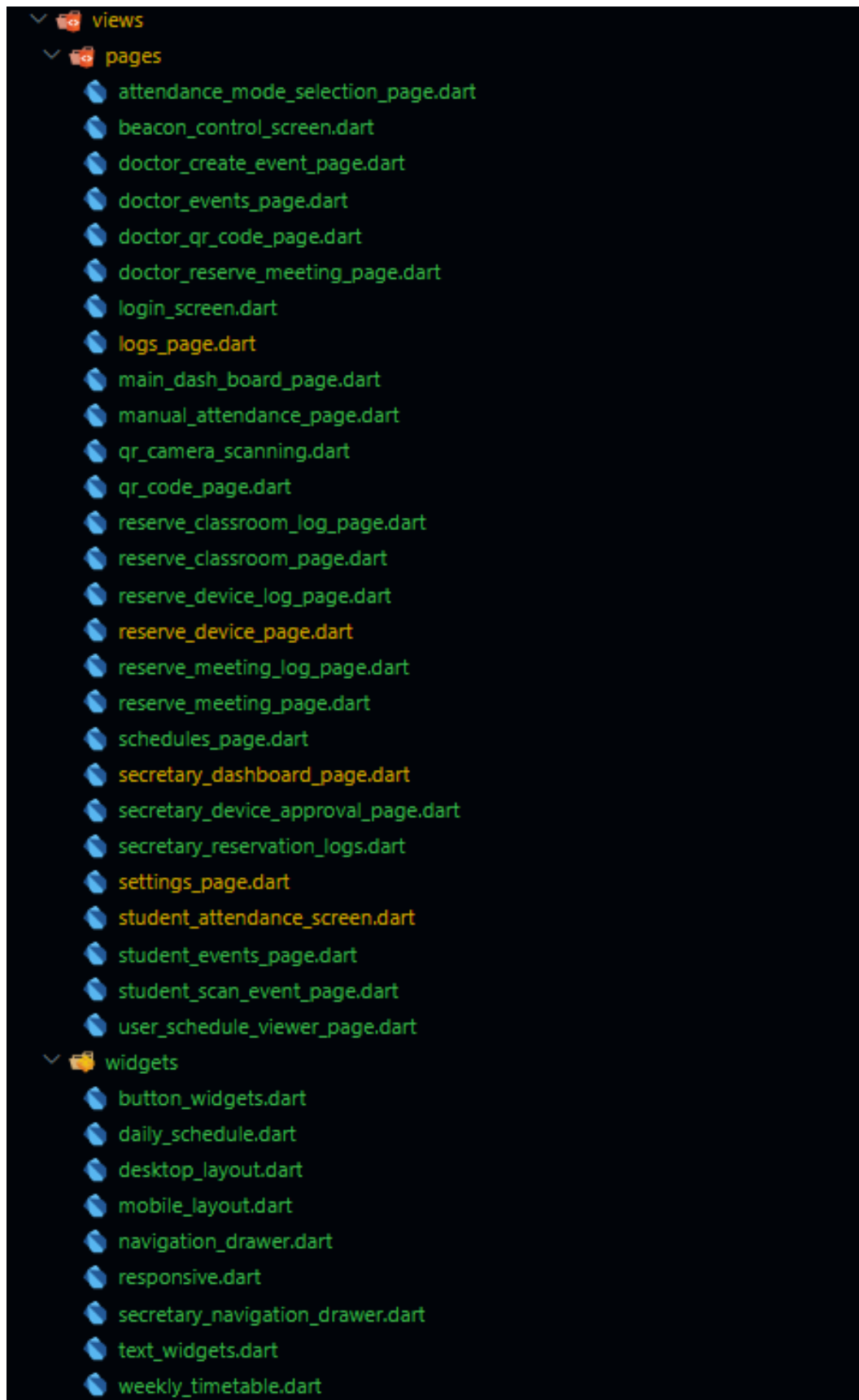


Figure 4.2: Views layer organization showing pages and reusable widgets used across the application.

4.2.3 Controllers

The controllers layer contains the business logic and service classes that connect the UI to backend systems. It includes nine main services: `auth_controller.dart` (authentication and sessions), `reservation_service.dart` (device booking), `classroom_reservation_service.dart` (classroom reservations and conflict detection), `meeting_reservation_service_v2.dart` (meeting requests), `event_service.dart` (events and QR-based workflows), `schedule_service.dart` (schedule retrieval and management), `notification_controller.dart` (OneSignal notifications), `wifi_service.dart` (BLE beacon detection for attendance), and `offline_storage_service.dart` (local caching and offline sync). Controllers perform validation, enforce rules (e.g., schedule conflicts), communicate with Supabase/Firebase, and return structured results to the views to ensure a strict separation between UI and application logic.

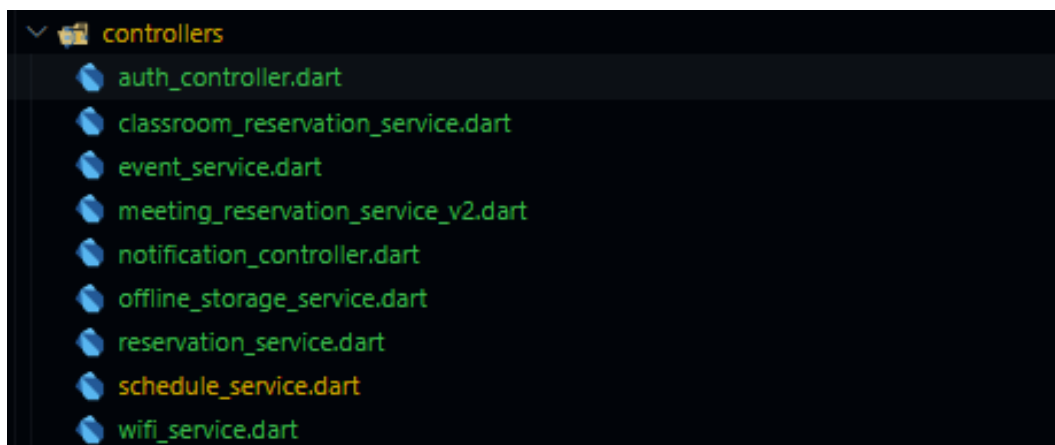


Figure 4.3: Controllers (service layer) showing core services and their interaction with Supabase/Firebase and system features.

4.3 Implementation of the Raspberry Pi and BLE Services

4.3.1 Attendance Logic

Figure 4.4 illustrates the attendance processing and positioning logic executed on the Raspberry Pi. The core intelligence of the system is hosted on the edge processor, where a hybrid classification engine retrieves raw attendance data from Supabase but processes it locally to ensure low latency and real-time responsiveness.

To address the *Corridor Problem*, where wireless signals may bleed through walls, a multi-stage decision algorithm was implemented. The system first applies a *Gap Logic* check that compares the relative signal strength of Wi-Fi access points, which typically penetrate

walls, against the BLE beacon signal, which is significantly attenuated by physical barriers. If Wi-Fi signal strength is high while the beacon signal is weak, the system immediately forces an *Outside* classification. For ambiguous cases where threshold-based rules are insufficient, a Weighted K-Nearest Neighbors (KNN) algorithm is applied using training data stored locally in a SQLite database. The algorithm computes the Euclidean distance between the live student fingerprint and the stored training vectors, applying a higher weight to the Bluetooth signal to prioritize proximity over environmental noise. This hybrid approach ensures robust and accurate attendance classification even across different smartphone models and signal conditions.

```

def main():
    # Initialize database
    db = SQLiteDB('attendance.db')
    db.create_tables([TrainingData, StudentData])

    # Load training data
    training_data = db.get_training_data()

    # Initialize KNN model
    knn = KNeighborsClassifier()
    knn.fit(training_data)

    # Main processing loop
    while True:
        # Get live signal data
        signal_data = get_signal_data()

        # Predict location
        prediction = knn.predict(signal_data)

        # Update backend
        update_backend(prediction)

        # Sleep for 10 seconds
        time.sleep(10)

if __name__ == '__main__':
    main()

```

(a) Main processing loop handling training data, prediction logic, and backend updates.

```

def process_attendance():
    # Feature extraction
    features = extract_features(signal_data)

    # Signal validation
    if not validate_signal(features):
        return 'Invalid Signal'

    # Decision logic
    prediction = knn.predict(features)

    # Update attendance
    update_attendance(prediction)

    return prediction

def validate_signal(features):
    # Check if signal is valid
    return features['signal_strength'] > 0.5

```

(b) Attendance processing pipeline showing feature extraction, signal validation, and decision logic.

Figure 4.4: Attendance logic implementation on the Raspberry Pi using BLE beacon signals, Wi-Fi fingerprinting, and local weighted KNN classification.

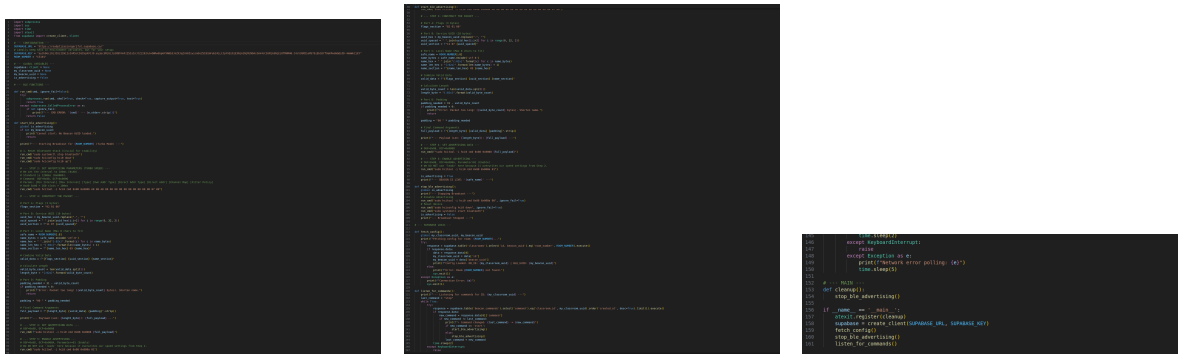
4.3.2 Bluetooth Low Energy (BLE) Connection

Figure 4.5 illustrates the Bluetooth Low Energy (BLE) advertiser implementation deployed on the Raspberry Pi for classroom presence broadcasting. This module is responsible for emitting a reliable and low-latency BLE signal that can be consistently detected by students' mobile devices during attendance sessions.

The broadcasting service is implemented in Python and interfaces directly with the Linux Bluetooth stack using the low-level `hcitool` and `hciconfig` system utilities. Higher-level BLE libraries were intentionally avoided to reduce latency and eliminate internal timing constraints. Instead of relying on standard BLE advertising modes, a custom *Turbo Mode* was implemented by manually constructing raw hexadecimal HCI (Host Controller Interface) command packets. This approach allows the advertising interval to be forced down to 100 ms, which is significantly faster than the default BLE advertising intervals.

Before each broadcasting session, the Bluetooth controller is explicitly reset to ensure a clean and stable state. The advertiser then injects a payload containing the classroom-specific UUID, beacon flags, and a shortened local name representing the room number. Packet length is carefully calculated, padded if necessary, and validated to comply with BLE constraints. Once configured, advertising is enabled using direct HCI commands, allowing the Raspberry Pi to behave indistinguishably from a commercial iBeacon. This guarantees immediate signal detection by student devices without instability or delayed discovery.

In addition, the Raspberry Pi continuously listens for start and stop commands issued from the backend via Supabase. These commands allow centralized control of BLE broadcasting per classroom, enabling or disabling attendance sessions dynamically without physical access to the device.



(a) Initialization and configuration of the BLE broadcasting service. (b) Construction of raw HCI packets and Turbo Mode advertising setup. (c) Backend-controlled BLE start/stop command listener.

Figure 4.5: BLE advertiser implementation on the Raspberry Pi using low-level HCI commands, Turbo Mode broadcasting, and backend-controlled session management.

4.4 User Interface

4.4.1 Digital ID and Authentication

Figure 4.6 illustrates the Digital Identity and Authentication flow in the BZU Assistant system. This module serves as the main entry point to the application, where users authenticate using their university Personal ID and password through the login interface. Upon successful authentication, the system identifies the user role and automatically loads the corresponding role-based dashboard. Students are directed to the student dashboard, instructors to the instructor dashboard, and secretaries to the secretary dashboard, ensuring that each user accesses only the features and permissions relevant to their role.

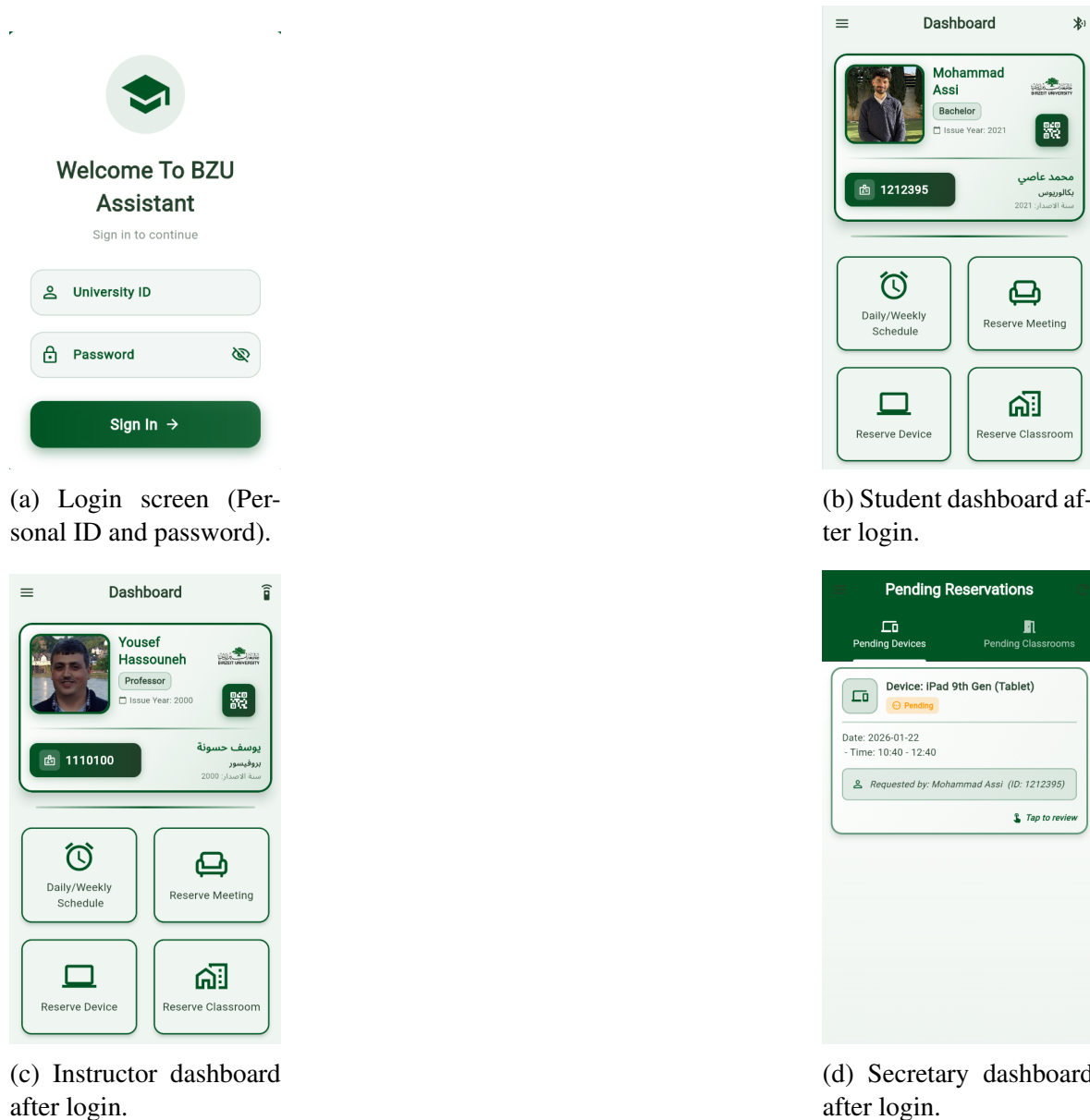
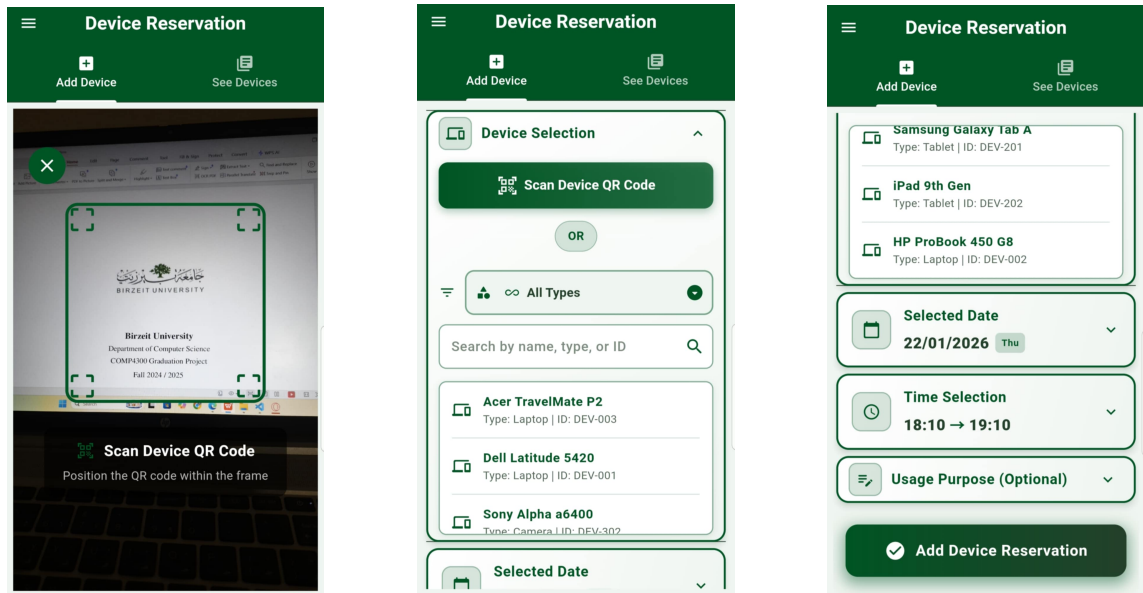


Figure 4.6: Digital identity authentication flow and role-based dashboards in BZU Assistant.

4.4.2 Device Reservation

Figure 4.7 presents the device reservation workflow in the BZU Assistant application. This feature allows students and instructors to request university-owned equipment such as laptops, tablets, and cameras directly through the mobile app. Users can initiate a reservation by scanning the device QR code or by manually searching for devices using the device name, type, or unique identifier. Filtering options are provided to simplify device discovery, and the system ensures that only available devices can be selected for the requested time slot.



(a) QR code scanning for device identification.

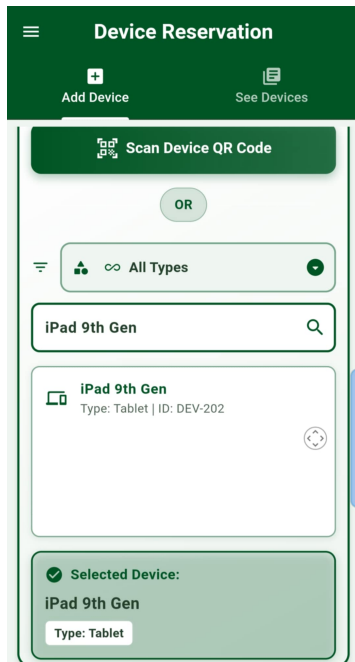
(b) Device selection interface.

(c) Reservation details and submission form.

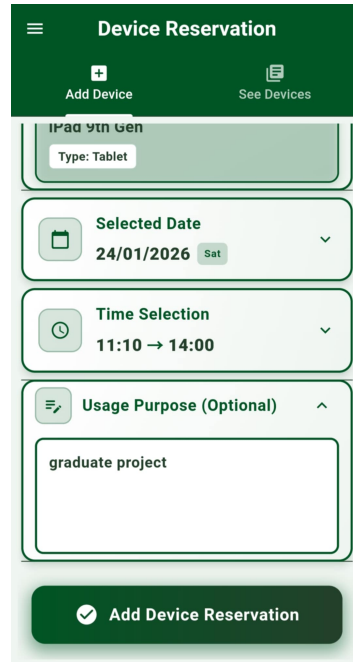
Figure 4.7: Device reservation workflow in the BZU Assistant application.

Device Reservation Workflow and Status Handling

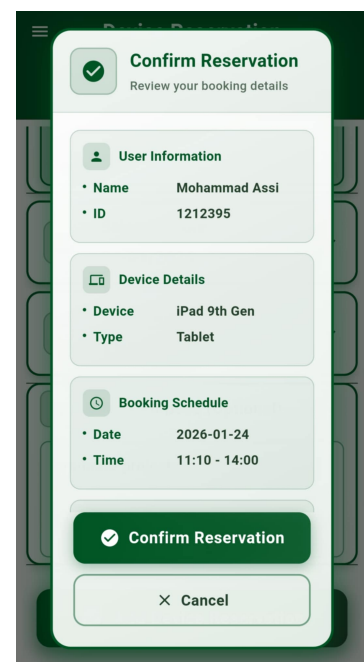
Figure 4.8 illustrates the complete device reservation workflow after a device is selected. The user specifies the reservation date and time range, after which the system validates the input, checks for scheduling conflicts, and prepares the reservation request. Once submitted, the reservation is created with a *Pending* status and a notification is automatically sent to the secretary for approval. This workflow prevents double booking and ensures controlled access to shared university devices.



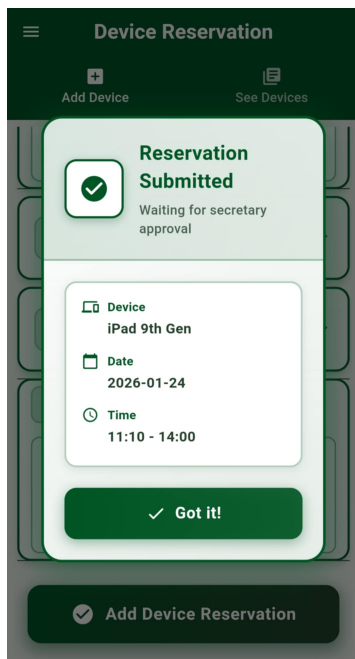
(a) Manual device search and selection.



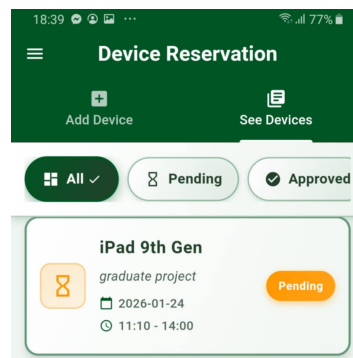
(b) Reservation form with date and time selection.



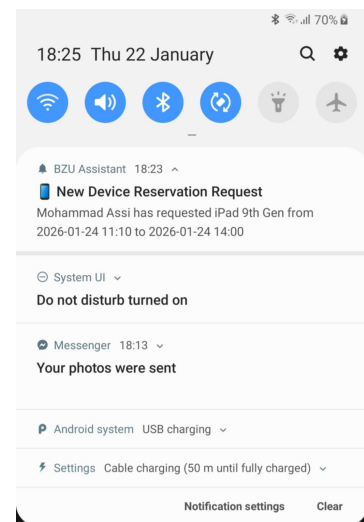
(c) Reservation confirmation dialog.



(d) Submission success dialog.



(e) User reservation list with status tracking.



(f) Secretary push notification for approval.

Figure 4.8: End-to-end device reservation workflow and status management in BZU Assistant.

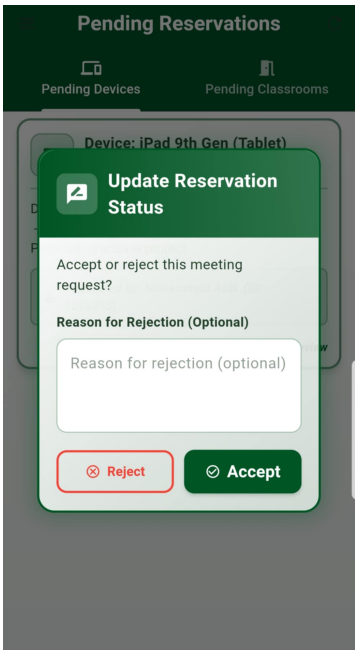
Secretary Approval Workflow for Device Reservation

Figure 4.9 illustrates the secretary-side workflow for handling device reservation requests. The secretary accesses the *Pending Devices* page to review incoming requests. Upon se-

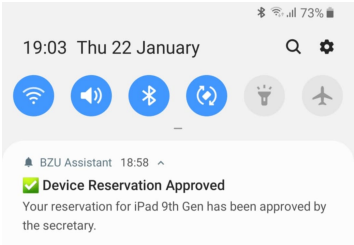
lecting a pending reservation, a decision dialog is displayed, allowing the secretary to either approve or reject the request. In the case of rejection, an optional justification message can be provided. Once a decision is made, the requesting user receives a push notification indicating whether the reservation was approved or rejected, and the reservation status is updated accordingly within the system.



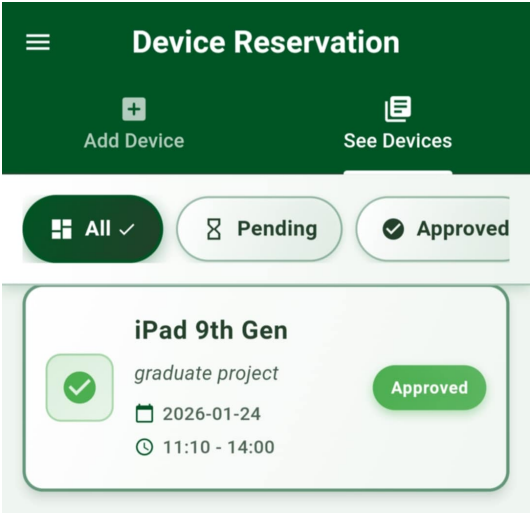
(a) Pending device reservation details (secretary view).



(b) Reservation approval or rejection dialog.



(c) User notification after secretary decision.

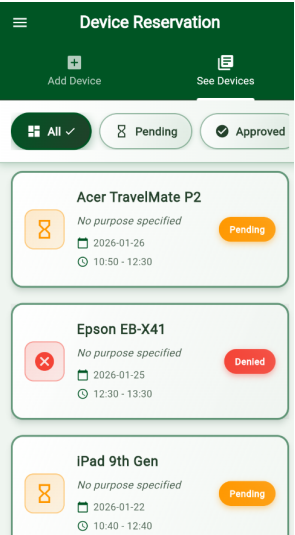


(d) Updated reservation status in user reservation list.

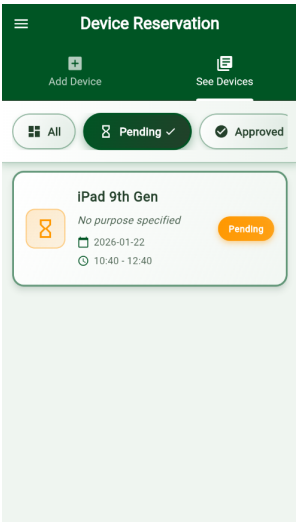
Figure 4.9: Secretary approval workflow for device reservations in the BZU Assistant application.

See Devices – Reservation Status Tracking

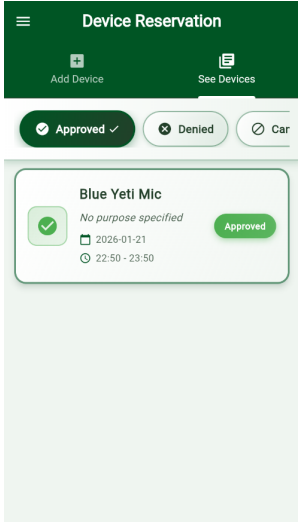
Figure 4.10 illustrates the *See Devices* page, which allows users to view and track all device reservation requests within a single interface. Each reservation is displayed with its current status, including *Pending*, *Approved*, *Denied*, *Canceled*, and *Completed*. Filtering options enable users to quickly narrow reservations by status, allowing efficient monitoring of reservation progress, review of past requests, and clear understanding of each reservation outcome without requiring direct administrative interaction.



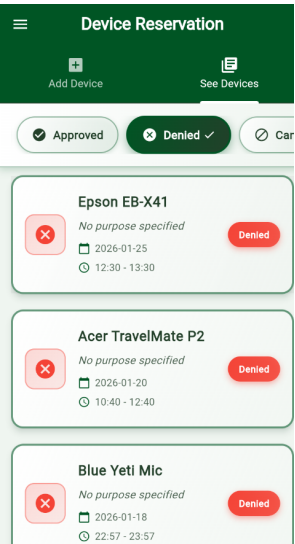
(a) All device reservations view.



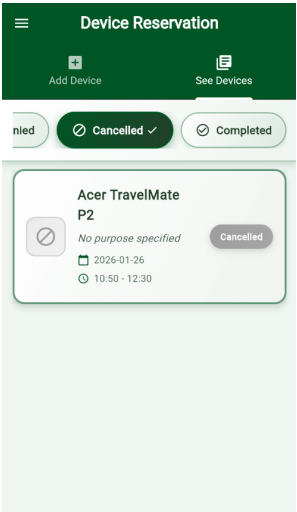
(b) Pending reservations.



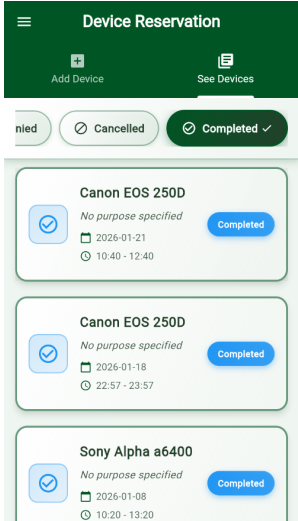
(c) Approved reservations.



(d) Denied reservations.



(e) Canceled reservations.

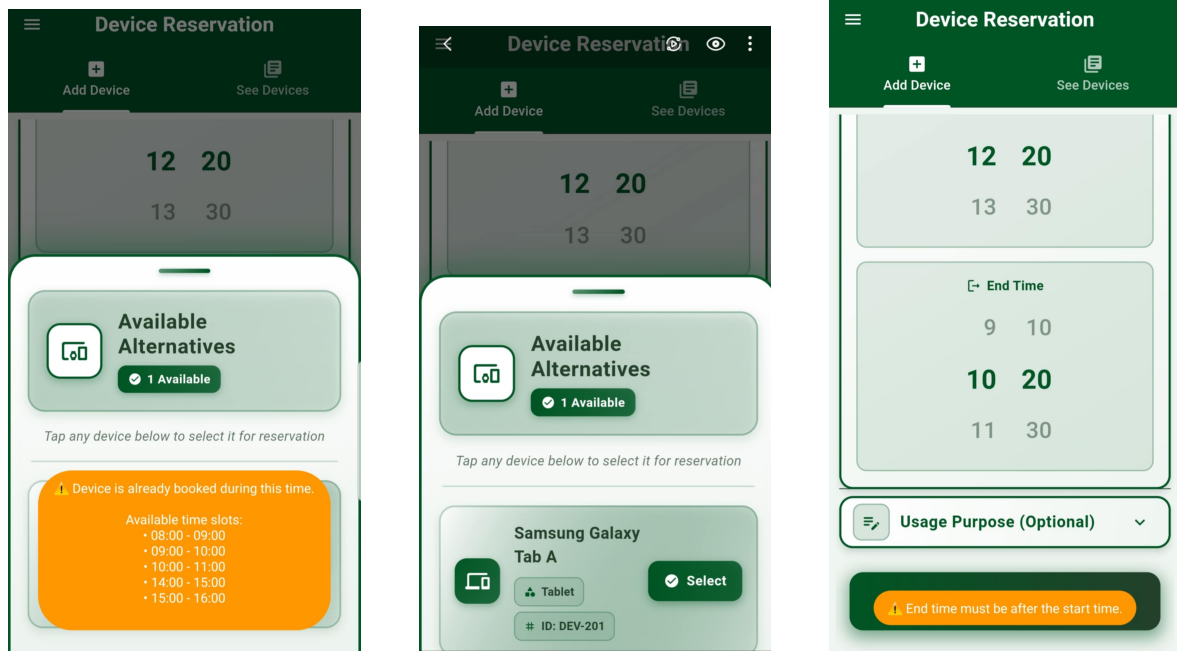


(f) Completed reservations.

Figure 4.10: Device reservation status tracking and filtering in the *See Devices* page.

Device Reservation Conflict Handling and Validation

Figure 4.11 illustrates how the system handles conflict and validation cases during the device reservation process. When a user attempts to reserve a device that is already booked for the selected time range, the system first suggests alternative available time slots. If the conflict persists, the system then displays alternative devices belonging to the same category. Additionally, input validation is enforced, and an error notification is shown when the user enters an invalid time range.



(a) Alternative available time slots for a booked device.

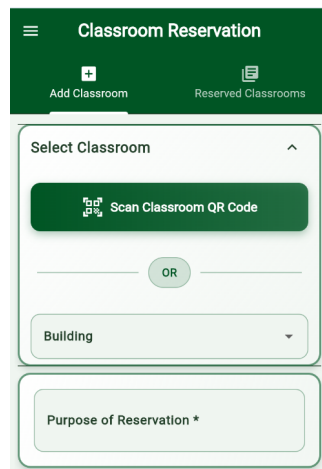
(b) Alternative devices from the same category.

(c) Invalid time range validation message.

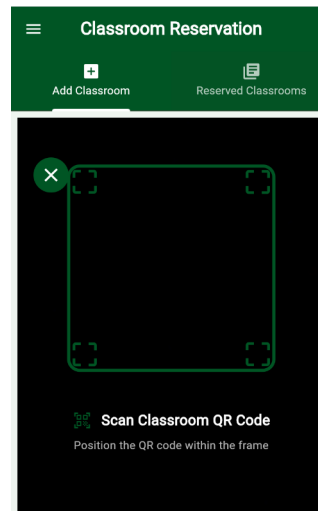
Figure 4.11: Conflict handling and validation feedback during device reservation in the BZU Assistant application.

4.4.3 Classroom Reservation

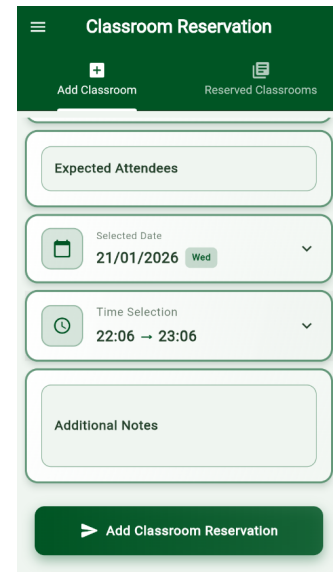
Figure 4.12 illustrates the classroom reservation workflow in the BZU Assistant application. This feature allows students and instructors to request classrooms or laboratories for exams, makeup lectures, and special academic activities. Users can select a classroom either by scanning the classroom QR code or by manually choosing the building and classroom. After selection, the user specifies the reservation purpose, expected number of attendees, date, and time range, with optional additional notes. The system validates classroom availability and submits the request with a *Pending* status for secretary approval, ensuring controlled access and preventing scheduling conflicts.



(a) Classroom selection by building and room.



(b) Classroom selection using QR code scanning.



(c) Reservation details including date, time, capacity, purpose, and notes.

Figure 4.12: Classroom reservation main workflow screens in the BZU Assistant application.

Classroom Reservation Request Completion

Figure 4.13 illustrates the completion and submission workflow of a classroom reservation request. After selecting a classroom, the user specifies the purpose of use, expected number of attendees, reservation date, time range, and any additional notes. The system displays detailed classroom information, including room type, capacity, floor, and available equipment, allowing the user to verify suitability before submission. Once the request is submitted, the system confirms successful submission and assigns the reservation a *Pending* status, indicating that the request is awaiting secretary review and approval.

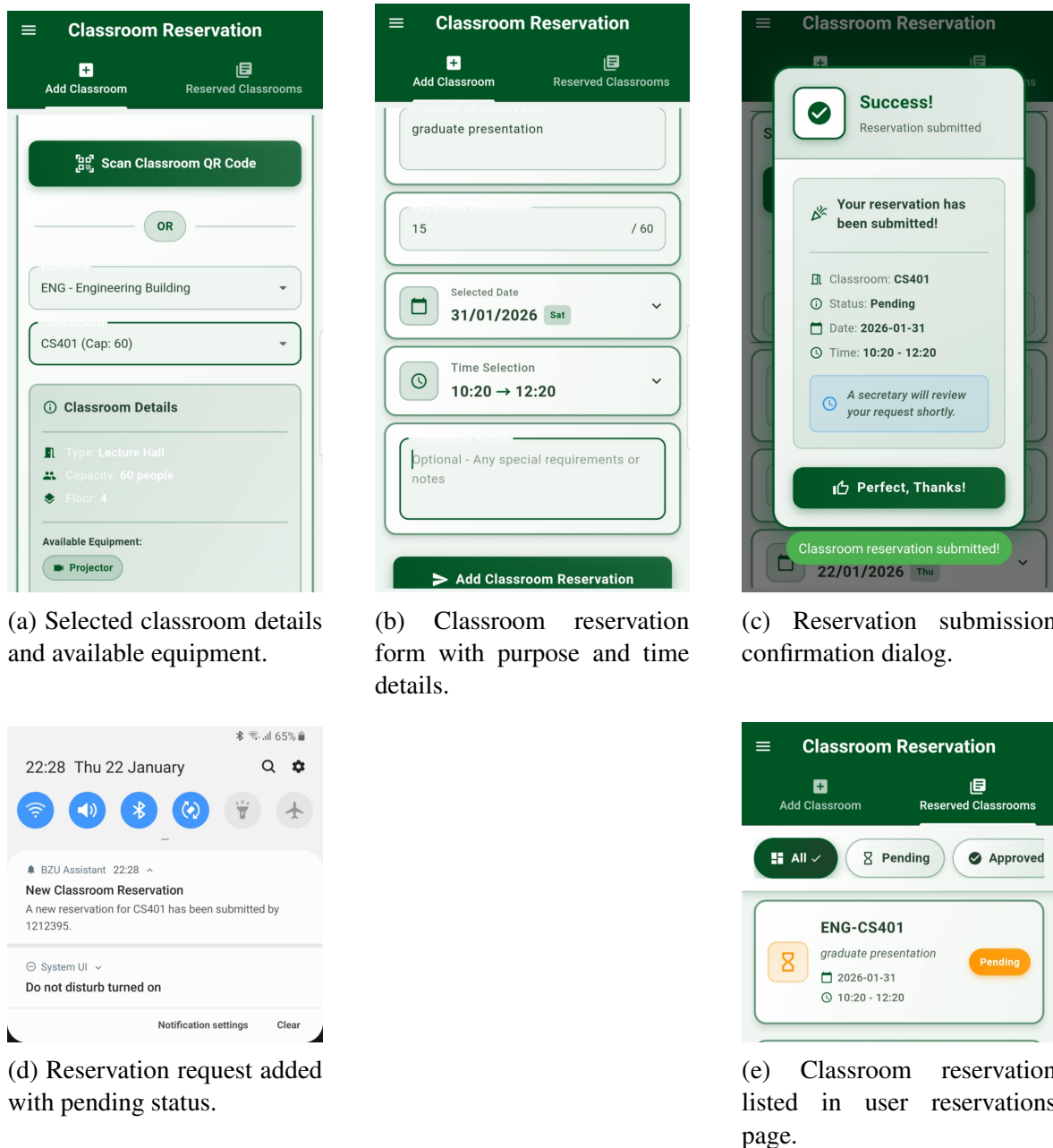


Figure 4.13: Classroom reservation request completion and submission workflow in the BZU Assistant application.

Secretary Approval Workflow for Classroom Reservation

Figure 4.14 illustrates the secretary-side workflow for handling classroom reservation requests. The secretary accesses the *Pending Classrooms* page, which lists all classroom reservation requests awaiting review. Upon selecting a request, a decision dialog is displayed, allowing the secretary to either approve or reject the reservation. In the case of rejection, an optional justification message can be provided to the requesting user. Once a decision is made, the user receives a push notification indicating whether the request has been approved or denied, and the reservation status is updated accordingly within the system.

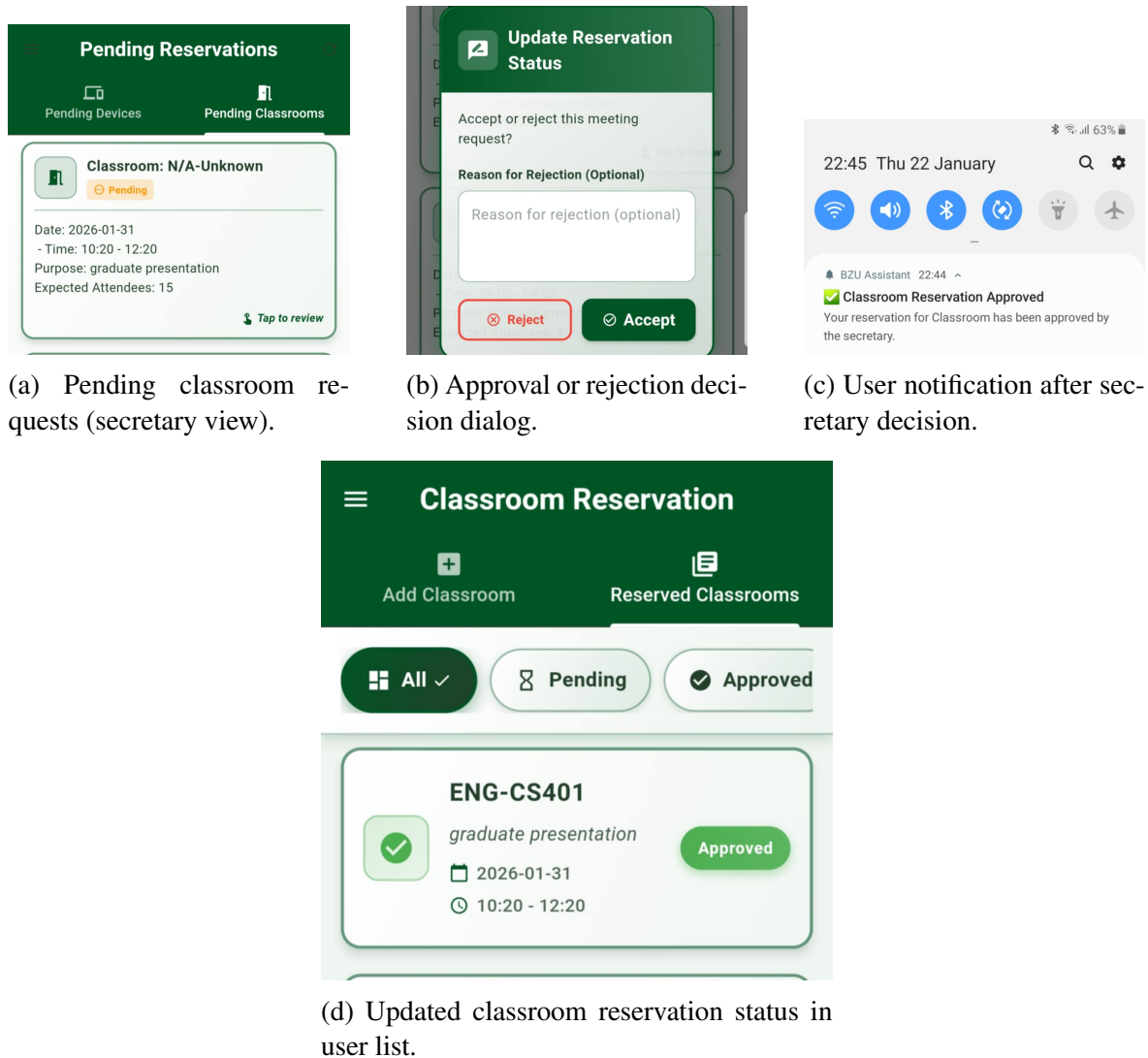
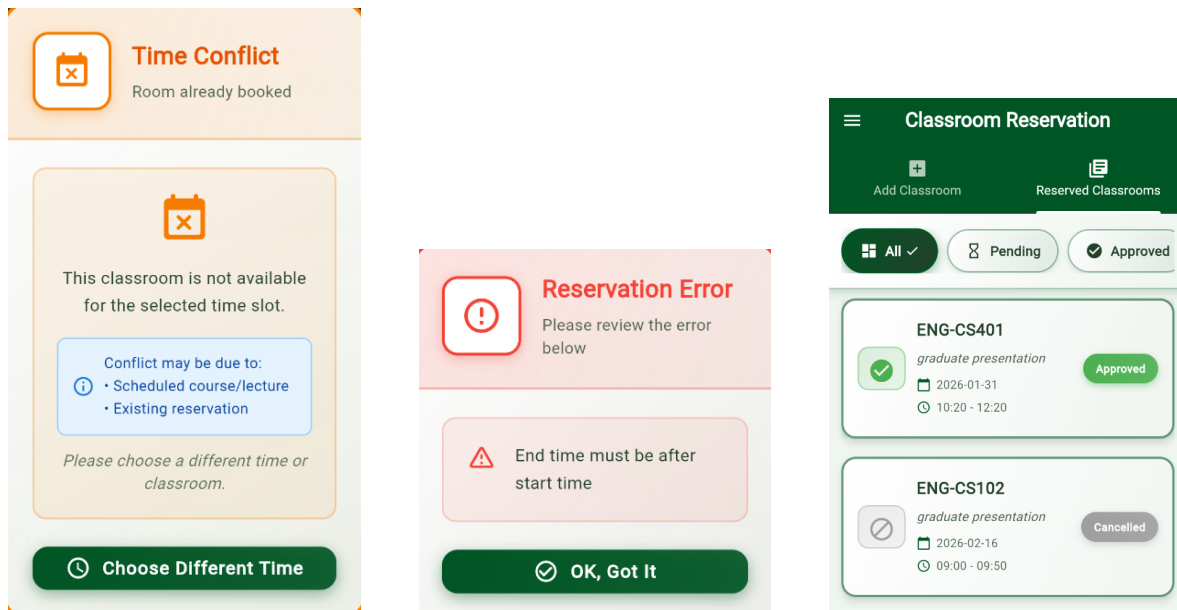


Figure 4.14: Secretary approval workflow for classroom reservations in the BZU Assistant application.

Classroom Reservation Conflict Handling and Status Tracking

Figure 4.15 illustrates how the system handles conflict and validation scenarios during classroom reservation, in addition to providing reservation status tracking. When a user attempts to reserve a classroom that is already booked or has a scheduled lecture at the selected time, a time-conflict dialog is displayed. If the user enters an invalid time range, the system presents a validation error dialog. The *Reserved Classrooms* page allows users to view all their classroom reservation requests along with their current statuses, including *Pending*, *Approved*, *Denied*, *Canceled*, and *Completed*.



(a) Time conflict dialog for an already reserved classroom.

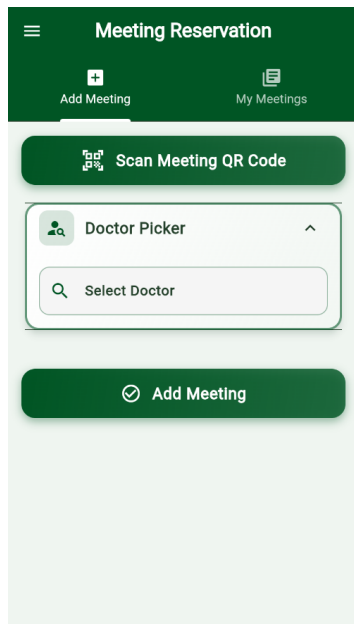
(b) Invalid time range validation dialog.

(c) Reserved classrooms page with status tracking.

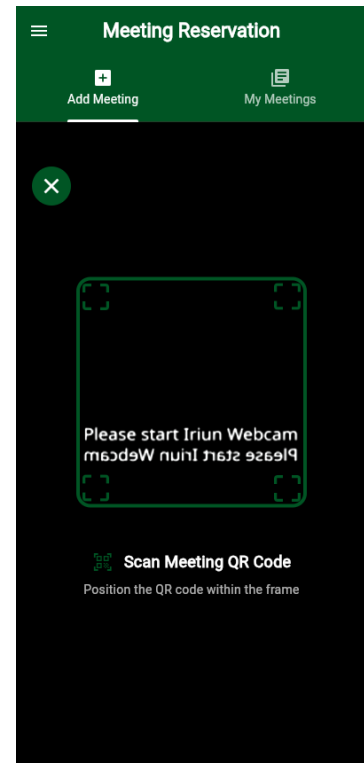
Figure 4.15: Conflict handling, validation feedback, and reservation status tracking for classroom reservations in the BZU Assistant application.

4.4.4 Reserve Meeting

Figure 4.16 illustrates the meeting reservation main screen in the BZU Assistant application. This feature allows students to reserve meetings with instructors during their office hours. Users can initiate a reservation either by selecting an instructor from the list or by scanning the instructor's QR code, which retrieves the available office hour slots and prepares the reservation process.



(a) Meeting reservation main interface.

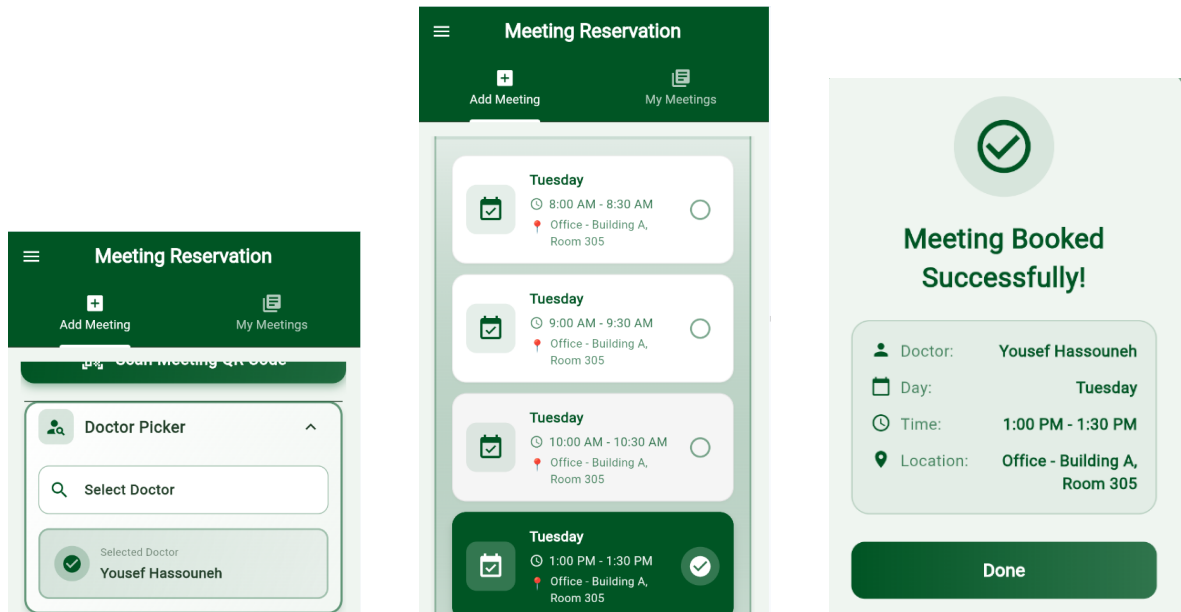


(b) QR code scanning to retrieve instructor office hour slots.

Figure 4.16: Meeting reservation main screen using direct selection and QR-based instructor identification.

Meeting Time Slot Selection and Confirmation

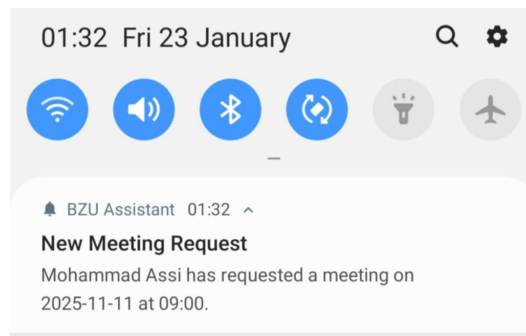
Figure 4.17 illustrates the workflow of reserving an office hour meeting after selecting an instructor. The system displays the available office hour time slots associated with the selected instructor, allowing the student to choose a suitable slot. The selected time slot is validated to prevent scheduling conflicts. Once confirmed, the meeting request is submitted and a success confirmation dialog is displayed, summarizing the meeting details such as instructor name, date, time, and location. The instructor is then notified of the new meeting request.



(a) Instructor selection during meeting reservation.

(b) Office hour time slot selection.

(c) Meeting reservation success confirmation.



(d) Instructor notification for a new meeting request.

Figure 4.17: Meeting reservation workflow from instructor selection to successful booking and notification.

Instructor Meeting Reservation Management

Figure 4.18 illustrates the instructor-side workflow for managing office hour meeting reservations in the BZU Assistant application. Each instructor is assigned a unique QR code that exposes their available office hours to students upon scanning. From the instructor interface, submitted meeting requests are organized into dedicated tabs, allowing instructors to review pending requests efficiently. For each request, the instructor can inspect the meeting details, including student name, date, time, and location, and then either approve or reject the request through confirmation dialogs. Once a decision is made, the system updates the meeting status in real time and sends a push notification to the student. Approved meetings are subsequently listed in the upcoming meetings view, ensuring clear visibility of confirmed appointments and consistent lifecycle management.

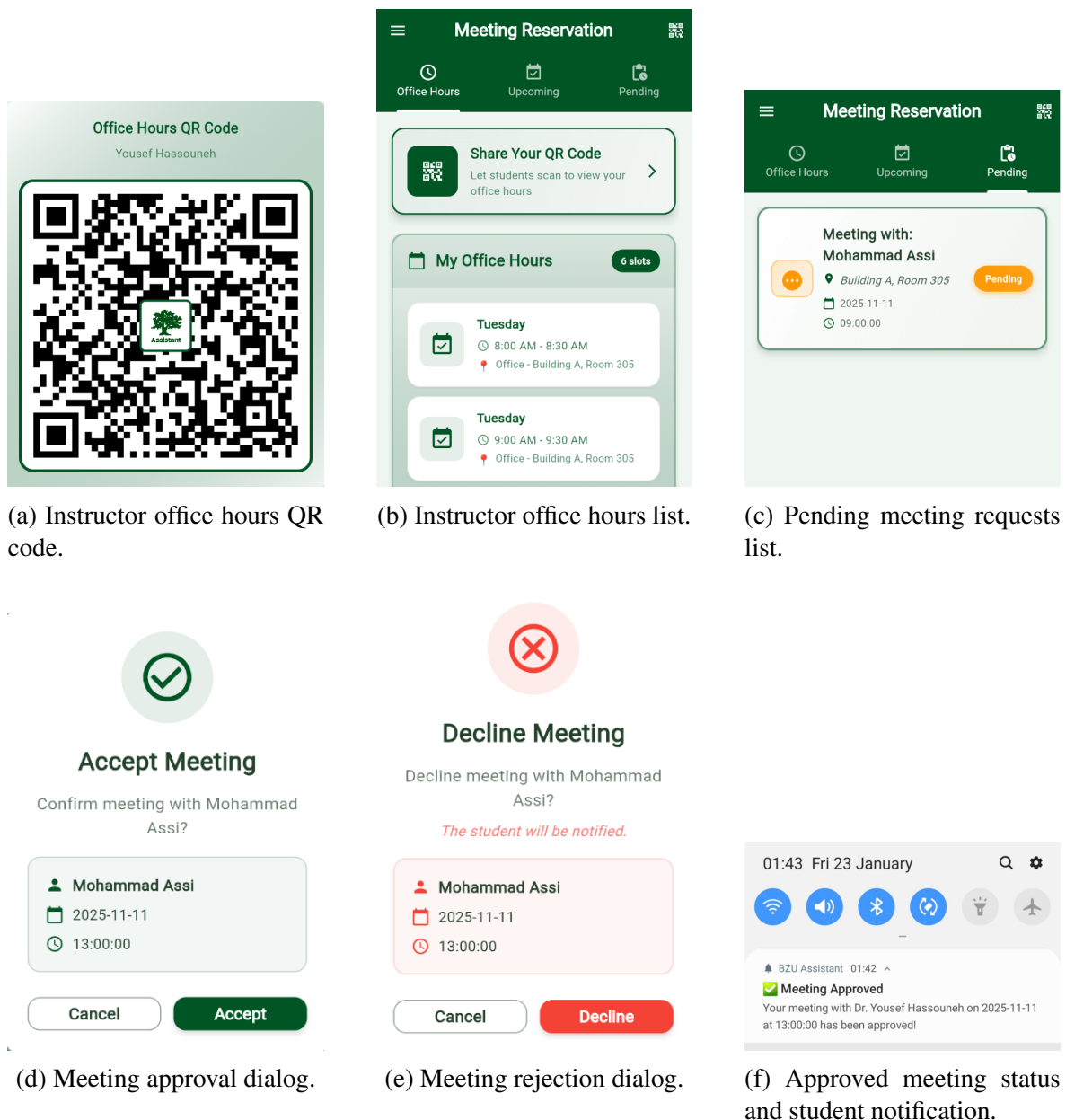


Figure 4.18: Instructor-side meeting reservation workflow showing QR-based office hours sharing, pending request review, approval and rejection actions, and final status updates with student notifications.

My Meetings – Status Tracking

Figure 4.19 illustrates the *My Meetings* page, which enables students to view and track all office hour meeting requests they have submitted. Each meeting entry displays essential information, including the instructor name, meeting location, date, time, and current request status (*Pending*, *Approved*, or *Declined*). Visual status indicators allow students to quickly identify the outcome of each request, providing clear visibility into upcoming meetings as well as rejected requests without requiring direct communication with the instructor.

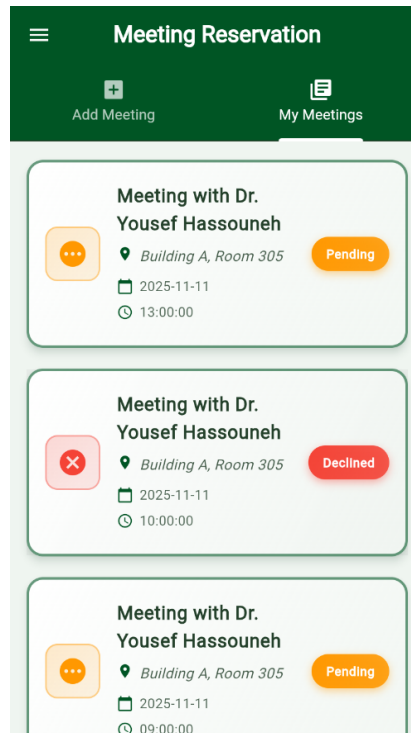


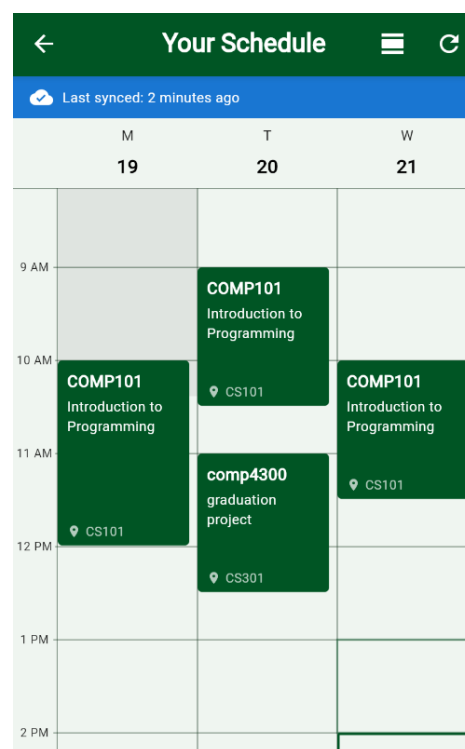
Figure 4.19: My Meetings page showing submitted meeting requests and their current statuses.

4.4.5 Daily and Weekly Schedule

Figure 4.20 illustrates the daily and weekly schedule feature in the BZU Assistant application. When the user opens the Daily/Weekly Schedule page, the system displays the daily schedule by default, showing detailed course information including course number, course name, instructor, classroom, and start–end time. Users can switch to the weekly schedule view by selecting the toggle button located next to the refresh icon, which presents all scheduled courses distributed across the academic week.



(a) Daily schedule view displaying courses for a selected day.



(b) Weekly schedule view showing courses across the academic week.

Figure 4.20: Daily and weekly schedule visualization in the BZU Assistant application.

4.4.6 Academic Events

Figure 4.21 illustrates the main instructor-facing interfaces of the Academic Events feature. The workflow begins from the instructor dashboard, where the *Manage Events* option is available as a dedicated feature entry point. This ensures quick access to event-related functionality without navigating through unrelated modules.

The event creation interface allows instructors to define all essential event details, including the event title, description, location, date, and start/end times. The layout is form-based, with clearly labeled input fields and a dedicated action button for generating the event and its corresponding QR code. The separation between event details and date/time selection improves readability and reduces input errors, while the consistent visual styling aligns with the overall application theme.

Once events are created, instructors can view and manage them through the *My Events* page. This page displays a list of all created events, showing key metadata such as event title, location, date, time range, and scan count. Each event card provides quick access to actions such as viewing the QR code or deleting the event. Past events are visually distinguished to help instructors differentiate between upcoming and completed activities, supporting better event organization and tracking.

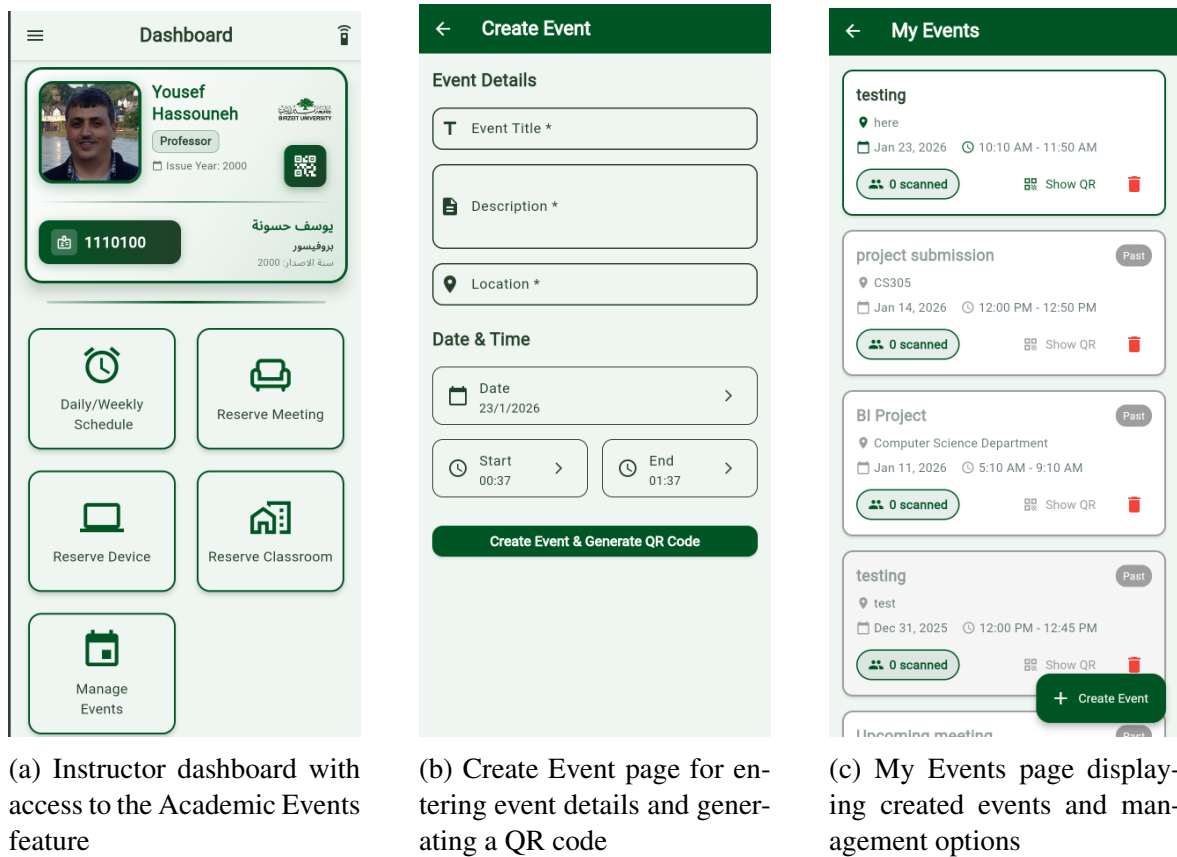


Figure 4.21: Academic Events user interface for instructors

The Academic Events user interface for instructors provides a clear and guided workflow for creating events, validating inputs, generating QR codes, and managing created events. As illustrated in Figure 4.22, the process begins on the `Create Event` page, where the instructor fills in the required event details including the event title, description, location, date, and start/end times. Date selection is handled through a calendar dialog, while time selection is performed using dedicated start and end time pickers. During data entry, the system performs immediate validation; for example, if the end time is earlier than the start time, a toast message is displayed to inform the instructor of the invalid time range and prevent submission. Once all fields are valid, the instructor submits the form, and the system generates a dedicated QR code for the event. This QR code is displayed in a modal dialog, allowing the instructor to save it directly to the device gallery for sharing with students. After successful creation, the event is automatically added to the `My Events` page, where instructors can view all their events, see the number of student scans, re-open or download the QR code, and manage past or upcoming events. This interface ensures a smooth, error-aware, and efficient event creation experience while maintaining real-time synchronization between event creation, QR code generation, and event listing.

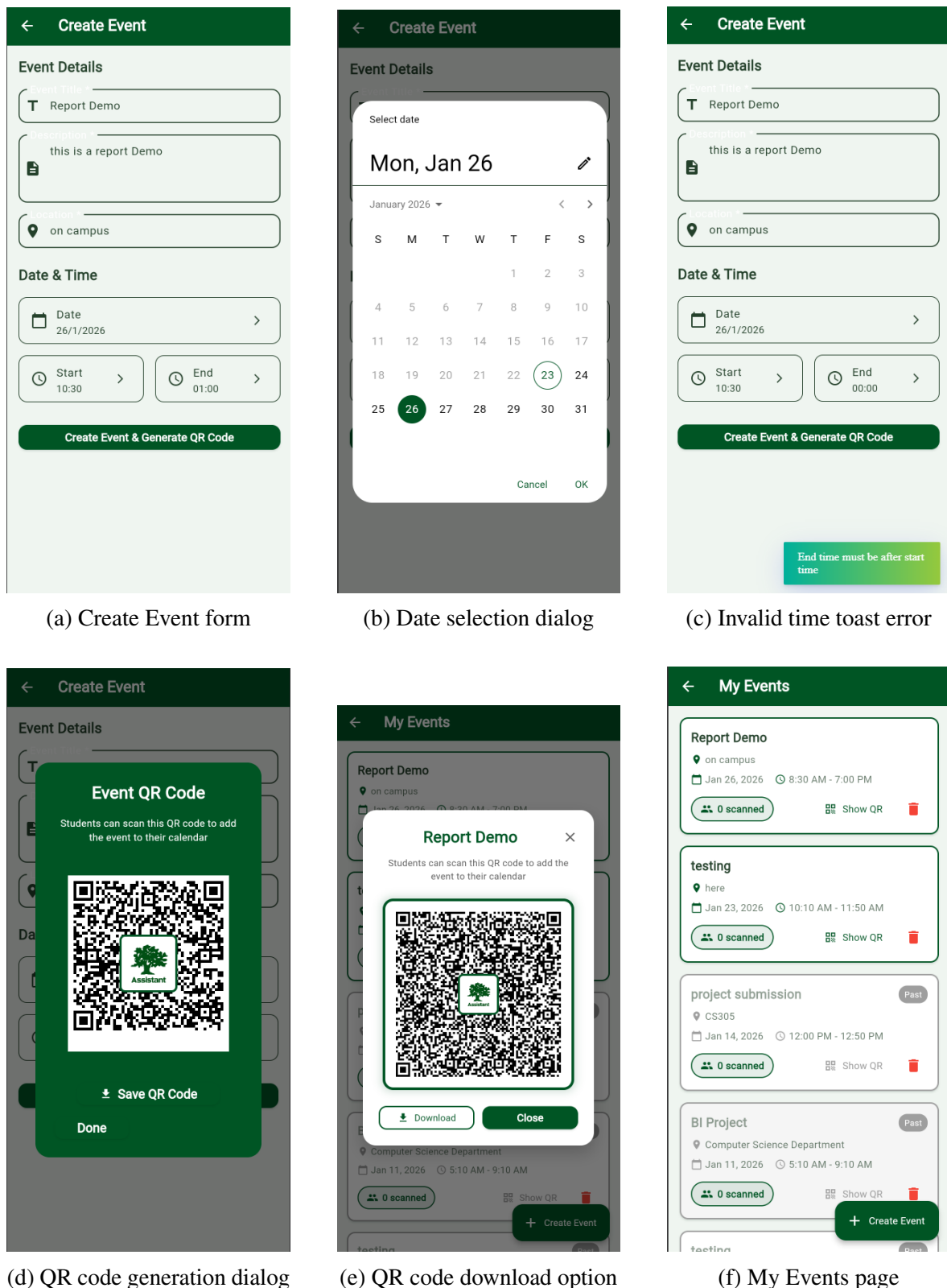
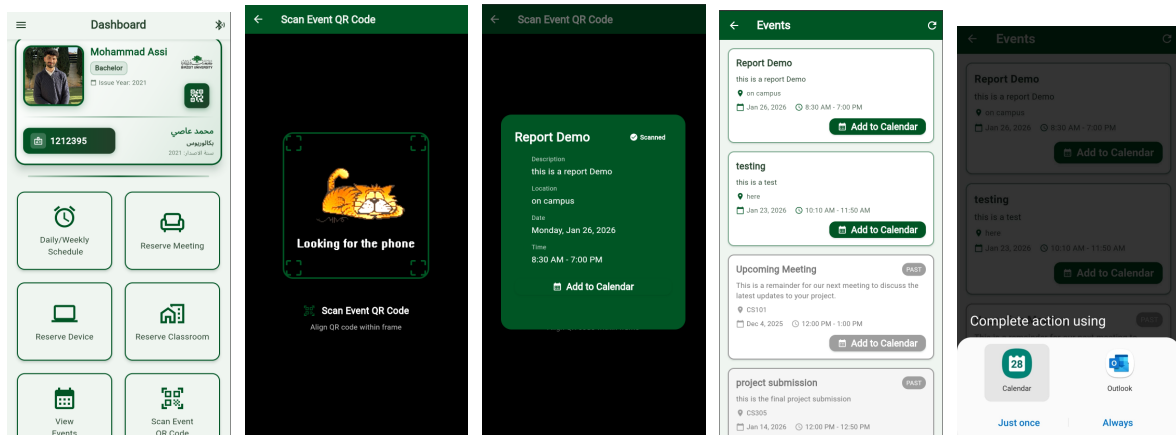


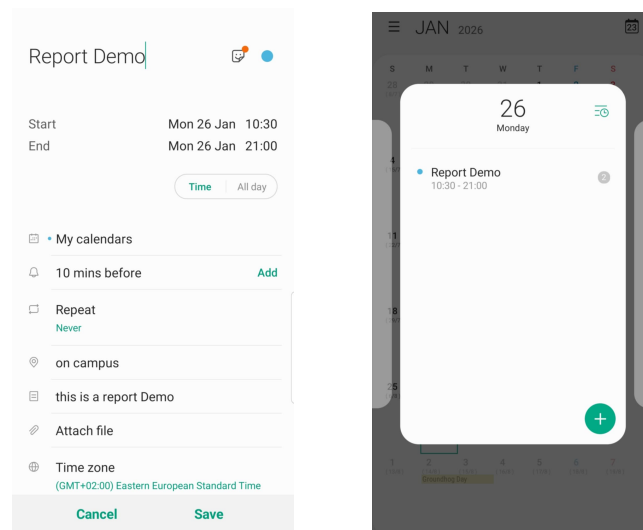
Figure 4.22: Instructor-side Academic Events user interface flow

On the student side, the Academic Events feature provides a simple and intuitive flow for discovering events, viewing event details, and adding events to the personal calendar. As shown in Figure 4.23, the process starts from the student dashboard, where the user se-

lects the `Scan Event QR Code` option. This action opens a dedicated scanning interface that activates the device camera and guides the student to align the event QR code within a highlighted frame. Once the QR code is successfully detected and validated, the system immediately displays a confirmation dialog containing the event details, including title, description, location, date, and time, along with a visual indicator confirming that the event has been scanned. At this stage, the student can add the event directly to the device calendar using the `Add to Calendar` button, which launches the native calendar application and pre-fills all event information. After successful scanning, the event is automatically stored in the student's event history and appears in the `View Events` page. This page presents a chronological list of all scanned events, clearly distinguishing upcoming and past events, and allows students to repeatedly access event information or re-add events to their calendar if needed. This flow ensures that students can seamlessly transition from QR code scanning to event tracking and calendar integration, providing a smooth and reliable user experience for managing academic events.



(a) Student dashboard entry point (b) Scan Event QR Code interface (c) Scanned event details dialog (d) Events page after adding the new calendar event (e) Adding event to calendar



(f) Calendar page for adding event (g) The event after added to calendar

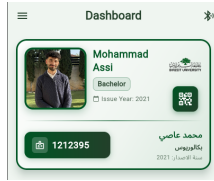
Figure 4.23: Student-side Academic Events user interface flow

4.4.7 Schedule Sharing via Digital ID QR Code

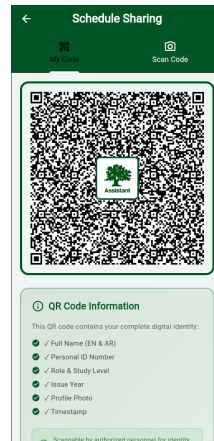
The schedule sharing user interface allows users to securely share and view academic schedules through Digital ID QR codes in a clear and guided interaction flow. As shown in Figure 4.24, the process begins from the dashboard where each user's digital ID card displays a dedicated QR code icon. When the user taps this icon, the system navigates to the *ScheduleSharing* screen and automatically opens the *My Code* tab, which presents the user's unique Digital ID QR code. This QR code visually encodes the user's academic identity, including full name in both English and Arabic, personal ID number, role, study level, issue year, profile photo reference, and a generation timestamp, accompanied by an information panel explaining the embedded data and clarifying that the code is scannable only by

authorized personnel.

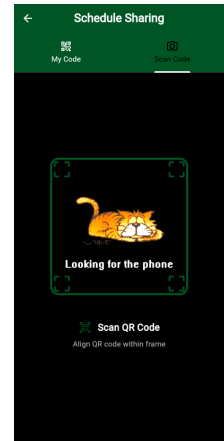
The interface also provides a *Scan Code* tab that activates a camera-based scanning screen. When another authorized user scans a Digital ID QR code, the system displays a verification dialog showing the scanned user's identity details along with a confirmation indicator and a clear *View Schedule* action. Selecting this option navigates to the scanned user's schedule page, where the schedule can be explored in both weekly calendar and daily timeline views using the same visual structure as the native schedule feature. To support continuity and reuse, every successfully scanned schedule is stored and listed in the *Scanned Schedules* page, allowing users to revisit previously scanned schedules without rescanning the QR code. This interface design ensures intuitive navigation between QR display, scanning, identity verification, and schedule visualization while maintaining consistency, clarity, and ease of access across all stages of the schedule sharing workflow.



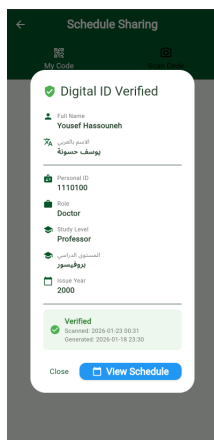
(a) Dashboard with Digital ID QR icon



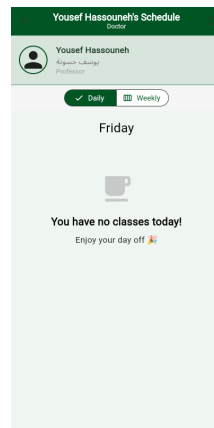
(b) My code interface.



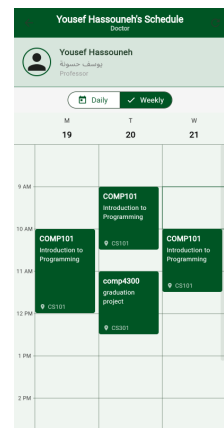
(c) Scan QR code interface



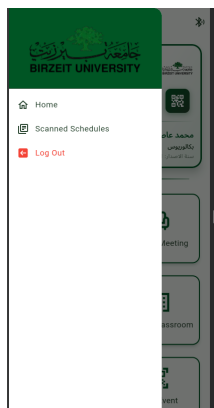
(d) Digital ID verification info with View Schedule option



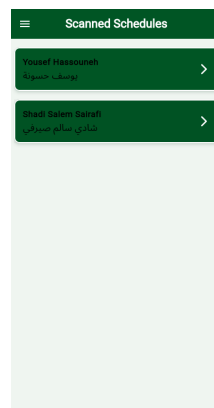
(e) Daily Schedule view



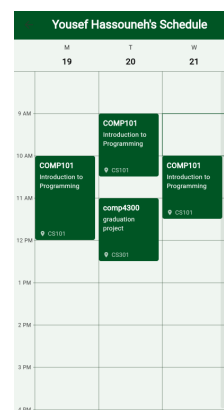
(f) Weekly schedule view



(g) Navigation drawer access



(h) Scanned schedules list



(i) The schedule from scanned schedules list

Figure 4.24: Schedule Sharing user interface flow using Digital ID QR codes

4.4.8 Register Attendance

The subsection describes the attendance registration feature in the BZU Assistant application, which supports both manual and automatic attendance modes. The workflow is designed to give instructors flexible control over attendance tracking depending on classroom conditions and technical availability.

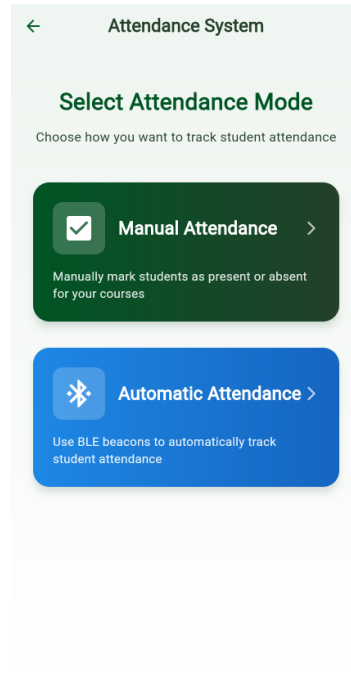
Manual Attendance

Figure 4.25 illustrates the manual attendance registration workflow from the instructor's perspective. The process begins when the instructor selects the attendance feature from the dashboard and clicks the beacon command icon located at the top-right corner. The system then displays the attendance mode selection screen, where the instructor chooses the *Manual Attendance* option.

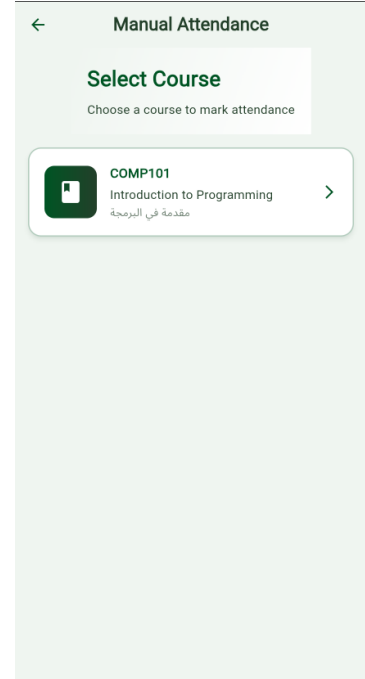
After selecting manual attendance, the system presents a list of courses taught by the instructor. Once a course is selected, the application displays the list of students enrolled in that course along with the selected lecture date. The instructor manually marks students as present or absent using selection checkboxes and confirms the process by saving the attendance record. This approach provides a reliable fallback mechanism when automatic attendance is unavailable or requires manual verification.



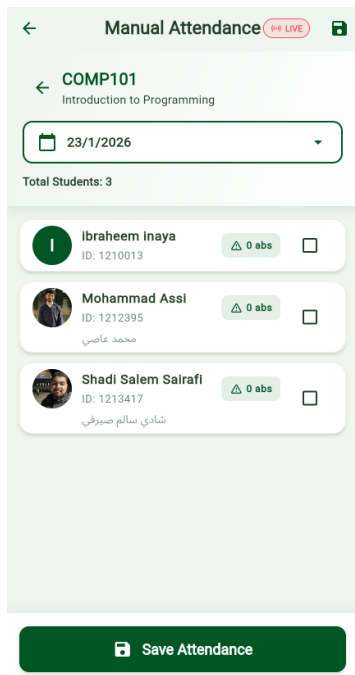
(a) Instructor dashboard with access to the attendance feature.



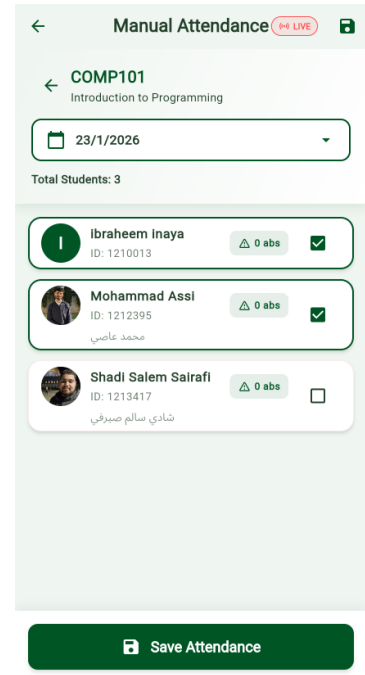
(b) Attendance mode selection screen.



(c) Course selection for manual attendance.



(d) Student list with attendance selection options.



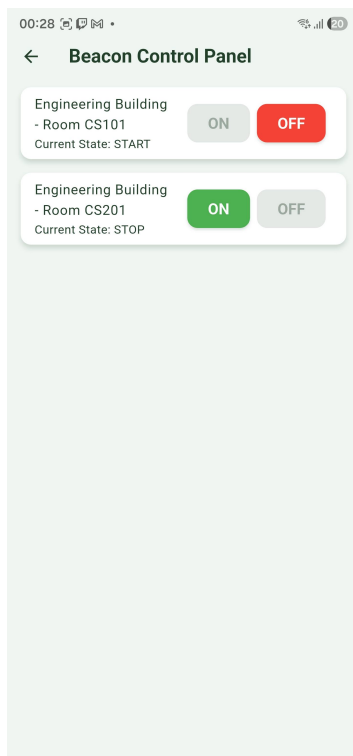
(e) Saved attendance after manual selection.

Figure 4.25: Manual attendance registration workflow for instructors in the BZU Assistant application.

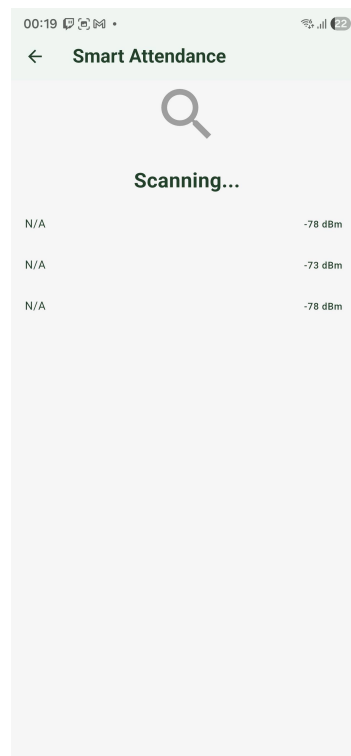
Automatic Attendance

Figure 4.26 illustrates the automatic attendance workflow based on Bluetooth Low Energy (BLE) beacons. After accessing the attendance feature, the instructor selects the *Automatic Attendance* option from the attendance mode selection screen. The instructor then starts the attendance session by pressing the *ON* button in the Beacon Control Panel, which activates the BLE beacon for the corresponding classroom.

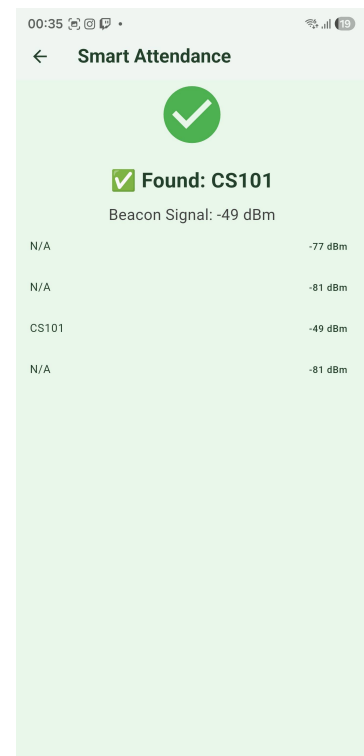
Once the attendance session is active, students open the application dashboard and enable Bluetooth by selecting the Bluetooth icon located at the top-right corner. The system automatically scans for nearby classroom beacons and measures the received signal strength. When the classroom beacon is detected with sufficient signal strength, the system confirms the classroom location and registers the student as present without any manual interaction. This process ensures fast, contactless, and reliable attendance tracking while minimizing instructor intervention.



(a) Beacon control panel for starting and stopping attendance sessions.



(b) Automatic attendance scanning interface on the student device.



(c) Successful classroom beacon detection and attendance confirmation.

Figure 4.26: Automatic attendance workflow using BLE beacons, instructor-controlled sessions, and student-side Bluetooth scanning.

Chapter 5

Testing

The chapter presents the testing activities conducted to verify the correctness, reliability, and functional completeness of the BZU Assistant system. The primary objective of testing was to ensure that each system feature operates according to the specified functional requirements and behaves consistently under expected usage scenarios. A structured test case-based approach was adopted, where core use cases were validated through predefined test scenarios, execution steps, and expected outcomes. The testing process focused on critical system modules, including digital identity authentication, automated attendance tracking, device and classroom reservation, office hours management, and academic event handling. The results of these tests provide evidence that the system meets its design objectives and is suitable for deployment within a real university environment.

5.1 Device Reservation Testing

The section presents the test case execution results for the Device Reservation feature of the BZU Assistant system. The executed test cases validate the core reservation workflow, including device selection, availability checking, conflict prevention, and the secretary approval or denial process. The test scenarios cover successful reservations, reservation conflicts due to unavailable devices, and denied requests, ensuring that the system enforces reservation rules and maintains consistent device availability states.

Table 5.1: Test Case Execution – Reserve Device (Successful Reservation)

Test Case Title		Reserve Available Device – Approved			
Test Date		1/18/2026			
Test Scenario		Device Reservation			
Test Designed By		Mohammad Assi			
Test Description		User successfully reserves an available device and gets approval.			
Pre-Conditions		- User is authenticated. - Device is available for the selected time slot. - Secretary account exists.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Search for a device	Available devices are listed	As expected	Pass	
2	Select device and time range	System validates availability	As expected	Pass	
3	Submit reservation request	Request stored as Pending	As expected	Pass	
4	Secretary approves request	Status changes to Approved	As expected	Pass	Notification sent

Table 5.2: Test Case Execution – Reserve Device (Conflict)

Test Case Title		Reserve Device Already Reserved			
Test Date		1/18/2026			
Test Scenario		Device Reservation			
Test Designed By		Mohammad Assi			
Test Description		User attempts to reserve a device already booked			
Pre-Conditions		- User is authenticated. - Device is reserved during requested time slot.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Search for device	Device details displayed	As expected	Pass	
2	Select reserved time slot	System detects conflict	As expected	Pass	
3	Submit reservation request	Request rejected with message	As expected	Pass	Alternatives suggested

Table 5.3: Test Case Execution – Reserve Device (Denied)

Test Case Title		Reserve Device – Denied by Secretary			
Test Date		1/18/2026			
Test Scenario		Device Reservation			
Test Designed By		Mohammad Assi			
Test Description		Reservation request is denied by secretary			
Pre-Conditions		- User is authenticated. - Device is available. - Secretary account exists.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Submit reservation request	Request stored as Pending	As expected	Pass	
2	Secretary reviews request	Decision options displayed	As expected	Pass	
3	Secretary denies request	Status updated to Denied	As expected	Pass	Reason provided
4	User receives notification	Denial notification shown	As expected	Pass	

5.2 Classroom Reservation Testing

The section describes the test case execution results for the Classroom Reservation feature of the BZU Assistant system. The test cases focus on validating classroom availability checks, time-slot conflict detection, and the secretary approval or denial workflow. The executed scenarios ensure that the system prevents double booking, correctly updates classroom reservation statuses, and maintains accurate classroom availability throughout the reservation life-cycle.

Table 5.4: Test Case Execution – Reserve Classroom (Successful Reservation)

Test Case Title		Reserve Available Classroom – Approved			
Test Date		1/18/2026			
Test Scenario		Classroom Reservation			
Test Designed By		Mohammad Assi			
Test Description		User successfully reserves an available classroom and gets approval.			
Pre-Conditions		- User is authenticated. - Classroom is available for the selected time slot. - Secretary account exists.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Search for a class-room	Classrooms are listed	As expected	Pass	
2	Select classroom and time range	System validates availability	As expected	Pass	
3	Submit reservation request	Request stored as Pending	As expected	Pass	
4	Secretary approves request	Status changes to Approved	As expected	Pass	Notification sent

Table 5.5: Test Case Execution – Reserve Classroom (Conflict)

Test Case Title		Reserve Classroom Already Reserved			
Test Date		1/18/2026			
Test Scenario		Classroom Reservation			
Test Designed By		Mohammad Assi			
Test Description		User attempts to reserve a classroom already booked			
Pre-Conditions		- User is authenticated. - Classroom is reserved during requested time slot.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Search for classroom	Classroom details displayed	As expected	Pass	
2	Select reserved time slot	System detects conflict	As expected	Pass	
3	Submit reservation request	Request rejected with message	As expected	Pass	

Table 5.6: Test Case Execution – Reserve Classroom (Denied)

Test Case Title		Reserve Classroom – Denied by Secretary			
Test Date		1/18/2026			
Test Scenario		Classroom Reservation			
Test Designed By		Mohammad Assi			
Test Description		Reservation request is denied by secretary			
Pre-Conditions		- User is authenticated. - Classroom is available. - Secretary account exists.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Submit reservation request	Request stored as Pending	As expected	Pass	
2	Secretary reviews request	Decision options displayed	As expected	Pass	
3	Secretary denies request	Status updated to Denied	As expected	Pass	Reason provided
4	User receives notification	Denial notification shown	As expected	Pass	

5.3 Office Hours (Book Meeting) Testing

The section presents the test case execution results for the Office Hours (Book Meeting) feature of the BZU Assistant system. The executed test cases validate the end-to-end meeting request workflow, including instructor discovery, time-slot selection, conflict validation, request submission with *Pending* status, and instructor approval or denial with user notification.

Table 5.7: Test Case Execution – Book Meeting (Approved)

Test Case Title		Book Office Hours Meeting – Approved			
Test Date		1/18/2026			
Test Scenario		Office Hours (Book Meeting)			
Test Designed By		Mohammad Assi			
Test Description		Student submits a meeting request and instructor approves it			
Pre-Conditions		- Student is authenticated. - Instructor has published office-hour time slots. - Selected time slot is available (no overlapping pending/approved meetings).			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Select instructor	Instructor office hours displayed	As expected	Pass	
2	Select time slot	System validates slot availability	As expected	Pass	
3	Submit meeting request	Request created with status Pending	As expected	Pass	Instructor notified
4	Instructor approves request	Status updated to Approved	As expected	Pass	Student notified

Table 5.8: Test Case Execution – Book Meeting (Time Slot Conflict)

Test Case Title		Book Meeting on Already Booked Slot			
Test Date		1/18/2026			
Test Scenario		Office Hours (Book Meeting)			
Test Designed By		Mohammad Assi			
Test Description		Student attempts to book a time slot that conflicts with an existing meeting			
Pre-Conditions		- Student is authenticated. - Instructor has office-hour slots. - Selected slot overlaps with an existing pending/approved meeting.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Select instructor and slot	System checks slot availability	As expected	Pass	
2	Submit meeting request	Request rejected due to conflict	As expected	Pass	Error message shown

Table 5.9: Test Case Execution – Book Meeting (Denied)

Test Case Title		Book Office Hours Meeting – Denied by Instructor			
Test Date		1/18/2026			
Test Scenario		Office Hours (Book Meeting)			
Test Designed By		Mohammad Assi			
Test Description		Student submits a meeting request and instructor denies it			
Pre-Conditions		- Student is authenticated. - Instructor has office-hour slots. - Selected time slot is valid and available at submission time.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Select instructor and time slot	System validates slot availability	As expected	Pass	
2	Submit meeting request	Request created with status Pending	As expected	Pass	Instructor notified
3	Instructor denies request	Status updated to Denied	As expected	Pass	Optional reason shown
4	Student receives notification	Denial notification displayed	As expected	Pass	

5.4 Schedule Viewing (Daily / Weekly) Testing

The section presents the test case execution results for the Daily and Weekly Schedule Viewing feature of the BZU Assistant system. The test cases validate correct schedule retrieval, view switching between daily and weekly modes. These tests ensure that users can reliably access accurate academic schedules.

Table 5.10: Test Case Execution – Schedule Viewing (Daily View)

Test Case Title		View Daily Academic Schedule			
Test Date		1/19/2026			
Test Scenario		Daily Schedule Viewing			
Test Designed By		Mohammad Assi			
Test Description		User views academic schedule for a specific day			
Pre-Conditions		- User is authenticated. - User has registered courses with scheduled sessions.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open schedule module	Daily schedule interface displayed	As expected	Pass	It takes longer time

Table 5.11: Test Case Execution – Schedule Viewing (Weekly View)

Test Case Title	View Weekly Academic Schedule				
Test Date	1/19/2026				
Test Scenario	Weekly Schedule Viewing				
Test Designed By	Mohammad Assi				
Test Description	User views academic schedule for the entire week				
Pre-Conditions	- User is authenticated. - User has multiple courses scheduled across the week.				
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open schedule module	Schedule interface displayed	As expected	Pass	
2	Switch to weekly view	Weekly timetable displayed	As expected	Pass	
3	Scroll through week	All courses visible and ordered	As expected	Pass	

5.5 Academic Event Creation Testing

The section presents the test case execution results for the Academic Event Creation feature of the BZU Assistant system. The test cases validate event creation by instructors, QR code generation, and successful calendar integration on student devices. These scenarios ensure that academic events are created correctly, shared securely, and reflected accurately in student calendars.

Table 5.12: Test Case Execution – Academic Event Creation (Successful)

Test Case Title		Create Academic Event Successfully			
Test Date		1/19/2026			
Test Scenario		Academic Event Creation			
Test Designed By		Mohammad Assi			
Test Description		Instructor creates a valid academic event			
Pre-Conditions		Instructor is authenticated. Instructor has permission to create events.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open event creation form	Event form displayed	As expected	Pass	
2	Enter event details	Inputs validated successfully	As expected	Pass	
3	Submit event	Event saved successfully	As expected	Pass	
4	Generate QR code	QR code displayed for event	As expected	Pass	

Table 5.13: Test Case Execution – Academic Event Creation (Invalid Input)

Test Case Title		Create Academic Event with Invalid Data			
Test Date		1/19/2026			
Test Scenario		Academic Event Creation			
Test Designed By		Mohammad Assi			
Test Description		Instructor attempts to create an event with invalid input			
Pre-Conditions		Instructor is authenticated.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open event creation form	Event form displayed	As expected	Pass	
2	Enter invalid event data	Validation errors shown	As expected	Pass	
3	Submit event	Event not created	As expected	Pass	

Table 5.14: Test Case Execution – Academic Event (Calendar Integration)

Test Case Title		Add Academic Event to Calendar via QR Code			
Test Date		1/19/2026			
Test Scenario		Academic Event Calendar Integration			
Test Designed By		Mohammad Assi			
Test Description		Student adds academic event to calendar by scanning QR code			
Pre-Conditions		Event exists. Student is authenticated. Student device supports calendar integration.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Scan event QR code	Event data decoded	As expected	Pass	
2	Confirm calendar ad- dition	Event added to calendar	As expected	Pass	
3	View calendar entry	Event details dis- played	As expected	Pass	Correct time/date

Table 5.15: Test Case Execution – Download Event QR Code Image

Test Case Title		Download Event QR Code Image from Event Manager			
Test Date		1/19/2026			
Test Scenario		Event Management – QR Code Download			
Test Designed By		Fadi Halaweh			
Test Description		Doctor downloads the QR code image of an academic event from the Event Manager page			
Pre-Conditions		Doctor is authenticated. Doctor has access to Event Manager page. Academic event exists. Device has sufficient storage permissions.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open Event Manager page	Event Manager page is displayed	As expected	Pass	
2	Select event and click “View Event”	Event details page opens	As expected	Pass	
3	Click “Download”	QR code image is downloaded and success message appears	As expected	Pass	
4	Verify stored image location	Image stored in Pictures/BZU_QRcodes directory	As expected	Pass	Correct directory
5	Open device gallery	Downloaded QR code image is visible in gallery	As expected	Pass	Image displayed correctly

5.6 Attendance Tracking Testing

The section presents the test case execution results for the Attendance Tracking feature of the BZU Assistant system. The test cases validate automatic and manual attendance recording, proper handling of BLE beacon connectivity, and accurate persistence of attendance data. These tests ensure that instructors can reliably monitor student attendance, that automatic attendance mechanisms function correctly, and that fallback procedures are in place when technical issues arise.

Table 5.16: Test Case Execution – Manual Attendance Using BLE Control Page

Test Case Title	Record Manual Attendance via Control Classroom Beacons				
Test Date	1/22/2026				
Test Scenario	BLE Attendance Management				
Test Designed By	Fadi Halaweh				
Test Description	Instructor manually records student attendance using the Control Classroom Beacons page				
Pre-Conditions	Instructor is authenticated. Instructor has access to Control Classroom Beacons page. Class session is active. Room “CS101” exists and is configured with BLE beacons. Student “Mohammad Assi” is enrolled in the course.				
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open Control Classroom Beacons page	Control page is displayed	As expected	Pass	
2	Select “Manual Attendance” option	Manual attendance mode enabled	As expected	Pass	
3	Select room “CS101”	Room CS101 is selected successfully	As expected	Pass	BLE room active
4	Select student “Mohammad Assi”	Student is marked as present	As expected	Pass	Student selected
5	Click “Save Attendance”	Attendance record is saved successfully	As expected	Pass	Confirmation message shown
6	Verify attendance record	Student attendance appears in session records	As expected	Pass	Data stored correctly

Table 5.17: Test Case Execution – Start Automatic Attendance via BLE Beacon

Test Case Title		Start Automatic Attendance via Control Classroom Beacons			
Test Date		1/22/2026			
Test Scenario		BLE Automatic Attendance Activation			
Test Designed By		Fadi Halaweh			
Test Description		Instructor activates BLE beacon to start automatic attendance collection			
Pre-Conditions		Instructor is authenticated. Instructor has access to Control Classroom Beacons page. Room “CS101” exists and is configured with BLE beacon. BLE beacon device is powered and connected to the system.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open Control Classroom Beacons page	Control page is displayed	As expected	Pass	
2	Select “Automatic Attendance” option	Automatic attendance mode enabled	As expected	Pass	
3	Select room “CS101”	Room CS101 is selected successfully	As expected	Pass	BLE room ready
4	Press “ON” button	Button color changes to green	As expected	Pass	Visual confirmation
5	Verify system message	Message “Start command has been sent” is displayed	As expected	Pass	Status notification shown
6	Verify BLE beacon status	BLE beacon enters listening mode	As expected	Pass	Ready to collect student data

Table 5.18: Test Case Execution – Stop Automatic Attendance via BLE Beacon

Test Case Title		Stop Automatic Attendance via Control Classroom Beacons			
Test Date		1/22/2026			
Test Scenario		BLE Automatic Attendance Deactivation			
Test Designed By		Fadi Halaweh			
Test Description		Instructor stops BLE beacon to end automatic attendance collection			
Pre-Conditions		Instructor is authenticated. Instructor has access to Control Classroom Beacons page. Room “CS101” exists and is configured with BLE beacon. Automatic attendance is currently active.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Open Control Classroom Beacons page	Control page is displayed	As expected	Pass	
2	Select “Automatic Attendance” option	Automatic attendance mode visible	As expected	Pass	Active state shown
3	Select room “CS101”	Room CS101 is selected	As expected	Pass	Beacon active
4	Press “OFF” button	Button color changes from green to default	As expected	Pass	Visual feedback
5	Verify system message	Message “Stop command has been sent” is displayed	As expected	Pass	Status notification shown
6	Verify BLE beacon status	BLE beacon exits listening mode	As expected	Pass	Attendance stopped
7	Verify attendance session closure	Attendance session is closed and saved	As expected	Pass	Data stored correctly

Table 5.19: Test Case Execution – Automatic BLE Beacon Shutdown After Session End

Test Case Title	Automatic Stop of BLE Attendance After Session End				
Test Date	1/22/2026				
Test Scenario	BLE Automatic Attendance Deactivation				
Test Designed By	Fadi Halaweh				
Test Description	System automatically stops BLE beacon when the scheduled session time ends				
Pre-Conditions	Instructor is authenticated. Instructor has access to Control Classroom Beacons page. Room “CS101” exists and is configured with BLE beacon. Automatic attendance is active. Class session has a predefined end time.				
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	Start automatic attendance for room “CS101”	BLE beacon enters listening mode	As expected	Pass	Session active
2	System monitors session time	Remaining session time tracked	As expected	Pass	Timer running
3	Session end time is reached	End-of-session event triggered	As expected	Pass	Scheduled timeout
4	System sends stop command to BLE beacon	Stop command issued automatically	As expected	Pass	No instructor action
5	Verify BLE beacon status	BLE beacon exits listening mode	As expected	Pass	Attendance stopped
6	Verify attendance data persistence	Attendance data is saved and session closed	As expected	Pass	Data stored correctly

5.7 Schedule Sharing

The section presents the test case execution results for the Schedule Sharing feature of the BZU Assistant system. The test cases validate that users can successfully share their academic schedules with other users, configure appropriate sharing permissions, and ensure that shared schedules are accurately reflected across different devices. These tests ensure that students and instructors can reliably disseminate and access schedule information, promoting better coordination and collaboration within the academic environment.

Table 5.20: Test Case Execution – Student shares Schedule with another Student Via QR-Code (Successful)

Test Case Title		Schedule Sharing between Students			
Test Date		1/19/2026			
Test Scenario		Schedule Sharing			
Test Designed By		Fadi Halaweh			
Test Description		Student shares his Schedule with another Student			
Pre-Conditions		Both Students are authenticated. Both Students have permission to share Schedule.			
Test Case Execution					
Step ID	Step Description	Expected Result	Actual Result	Pass / Fail	Additional Notes
1	First student open Schedule Sharing from the home Screen	QR Code displayed	As expected	Pass	
2	Second Student opens schedule sharing	Inputs validated successfully	As expected	Pass	
3	Second Student scans the QRCode of the First Student	First Student personal information shown	As expected	Pass	
4	Second student clicks View Schedule	a Weekly schedule of the First student is shown	As expected	Pass	

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presented the design and implementation of BZU Assistant, an Android-based digital identification and academic service platform developed for Birzeit University. The system was designed to modernize and streamline essential academic and administrative processes by replacing fragmented, paper-based, and manual workflows with an integrated mobile-first solution.

The application successfully integrates multiple core services, including digital identity management, automated attendance tracking using Bluetooth Low Energy (BLE), QR code-based device and classroom reservations, office hours scheduling, academic schedule visualization, and academic event creation with calendar integration. By leveraging Flutter for mobile development, Supabase for backend services, and Raspberry Pi devices for BLE-based automation, the system demonstrates a practical and scalable smart campus architecture.

Testing and evaluation confirmed that the system meets its functional and non-functional requirements, providing reliable performance, secure authentication, and an intuitive user experience for students, instructors, and secretaries. The results show that an Android-centric approach combined with BLE and QR-based interactions can significantly reduce administrative overhead, improve accuracy, and enhance coordination across academic workflows.

Overall, BZU Assistant achieves its stated objectives and provides a solid foundation for future expansion. The project demonstrates the feasibility of deploying a real-world smart campus solution that aligns with Birzeit University's digital transformation goals.

6.2 Future Work

Although the BZU Assistant system provides a comprehensive set of academic and administrative services, several enhancements can be considered for future development to improve

scalability, usability, and system intelligence.

1. iOS Platform Support

Extend the application to support iOS devices, enabling broader accessibility for university members while addressing platform-specific limitations related to Bluetooth Low Energy (BLE) background execution.

2. Advanced Attendance Analytics

Introduce analytical dashboards for instructors and administrators that provide attendance statistics, trends, low-attendance alerts, and exportable reports to support academic decision-making.

3. Enhanced BLE Localization Accuracy

Improve attendance detection accuracy by integrating advanced BLE fingerprinting techniques, adaptive RSSI calibration, or machine learning–based localization methods to mitigate environmental interference.

4. Integration with University Information Systems

Integrate the system with existing university platforms such as course registration, grading systems, and learning management systems to enable automated data synchronization and reduce manual data handling.

5. Offline and Fault-Tolerant Operation

Add offline support for selected system features, allowing attendance data and reservations to be temporarily cached during network outages and synchronized automatically once connectivity is restored.

6. Security and Privacy Enhancements

Strengthen system security by introducing multi-factor authentication, encrypted QR code payloads, and detailed audit logs to enhance data protection and system trustworthiness.

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Appendix A

Use Case Specification

The appendix presents supplementary material that supports the main body of the report. It includes the complete specifications of the remaining use case descriptions that were not fully detailed in Chapter 3, as well as additional diagrams and images related to system functionality and workflows. These materials are provided for reference purposes to enhance clarity and ensure completeness of the system documentation.

Table A.1: UC-04 Office Hours Management

Use Case ID	UC-04
Use Case Name	Office Hours Management
Created By	Mohammad Assi
Date Created	1/5/2026
Primary Actor	Instructor, Student
Secondary Actors	
Trigger	The instructor defines office hour availability, or a student attempts to book an office hour appointment.
Description	BZU Assistant enables instructors to create, manage, and publish their office hour schedules through the BZU Assistant system. Students can view available office hour slots and book appointments either by scanning a QR code shared by the instructor or by searching for the instructor within the application. The system ensures conflict-free scheduling and records all appointments.
Preconditions	1. The instructor is authenticated in the BZU Assistant application. 2. The student is authenticated in the BZU Assistant application. 3. The instructor has defined at least one office hour slot.
Postconditions	1. An office hour appointment is successfully scheduled or updated. 2. Both instructor and student can view the appointment details in their schedules.
Normal Flow	1. The instructor opens the BZU Assistant application. 2. The instructor creates or updates office hour availability. 3. The system publishes the available office hour slots. 4. The student scans the instructor's QR code or searches for the instructor. 5. The student selects an available time slot. 6. The system confirms the booking and records the appointment.
Alternative Flows	A1. Slot Already Booked 1. The student selects an office hour slot that has been booked. 2. The system displays an unavailability message and prompts the student to choose another slot. A2. Appointment Cancellation 1. The instructor or student selects an existing appointment. 2. The appointment is cancelled. 3. The system updates the availability accordingly.
Exceptions	E1. Schedule Conflict The system prevents overlapping office hour appointments. E2. Backend Service Unavailable The system fails to process the request and notifies the user.
Priority	Medium
Other Information	1. Office hours can be accessed via QR code or in-app search. 2. The system maintains a complete history of office hour appointments.
Assumptions	1. Instructors regularly maintain accurate office hour availability. 2. Students book appointments within published office hour times.

Table A.2: UC-05 Schedule Viewing and Sharing

Use Case ID	UC-05
Use Case Name	Schedule Viewing and Sharing
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Student, Instructor
Secondary Actors	
Trigger	The user opens the schedule section of the application or scans a shared schedule QR code.
Description	BZU Assistant enables students and instructors to view their personalized academic schedules, including lectures, office hours, reservations, and events. Users can also share their schedules with others through dynamically generated QR codes, enabling quick access without manual searching. The system ensures that displayed schedules are always synchronized with backend data.
Preconditions	1. The user is authenticated in the BZU Assistant application. 2. The user has an existing academic schedule stored in the system.
Postconditions	1. The user's schedule is displayed successfully. 2. A schedule QR code is generated and can be shared with other users (if requested).
Normal Flow	1. The user opens the BZU Assistant application. 2. The user navigates to the schedule section. 3. The system retrieves the user's daily or weekly schedule. 4. The system displays the schedule in a structured view. 5. The system generates a QR code representing the schedule.
Alternative Flows	A1. Schedule Viewed via QR Code 1. A user scans a schedule QR code located on digital ID QR code. 2. The system retrieves the shared schedule data. 3. The schedule is displayed in read-only mode. A2. No Scheduled Items 1. The user opens the schedule section. 2. The system displays an empty schedule message.
Exceptions	E1. Backend Data Unavailable The system fails to retrieve schedule data and displays an error message. E2. Invalid or Expired QR Code The system notifies the user that the schedule link is no longer valid.
Priority	Medium
Other Information	1. Schedules can be viewed in daily or weekly formats. 2. Shared schedules are read-only to preserve data integrity.
Assumptions	1. Schedule data is kept up to date by the backend system. 2. Users only share schedules with authorized individuals.

Table A.3: UC-06 Academic Event Creation and Calendar Integration

Use Case ID	UC-06
Use Case Name	Academic Event Creation and Calendar Integration
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Instructor, Student
Secondary Actors	Android Calendar
Trigger	The instructor creates an academic event and chooses to publish or share it with students.
Description	BZU Assistant enables instructors to create academic events such as lectures, make-up classes, announcements, or special sessions using the BZU Assistant system. Once created, the system generates a QR code that students can scan to view event details and add the event directly to their mobile calendar on Android devices.
Preconditions	1. The instructor is authenticated in the BZU Assistant application. 2. The instructor has permission to create academic events. 3. Students use Android devices with calendar access enabled.
Postconditions	1. The academic event is stored in the backend system. 2. Students can add the event to their mobile calendars.
Normal Flow	1. The instructor opens the BZU Assistant application. 2. The instructor selects the option to create a new academic event. 3. The instructor enters event details (title, date, time, location, description). 4. The system saves the event and generates a unique QR code. 5. The instructor shares the QR code with students. 6. A student scans the QR code using the application. 7. The system displays the event details and prompts calendar integration. 8. The event is added to the student's Android calendar.
Alternative Flows	A2. Calendar Permission Denied 1. The student denies calendar access permission. 2. The system displays a warning and skips calendar integration.
Exceptions	E1. Invalid Event Data The system rejects the event creation and prompts the instructor to correct the input. E2. Backend Service Unavailable The system fails to save the event and notifies the instructor.
Priority	Medium
Other Information	1. Events are read-only for students. 2. QR codes link securely to event metadata stored in the backend.
Assumptions	1. Instructors share event QR codes only with relevant students. 2. Students grant calendar access permissions when required.

Table A.4: UC-07 Notifications Management

Use Case ID	UC-07
Use Case Name	Notifications Management
Created By	Fadi Halaweh
Date Created	1/15/2026
Primary Actor	Student, Instructor
Secondary Actors	
Trigger	A system event occurs (e.g., attendance recorded, reservation update, office hour booking, or academic event creation), or a user updates notification preferences.
Description	BZU Assistant system manages the generation, delivery, and presentation of system notifications within the BZU Assistant application. Notifications inform users about important events such as attendance status, reservation confirmations or cancellations, office hour updates, and newly created academic events. Users can view notifications in-app and control notification preferences.
Preconditions	1. The user is authenticated in the BZU Assistant application. 2. Notification permissions are granted on the Android device.
Postconditions	1. Relevant notifications are delivered to the user. 2. Notifications are stored and accessible within the application.
Normal Flow	1. A system event triggers a notification. 2. The backend generates the notification message. 3. The notification is sent to the user's device. 4. The user receives the notification on the Android device. 5. The notification is stored in the in-app notification list.
Alternative Flows	A1. Notifications Disabled 1. The user disables notifications in application settings. 2. The system suppresses push notifications. 3. Notifications remain accessible within the application only. A2. User Views Notification History 1. The user opens the notifications section. 2. The system displays a list of past notifications.
Exceptions	E1. Notification Service Unavailable The system fails to deliver the notification and retries later. E2. Permission Denied The system displays a warning and limits notifications to in-app messages.
Priority	Medium
Other Information	1. Notifications may be delivered as push notifications or in-app messages. 2. Notification content is contextual and role-based.
Assumptions	1. Users allow notification permissions on their Android devices. 2. Backend notification services are operational and reliable.

Table A.5: UC-08 Reserve Classroom use case description

Use Case ID	UC-08
Use Case Name	Reserve Classroom
Created By	Mohammad Assi
Date Created	1/15/2026
Primary Actor	Student, Instructor
Secondary Actors	Secretary
Trigger	The user initiates a classroom reservation request by selecting a classroom or scanning its QR code.
Description	BZU Assistant enables the students and instructors to reserve university classrooms for exams, makeup lectures, or special academic events. The system validates classroom availability, checks for scheduling conflicts, and forwards the request to the secretary for approval or denial.
Preconditions	1. The user is authenticated in the BZU Assistant application. 2. The classroom is registered in the system. 3. The classroom is available for reservation during the selected time period.
Postconditions	1. The classroom reservation is successfully recorded in the system. 2. The classroom status is updated to reflect the reservation.
Normal Flow	1. The user opens the classroom reservation interface. 2. The user scans the QR code of a classroom, or selects a building and classrooms. 3. The system displays classroom details such as the floor, capacity, and available equipments. 4. The user selects a reservation date and time range and enters the purpose of reservation. 5. The system validates the reservation request and checks for time conflicts. 6. The user confirms the reservation request. 7. The system stores the reservation with status Pending and notifies the secretary. 8. The secretary reviews and approves the reservation request. 9. The system updates the reservation status to Approved and reserves the classroom.

Alternative Flows	A1. Classroom Unavailable 1. The user selects a classroom that is already reserved. 2. The system displays an unavailability message that the classroom is already reserved. A2. Reservation Cancellation 1. The user selects an existing reservation. 2. The user cancels the reservation. 3. The system updates the classroom status to available. A3. Reservation Denied 1. The secretary denies the reservation request. 2. The system updates the reservation status to Denied. 3. The system notifies the user of the decision.
Exceptions	E1. QR Code Invalid or Unreadable The system displays an error message and prompts the user to retry or search manually. E2. Backend Service Unavailable The reservation request fails and the system notifies the user to try again later.
Priority	Medium
Other Information	1. Reservations support classrooms using a unified workflow. 2. Conflicting reservations are prevented by the system.
Assumptions	1. QR codes are correctly assigned to classrooms.